The Geography of the Local Service Multiplier

Rachel Bryant[1]

University of Chicago

November 9, 2026

Please click here for the latest version.

Abstract. State and local governments spend approximately $20 billion a year on firm-specific subsidies, rationalized by the direct jobs these firms create and the indirect jobs their payroll supports. This paper studies, conditional on firm entry, the strength of the local service multiplier and how it depends on commuting and shopping networks in a city. Using employer-employee linked Census data, I find that about half of the entrant's new employees were not previously attached to the city labor market. Whether these workers migrate, commute from elsewhere, or were previously unemployed determines how much of the entrant's payroll becomes new local spending. I build a spatial equilibrium model across neighborhoods, calibrated to these rates of migration, commuting, and labor force participation, in which residents' choices of where to shop determine where their income is spent. In the model, the service multiplier is smaller than most existing estimates. For firm entries of the same relative size, the multiplier is about 2.7 times as large at the 90th percentile as at the 10th. Aggregate characteristics of the local economy explain only a small share of this variation, and much of the rest reflects the commuting and shopping flows around the entrant's specific location. Finally, I combine the model with a new dataset of subsidized firm entry events that records promised jobs, realized jobs, and the state and local portions of a subset of subsidy award. In most events, the added residential property tax from the entry covers the local share of the award, and residents' welfare gains exceed the award's full cost, assuming the firm would not have entered otherwise. However, the same subsidy can deliver substantially different welfare gains at other locations in the same city.

[1]I am grateful to my advisors Milena Almagro, Erik Hurst, and Esteban Rossi-Hansberg for their guidance, support, and encouragement. This paper benefited from comments from participants in the UEA North America Meeting, UEA Summer School at LSE and the Trade & Spatial Working Group at the University of Chicago. Any views expressed are those of the author and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 3321 (CBDRB-FY27-P3321-R13471). This research uses data from the Census Bureau's Longitudinal Employer-Household Dynamics Program, which was partially supported by the following National Science Foundation Grants SES-9978093, SES-0339191 and ITR-0427889; National Institute on Aging Grant AG018854; and grants from the Alfred P. Sloan Foundation. All errors are my own.

1 Introduction

State and local governments in the United States frequently provide subsidies to tradable firms as a development tool to boost local employment and increase economic activity. Recent estimates put this spending at roughly $20 billion per year over 2011–2021 (Hanson et al., 2025). In 2014, states spent between $5 and $216 per capita on business incentives, and in the typical state these outlays were close to 40 percent of corporate tax revenue (Slattery and Zidar, 2020).

Local officials argue that the benefits reach beyond the subsidized tradable firm because new workers spend at local service establishments such as restaurants and stores. In this narrative, the entrant firm's own hiring is the first step in a broader economic expansion. A city mayor expresses it this way:

> "A lot of them [new workers] live within the city of Reed City. So, they own homes here and pay property taxes. But even if they're not here inside the city, they're doing business here. They're supporting our grocery stores, they're supporting our gas stations. They're supporting our local restaurants."
>
> —Mayor of Reed City, Michigan

This paper asks whether that narrative holds by measuring how many service jobs follow the entry of a subsidized tradable firm and where those jobs appear. While these subsidized tradable firms are among the largest half a percent of establishments in the United States, they are small relative to the labor market of the county and commuting zone (CZ) they enter. The median subsidized firm promises over $70 million in new payroll, yet it employs less than one percent of its county's workers and approximately 0.16 percent of its commuting zone's labor force. Because of the scale mismatch, the impact of firm entry is obscured by county- and CZ-level economic trends.

To overcome this challenge, I conduct my analysis at the neighborhood level. I combine employer-employee linked Census microdata on where each entrant's hires previously lived and worked, visits residents make to service establishments, a new subsidy dataset that distinguishes promised from actual jobs and local from state subsidies for a subset of awards, and a quantitative general equilibrium model of Census tracts to ask three questions: Does the increase in service demand from the entrant's new hires lead to job gains and higher resident welfare? What mechanisms determine the scale of these gains? What proportion of the gains stays in the local jurisdiction paying for the subsidy, and how large is the financial leakage?

To help answer these questions, I trace the chain from the entrant's payroll to local service jobs. First, I use worker histories and earnings to approximate how much of the payroll becomes new income for CZ residents. Second, I observe where residents of each census tract travel to buy services using cell phone ping data. I determine what percentage of total household shopping trips to service firms is done at local establishments. Third, I present quasi-experimental evidence on service employment from a difference-in-differences design using a tract-level exposure measure motivated by the two aforementioned pieces of motivating evidence.

The number of promised jobs is not the correct metric for measuring an increase in local income and its subsequent service demand. Those are impacted by whether the hired workers are existing residents, commuters, or migrants. I find that 58 percent of hires are continuing residents of the CZ, 28 percent live outside it after hire, and 14 percent move in. Additionally, 50 percent previously worked in the CZ. An approximate accounting from the Census microdata shows that, for every $100 of entrant payroll, $28 goes to workers living outside the CZ and commuting in. Subtracting a further $40 in pre-hire earnings of workers who previously lived in the CZ leaves $32 in additional CZ resident income. About $15 comes from earnings paid to workers who migrate into the CZ, while $17 represents the net income gain among workers who lived there before hire.

National accounts data show that approximately 48 percent of after-housing consumer spending goes to services that are consumed locally. Routing income through the full matrix of shopping flows conflates the entrant's payroll with every other income shock in the CZ. To better isolate the relationship between the income of an entrant's workers and their contribution to the nearby service sector, I use cell phone ping data. This shows that 85 percent of shopping trips to these services stay within the shopper's home CZ. A third of visits end in the home tract or a neighboring tract, and another third end at a shopping hub.[2] I use the third of spending that ends in the home tract or its neighbors to identify the impact of the entrant, because that is the piece that varies across space.

This paper presents quasi-experimental evidence on how service employment responds across neighborhoods after a firm enters, using a difference-in-differences design in which tracts differ in their exposure to the spending of the entrant's new hires. To define the exposure metric, I locate the prior residential tracts of entrants' new workers and allocate their potential service spending over their home and adjacent tracts, reflecting the pattern of spending from the cell phone ping data that varies within a CZ. Within each event, I rank tracts by this exposure and compare the top and bottom quartiles of tracts before and after entry in a stacked event study. I restrict the sample to commuting zones with no other large entrant. Seven years after entry, the top quartile has gained 60 service jobs and $5 million in service payroll relative to the bottom quartile of exposed tracts. The response is concentrated in the neighborhoods where the entrant's workers live, and gains of this size in a small number of tracts are difficult to detect in county totals.

These three pieces of evidence each trace one link – worker sourcing, shopping patterns, and service employment across neighborhoods. However, they cannot account for the full general equilibrium response of employment, wages and rents. To study these equilibrium effects, I build a model in which each CZ is a set of census tracts and an outside location. Individuals of two skill types decide where to live, work, and shop. Individuals can commute from an outside location into the CZ, and migrants move to the CZ from a national outside option. In the model, there are both tradable and nontradable firms of either high- or low-skill intensity. I model firm entry

[2] I identify shopping hubs the way the urban literature identifies employment subcenters (Giuliano and Small, 1991; McMillen, 2001): a hub is a contiguous cluster of tracts that forms a local peak in service visits and receives at least 2 percent of its commuting zone's visits, with peaks within ten minutes' drive of one another merged into one hub (Appendix A.5).

as a productivity increase in the entrant's tract and sector, with the increase scaled so that employment rises by the firm's promised number of jobs. Individuals choose whether to work and in which sector. Therefore, the entrant affects service firms through the increased demand for services resulting from higher incomes and through higher wages and land rents. Additionally, individuals travel to consume services, incurring a cost that makes them sensitive to distance. These service trips can originate from both an individual's home and workplace. The migration, participation, sector choice, commuting, and shopping elasticities jointly determine how much of the entrant's payroll becomes local spending and where that spending lands. Tradable firms at other locations shrink as the entrant increases equilibrium wages and commercial rents, consistent with the modest employment decline in the tradable industries most exposed to the entrant's hiring I observe in the data. Service firms are monopolistically competitive and enter freely. Both residents and firms consume land, and residents earn rental income through ownership of residential land while commercial rents accrue to absentee landlords. The model yields welfare by skill, the fiscal cost of a job, and the share of the response that stays inside the jurisdiction that pays for it. The model isolates the effect of the entering firm, removing confounding trends that obscure the empirical comparisons at the county and CZ level.

I quantify the model separately for each of the firm entrant events in its CZ. Two of the recruitment elasticities, inbound commuting and labor participation, are calibrated by minimum distance to the hiring shares I observe in the employer-employee linked data. The migration and sector-choice elasticities are estimated with shift-share designs of industry growth instrumenting for cross-CZ variation in earnings, and the commuting and shopping elasticities are estimated from gravity equations on tract-pair flows. The remaining spending, cost, and supply parameters come from the national accounts, the shopping trip data, and the literature. Given these parameters, I invert the model to recover amenities, productivities, and fixed costs so that it reproduces each commuting zone's economy before entry. None of these steps directly uses the difference in service employment across tracts, so the event-study estimates are untargeted moments. To validate the model, I solve the equilibrium with and without each entrant, run the difference-in-differences design on the model's output, and compare the model-implied event-study coefficients with their empirical counterparts.

With the model quantified, I return to the three initial questions, taking into account the full general equilibrium effects. First, does the increase in service demand from the entrant's new hires lead to job gains and higher resident welfare? Across events, an entry adds a median of 74 jobs to the commuting zone for every 100 it promises. Competition for workers and floor space removes 49 jobs at tradable firms in other locations, but increased service demand supports 22 additional service jobs. Measured per net tradable job, as in the literature, the service multiplier is about 44 service jobs per 100. Of each \$100 of additional payroll at the entrant's site, residents' earnings rise by about \$40, compared with \$32 in the hired-cohort accounting, and about \$25 reaches service firms in the CZ, because the rest goes to housing, tradable goods, and service firms elsewhere. Residents' welfare rises by about \$7,600 a year per entrant job, or 8 percent of the average wage

in the entrant's tract and industry, because each hire gains only the new wage net of their outside option.

What determines the scale of these gains? Differences in entry size explain little. Across markets, the multiplier is about 2.4 times as large at the 90th percentile event as at the 10th, ranging from about 14 to 33 service jobs per 100 entrant jobs, and fixing every entry at 500 jobs removes only about a fifth of that variance. Aggregate characteristics of the local economy explain only about 21 percent of the variation within industries; market size alone explains about 8.5 percent, with smaller multipliers in larger markets. Within a county, commuting and shopping exposure on their own explain 77 percent of the variation in service-job gains across neighborhoods. The entrant's industry also matters. A high-skill-intensive entrant supports about 1.6 times as many service jobs per entrant job as a low-skill-intensive one and about 1.2 times as many per dollar of payroll, so higher pay explains about 63 percent of the gap. Much of the remaining variation comes from the particular site. Holding the promised jobs and the award fixed, I move each entry across up to ten alternative sites in its host county. The observed site ranks at the 40th percentile of residents' gain at the median, the site at the 90th percentile adds about 1.8 times as many service jobs as the site at the 10th, and the best alternative raises residents' welfare by 48 percent.

How much of the gain stays with the jurisdiction that pays for it? The award data I build separates the local share of each subsidy from the state share for about one fourth of entry events. In the majority of cases, it is the county that grants the subsidy. At the median, the county holds 28 percent of the CZ's residents and receives 60 percent of the gain in resident income, but it keeps 64 percent of the CZ's service job gain. The rest goes to the neighboring counties where the entrant's workers live and shop. The median annual tax expenditure is \$3,600 per net CZ job and \$3,000 per net host-county job. Median resident welfare per public dollar is \$2.8 across the CZ and \$0.86 in the host county. It is important to note that these are annual benefit comparisons conditional on entry, rather than an estimate of the marginal effect of providing the subsidy.

Taken together, these results connect the entrant's employment commitment to the payroll needed to fill its jobs, the spending that reaches local businesses, and the gains accruing to residents. They distinguish regional benefits from the return to the jurisdiction offering an award. I find that the spillovers officials promise exist but are smaller than the promised jobs imply, and that their size depends on who fills the jobs, where those workers live and shop, and the entrant's industry. Evaluating subsidies requires accounting for both the employment and welfare created throughout the local economy and the share of those gains received by the residents bearing their cost.

Contribution to the literature. This paper makes three contributions. First, it measures the local multiplier of individual subsidized plants and traces it from payroll to service employment. Estimates from industry-wide growth (Moretti, 2010; Van Dijk, 2016), from shift-share designs (Bartik, 1991; Bartik and Sotherland, 2019; Autor et al., 2013, 2021; Hershbein and Stuart, 2024; Yagan, 2019; Hornbeck and Moretti, 2024; Acemoglu and Restrepo, 2020), and from regional variation in military procurement (Nakamura and Steinsson, 2014) treat the county, the CZ, or the

state as a single point and do not observe who fills the new jobs (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Plant-opening and subsidized-firm designs (Greenstone et al., 2010; Patrick, 2016; Patrick and Partridge, 2022; Slattery and Zidar, 2020; Hanson and Rohlin, 2024) observe the firm but not its workers, and county employment net of the entrant is too noisy to recover the effect of a single firm whose hiring is a small fraction of total employment. Evaluations of larger place-based programs do find local gains. Manufacturing employment in Tennessee Valley Authority counties was still growing faster than in comparison counties four decades after federal transfers lapsed (Kline and Moretti, 2014b), and federal Empowerment Zone designation raised employment in zone neighborhoods by 12 to 21 percent (Busso et al., 2013), although California's enterprise zones did not raise employment (Neumark and Kolko, 2010). These programs fund infrastructure, tax credits, or block grants across a region or neighborhood rather than a single firm. I follow the entrant's hires to where they live and shop and add up the links in equilibrium. The results reconcile the two literatures. An entrant's payroll supports local service jobs, but only about half of its own jobs are net additions, because it crowds out other tradable producers. The service response is spread over the neighborhoods where its workers live and shop, and 36 percent of it lands outside the host county, rendering a county-wide comparison incomplete. Because much of the payroll goes to workers who already had jobs in the CZ or who live outside it, the response to a single plant depends on the new income its payroll creates rather than on its headcount.

Second, I show that how open a local market is depends on where its boundary is drawn, and I measure the multiplier at the boundary of the county that pays as well as at the boundary of the CZ. The model has the neighborhood structure of Ahlfeldt et al. (2015) and Tsivanidis (2026) and adds two margins that the entry question requires. Individuals shop from home and from work, as in the consumption-trip model of Miyauchi et al. (2021) and in line with smartphone evidence that consumption trips are short and concentrated (Davis et al., 2019; Agarwal et al., 2020; Oh and Seo, 2023). I add a sector choice where individuals switch between tradable and service employment, so that the entrant's wage pressure can directly impact service firms. Both factor demand and output demand operate in a CZ whose boundary workers and shoppers cross in both directions, with leakage measured from data. Monte et al. (2018) show that local employment elasticities depend on commuting openness. I add shopping openness, which differs from commuting openness at every boundary, and show that the county that pays for an award is considerably more open than the CZ that contains it. The model is solved for a single large employer entering fine geography in 95 actual CZs rather than across counties.

Third, I separate the part of the multiplier that depends on the firm type and the part that depends on the local characteristics of the market, namely the commuting and shopping openness and the housing supply elasticity. I measure each channel by comparing the CZ total to the county that finances the employment. Theory treats mobility as a continuous relocation margin (Kline and Moretti, 2014a; Busso et al., 2013; Fajgelbaum and Gaubert, 2020), and the incidence of state business taxes is divided among workers, landowners, and firm owners with aggregate elasticities (Suárez Serrato and Zidar, 2016, 2023). Garin (2025) argues that local job counts are uninformative

when the jobs go to migrants, commuters, or workers who would have been employed anyway, and the contrast between location-based and hiring-based incentives in enterprise zones (Neumark and Kolko, 2010; Freedman, 2013) shows that the concern is practical. I contribute to this analysis by showing that while award programs measure performance by the jobs a firm promises, most of the variation in what those jobs contribute locally depends on the commuting and shopping openness of the place as well as housing flexibility.

The rest of this paper is organized as follows. Section 2 describes the policy setting and the data. Section 3 presents three facts that follow the entrant's payroll to whom it hires, where those workers shop, and where service employment responds. Section 4 develops the model and Section 5 quantifies it. Section 6 measures the net employment and welfare effects of entry, shows what shopping, commuting, and migration add, and traces where the gains land within the commuting zone. Section 7 asks how much the multiplier varies across markets and across locations within a county, and what each job costs the county that subsidizes it. Section 8 concludes.

2 Data and Context

Firm-specific incentives are one of the largest and fastest-growing forms of place-based economic development policy in the United States. State and local governments lost roughly $19.50 billion per year in potential revenue from 2011 to 2021 in incentive packages for business, double the level of the prior decade (Hanson et al., 2025). Good Jobs First, a nonprofit that tracks subsidies, counts 21 megadeals worth a combined $19.20 billion in 2022 alone (Tarczynska, 2022). A typical package combines income tax credits, job training credits, grants, infrastructure spending, and property tax abatements.

Firm entry. I assemble awards from Good Jobs First, state development agencies, and county records. I match each subsidized firm to its establishment record in Data Axle, a commercial dataset of U.S. businesses, and verify the location of the subsidized site with the Google Maps API. This matching allows me to ensure the firm actually enters. I then observe the exact address of its entry and can assign it an industry code.[3] For approximately 40% of the awards, I also separate the portion offered locally, as property taxes forgone by the county and other local governments, from the portion paid by the state through credits, rebates, and grants. For the remaining awards, I simply retain the amount spent by the state. I measure the jobs a firm actually creates from state compliance reports and other documented employment records. After location site verification, the subsidy dataset contains 226 firm-site events (Appendix A.1). Among the 64 firm entry events with reported employment at least four years after award, 39 percent of firms reach their promised

[3]Good Jobs First is a nonprofit research center in Washington, DC, that tracks economic development subsidies. Its Subsidy Tracker database compiles company-specific awards from federal, state, and local programs using government disclosures and public records requests. Data Axle, formerly Infogroup, maintains a commercial establishment-level database of U.S. businesses compiled from public sources and verified by telephone. Its annual snapshots, available from 1997, record each establishment's name, location, industry, employment, sales, year established, and parent company.

employment, the median firm delivers 79 percent of the jobs promised, and 8 firms more than double their hiring promise (Figure A.1).

I focus on the entry and expansion of large, tradable firms. Consistent with the subsidy literature, I consider a firm "large" if it promises at least 500 new jobs, or 0.50% of the total employment in its commuting zone (CZ). I distinguish tradable industries, whose demand originates primarily in national or international markets, from nontradable industries, which mainly serve local demand (Moretti, 2010; Van Dijk, 2016). Throughout the paper, the service sector whose response I measure comprises retail, education, health, food services, and other services (NAICS 44–45, 61, 62, 722, and 81).[4]

For the motivating empirics, I examine all large tradable entries and expansions, not only subsidized ones. The labor demand and local spending shock depend on actual hires, whether or not the firm's entry decision was financed in part by subsidies. The subsidized firms are already a selected sample based on the firms' bargaining power and on the states that choose to disclose their awards (Slattery and Zidar, 2020). Using the Longitudinal Employer-Household Dynamics (LEHD) data, which link 95% of all private-sector workers to their employer, earnings, and residence, I identify large openings and expansions of tradable establishments in the 29 U.S. states for which I have data. I trace the hiring of each firm to the neighborhoods where its workers live, and then to the neighborhoods where they shop. I measure the response of nontradable employment, payroll, and establishments in those locations. Subsidized deals enter this sample when they meet the same size and industry screen I use to classify large, tradable firms in the public data.

The LEHD sample contains approximately 400 such entrants. I classify an entrant into high- and low-skill-intensive tradables.[5] Table 1 compares the two firm types. I find that high-skill-intensive tradable firms locate in substantially larger markets, with a median population of 1.21 million, compared with 291,000 for low-skill-intensive entrants. High-skill-intensive firms locate closer to the CZ center (defined as the group of tracts with the highest employment totals), with a median driving time of 18.2 minutes to the downtown area rather than 25.5 minutes. Lastly, median annual pay indicates that the entry of a high-skill-intensive firm delivers a much larger payroll shock, but its entry is less common in less densely populated labor markets.

Local markets and jurisdictional boundaries. I use the commuting zone (CZ) to define the local labor market, and set Census tracts as the unit for locations within it. I use CZs because they are contiguous county groupings (unlike MSAs) that maximize residence–workplace connections within each zone (Tolbert and Sizer, 1996), but neither a county nor a CZ is a closed economy. Neither one of these boundaries captures the full effects of a firm entry shock. Across entrants, the mean share of workers who live in the same county is 46%, and at the CZ boundary it rises to 78%.

[4] Because LODES and the Quarterly Workforce Indicators report only two-digit sectors, the public-data outcomes and the model's baseline use all of NAICS 72, which adds accommodation; the national spending shares in Section 3 also include it.

[5] A firm is classified as high skill when its four-digit industry had an above-median share of workers with a bachelor's degree in 2000 and as low skill otherwise.

Table 1: Characteristics of low- and high-skill entrants

	Low-skill entrants	High-skill entrants
Firms (disclosure-rounded)	200	200
Hires		
Annual pay of high-skill hires (2023 $)	57,420	130,900
Annual pay of low-skill hires (2023 $)	44,690	87,240
Site and labor market		
Driving time to CZ center (minutes)	25.54	18.17
CZ population	290,600	1,211,000
Annual CZ population growth, 3 years before entry	0.68	0.61

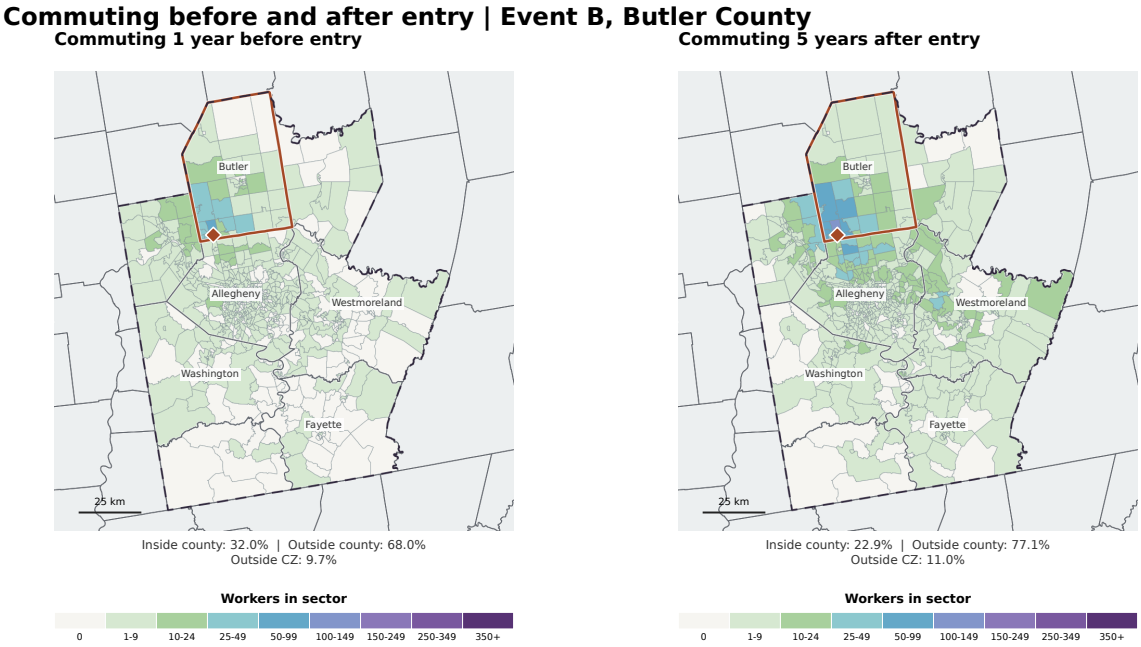
Notes: Released estimates from the LEHD entrant sample. Characteristics are medians across entrants of each type; entrant counts are disclosure-rounded. Pay refers to the entrant's hires by skill and is annual, in 2023 dollars. Proximity to the CZ center is measured in driving minutes. Population growth is the annual percentage change in the CZ from five to two years before entry.

Figure 1 shows both boundaries for one entry into a suburban county of a large CZ that shows a more extreme split between county and CZ. It maps where the people working in the entrant's tract live, one year before the firm enters and five years after. The entrant locates at the edge of its county, next to the CZ's central county. 77 percent of the added workers live outside the host county, but only 11 percent of the tract's workers live outside the CZ five years after entry. Section 6 discusses this entry using the model.

Shopping trips. To measure where residents shop, I use 2019 smartphone visit records from Veraset (Veraset, 2022). I weight each device equally within a month, so that the annual measure is a device-weighted distribution of visits from each home tract to each destination tract. Shopping is local: 84.80% of service visits stay within the shopper's home commuting zone, yet only 10.60 percent of within-CZ visits stay in the home tract (Table 3 in Section 3 and Appendix Table A.12), reflecting the need for a model with shopping trips to capture effects at granular scales. Because the visit data are observed in 2019, after all of my firm entry events, I use them to describe where entry-linked demand may be observable and to estimate shopping gravity. They are not used in the inversion of the model.

Prices, employment, and wages. Land data come from CoreLogic deed and assessor records over 2004–2020, which give residential and commercial repeat-sales and hedonic price indexes, building square footage, and property counts for each tract. Public employment outcomes come from LODES, which records jobs by workplace tract, resident workers by home tract, and the commuting flows between them. I use the Quarterly Workforce Indicators for county earnings by education that I allocate to tracts as workplace wages, and I use the County Business Patterns to provide establishment counts by county and industry. Appendix A documents each source and its construction.

Figure 1: Where the workers at an entrant's tract live, before and after entry



Notes: Public LODES origin–destination counts of primary jobs (JT01) in the broad All Other Services group (SI03) at the entrant's tract, by the worker's residence tract, one year before and five years after entry. These counts include every employer in that group in the tract. This firm entered a business park with the entry of other employers within successive years. The diamond marks the entrant's tract and the outlined area its host county. The denominator for the shares is all workers, so the Outside CZ is included in the Outside County share.

3 Motivating Facts

Fact 1. Only part of entrant payroll becomes additional local income. I use LEHD records for approximately 400 large tradable firm entries and expansions observed between 2004 and 2017, and I classify the new workers by their residence. The average entrant pays its workers about $170 million a year, in 2023 dollars, ranging from $74 million for low-skill-intensive entrants to $270 million for high-skill-intensive entrants.[6] Table 2 shows where these hires live and what they earn. Together these determine how much of new payroll becomes new local income.

Combining the hiring counts and earnings summarized in Table 2 gives an approximate account of where entrant payroll ends up. Of every $100 in new payroll, only about $32 is new income for the CZ's residents. About $28 goes to workers who live outside the CZ, and another $40 replaces what local hires earned at their previous jobs. Of the new $32, $15 is paid to workers who move into the CZ and $17 is the net earnings gain of hires who already lived there.[7]

[6]Total annual earnings of the entrants' hires, measured one year after hire, are $69 billion across about 400 entrants, a disclosure-rounded count with about 200 of each type.

[7]The earnings allocation is an approximation. It weights the joint residence–workplace individual counts by their median earnings, pooling prior industries and individual skills within entrant type and prior workplace. Then I scale the prior and after-hire allocations to the earnings totals within entrant type and post-hire residence. I assume hires with no observed prior job have zero prior income.

Table 2: Entrant hires, earnings, and payroll

A. Residence

	Percent of hires
Continuing CZ residents	57.79
Live outside the CZ	27.90
Migrated to the CZ	14.32
	Percent of continuing residents
No observed prior job	9.56

B. Earnings (2023 $ thousands)

	After hire	Increase
Continuing CZ residents	81.8	20.1
Live outside the CZ	84.2	26.6
Migrated to the CZ	87.1	27.8
No observed prior job	60.5	—

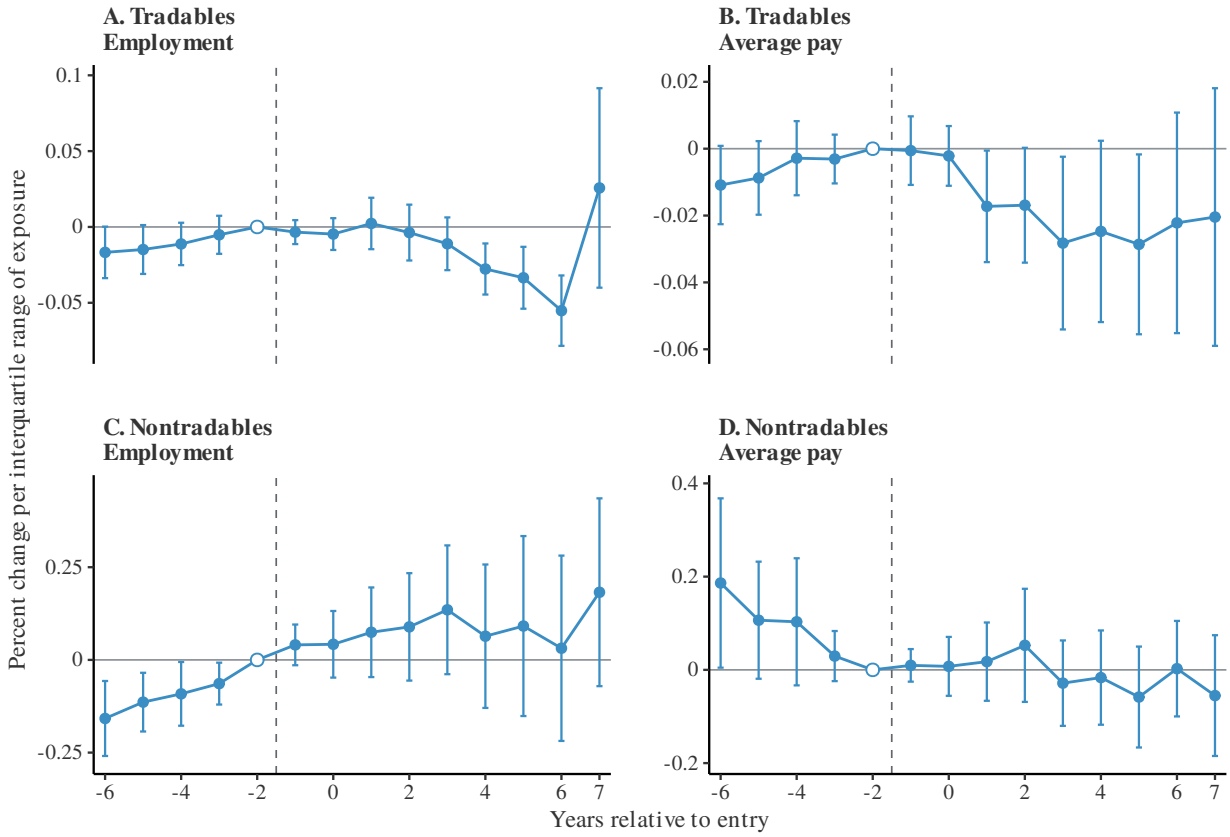
C. Payroll by residence one year after hire

	Percent of entrant payroll
Paid to residents inside the CZ	71.69
Paid to residents outside the CZ	28.31

Notes: Released LEHD tabulations pooling all firm entries and expansions. Panel A reports shares of hires by residence one year after hire. The no-observed-prior-job share uses continuing residents as its denominator. Panel B reports worker-weighted averages of the earnings medians, in thousands of 2023 dollars, for the Panel A groups. The outside row excludes the 1 percent of hires who lived in the CZ before hire and outside it after. After-hire earnings are annual earnings one year after hire; the increase is the difference between the after-hire and prior-year median earnings. Panel C uses group earnings totals by post-hire residence. Shares may not sum to 100 because of rounding.

The $40 that replaces hires' previous pay is not necessarily lost to the CZ. When a hire's former employer refills the vacated job with someone who was not working in the CZ or increases wages to retain employees, that pay is new income too, so the $32 of new income is a lower bound. To examine what happens to existing firms, I compare incumbent industries in the entrant's commuting zone with the same industries in five matched commuting zones. An industry's exposure is the entrant's employment multiplied by the share of national job-to-job movers who flow between that three-digit industry and the entrant's industry before entry, divided by the industry's baseline employment in the CZ. I estimate an event study of log employment and log average pay on this exposure interacted with event time and an indicator for the entrant's CZ, with event–CZ–industry, event–CZ–year, and event–industry–year fixed effects, so each coefficient compares more and less exposed industries in the entrant's CZ relative to the same industries in the matched CZs (Appendix A.4). Tradable industries more exposed to the entrant lose employment four to six years after entry, by up to 0.06 percent per interquartile range of exposure, while nontradable industries show no clear change (Figure 2). Appendix A.4 shows the raw employment and pay trajectories of the establishments directly exposed to the entrant.

Figure 2: Incumbent industries linked to the entrant through national worker flows



Notes: Predicted labor exposure uses national job-to-job flows between each incumbent three-digit industry and the entrant's industry, averaged over the three years before entry. Outcomes are log employment and log average pay for tradable (top row) and nontradable (bottom row) incumbent industries. Released log coefficients and 95 percent confidence intervals are multiplied by 100, so they read as approximate percent changes for one interquartile range of exposure. Year -2 is omitted, the dashed line falls between years -2 and -1, and each panel has its own scale. The release covers 80 CZs.

Fact 2. Local income is spent on services close to home. Fact 1 separates entrant payroll from new income for the CZ's residents. The next question is how much of that income reaches service firms in the CZ, and where. Nearly half of what households have left after housing goes to services that are mostly consumed locally.[8]

Workers will travel across a commuting zone for their job, but they are less willing to travel for groceries or a haircut. To directly compare commuting and shopping, I estimate, separately for each commuting zone c and each travel mode $m \in \{S, C\}$, a gravity equation for shopping visits (S) and commuting flows (C) between tracts i and j,

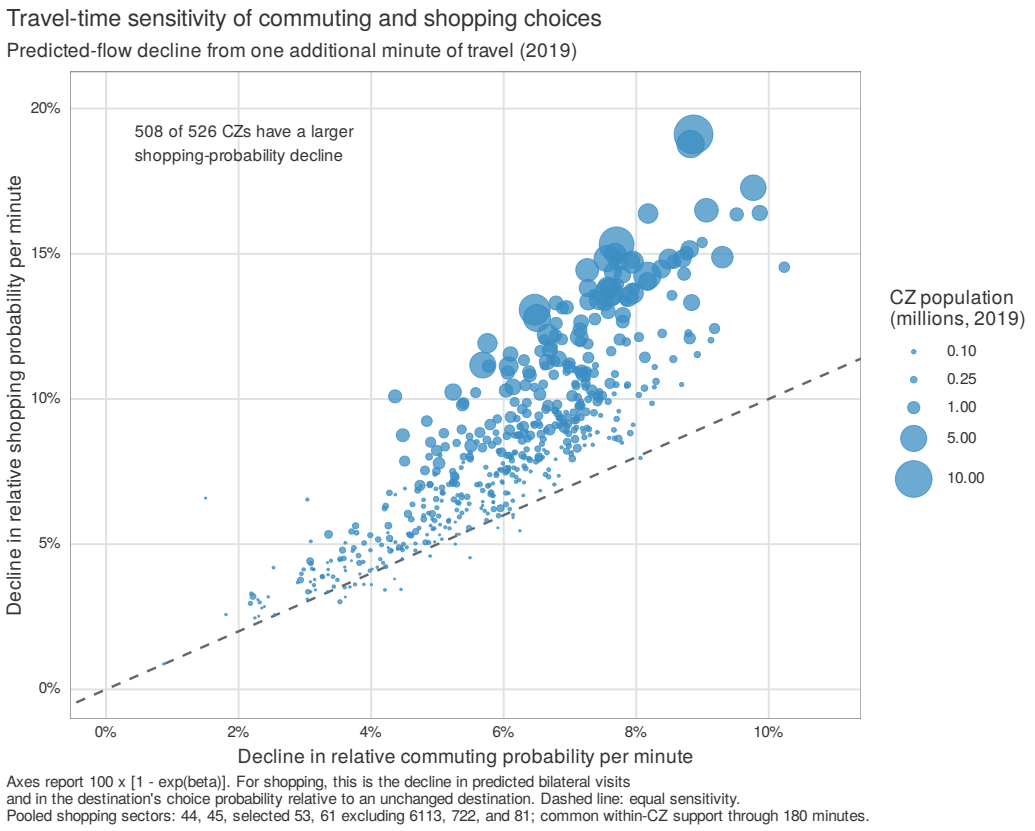
$$\mathbb{E}\left[F_{ijc}^{m}\right] = \exp\left(\alpha_{ic}^{m} + \delta_{jc}^{m} + \beta_{c}^{m}\, t_{ijc}\right), \qquad m \in \{S, C\}, \tag{1}$$

[8] In 2024, personal consumption of retail trade, education, health care, accommodation and food, and other services totaled $9.04 trillion, or 41.25% of disposable personal income and 48.14% of income left after housing. Author's calculations from the 2024 BEA PCE Bridge and NIPA Table 2.1; see Appendix A.5.

where F^S_{ijc} is 2019 device-weighted service visits from residents of i to establishments in j, F^C_{ijc} is the 2019 LODES count of workers living in i and working in j, and t_{ijc} is driving time in minutes. I estimate equation (1) by Poisson pseudo-maximum likelihood (Silva and Tenreyro, 2006). The origin effects α^m_{ic} absorb the number of residents and workers in each tract, and the destination effects δ^m_{jc} absorb workplace size and destination attractiveness.

Figure 3 plots the two estimates for the 526 commuting zones. Shopping decays faster than commuting in 97% of CZs. At the median, an additional minute of driving time reduces predicted shopping flow by 8.1% and predicted commuting flow by 6.1%, a difference of a third. With an increase of a ten-minute drive, shopping flows fall by 57% and commuting flows by 47%. The two coefficients are strongly correlated across places (0.86), so places in which commuting is sensitive to travel time also have concentrated shopping. These results show that a new workplace can pull employees from a larger area of the commuting zone while the consumption their earnings generate remains concentrated near a smaller set of residential neighborhoods.

Figure 3: Travel-time sensitivity of shopping and commuting across commuting zones



Notes: Each observation is a commuting zone. The horizontal axis is the percent decline in predicted commuting flow from one additional minute of driving time, and the vertical axis is the analogous decline in service visits, both computed as $100[1-\exp(\widehat{\beta}^m_c)]$ from the CZ-specific PPML estimates of equation (1). The dashed line denotes equal decay. The point size is proportional to 2019 population. Shopping pools NAICS 44–45, 61 excluding colleges, 722, and 81. Both trips use 2019 flows and 2020 OSRM driving times.

The same is true within counties. The median county holds 40 percent of its residents' jobs

but 53 percent of their shopping visits (Appendix Figure A.5). Within a CZ, pooling across service sectors, 11 percent of shopping visits end in the shopper's home tract and 24 percent in an adjacent tract, so over a third of service trips end within one tract boundary of a person's home tract. Nearly half of the rest goes to shopping hubs, clusters of destination tracts within ten minutes' drive of one another that each receive at least 2 percent of their CZ's service visits (Appendix A.5). Hubs receive 43.4 percent of within-CZ visits, and two-thirds of hub visits are to a hub outside the shopper's home neighborhood.[9]

In Fact 3, I compare service employment across tracts within a CZ, which requires shopping exposure that varies with where an entrant's hires live. Because residents throughout a CZ shop at the same hubs, spending routed through the full visit matrix ends in similar tracts irrespective of origin. Among the 85 firm entry events in CZs with more than one entry, destination alone explains 76.6 percent of the variation in that exposure. A design built on the full matrix picks up growth at each market's hubs but cannot attribute it to a particular entrant, and the 2019 matrix postdates the entries (Appendix A.6).[10] Instead, I use the local part that does vary. 34.4 percent of a tract's visits end in its own or an adjacent tract, against 3.2 percent if it shopped like its CZ as a whole (Appendix A.5). Fact 3 therefore assigns each worker's spending to their home neighborhood, 30 percent to the home tract (its share of local visits in Table 3) and 70 percent split equally among the adjacent tracts.

Table 3: Allocation of within-CZ service visits between the home neighborhood and hubs, 2019

Sector	Same tract	Adjacent tract	Nonlocal 2% hub
NAICS 44	12.2	26.1	26.4
NAICS 45	9.0	23.6	33.5
NAICS 722	9.6	23.1	30.5
NAICS 81	13.5	22.9	22.7
Combined core	*10.6*	*23.8*	*29.3*
NAICS 61	7.2	16.2	32.8

Notes: Entries are percentages of annual device-weighted visits, conditional on the destination remaining within the device's home CZ. The categories are mutually exclusive. The remaining visits go to other destinations within the home CZ. "Adjacent tract" excludes the home tract; "Nonlocal 2% hub" is a shopping hub that excludes both the home tract and its adjacent tracts. Sector definitions are in Appendix A.5.

Fact 3. Service employment grows near the residential locations of workers. The first two facts imply that an entrant's payroll should raise service employment in the neighborhoods

[9]The number of hubs rises with market size, with an elasticity with respect to population of about 0.43 under the 2 percent definition (Appendix A.5).

[10]The visit data are from 2019, after every entry in the sample. If shopping itself responds to entry, because new stores open near the entrant or near its workers' homes, exposure routed through hubs and the full network would partly reflect that response. The home-and-adjacent rule does not have this problem, because which tracts it ranks as most exposed depends only on where hires lived before entry and on which tracts border theirs.

where its workers live. I estimate this effect on service employment and payroll with a stacked event study of tracts (Cengiz et al., 2019; Deshpande and Li, 2019). Every tract in a stack is exposed to the same entry, so each stack compares more and less exposed tracts rather than treated and untreated ones. Because the two groups receive equal weight within each stack and every stack receives the same total weight, the pooled coefficient is the simple average of the stack-specific contrasts (Gardner, 2021; Wing et al., 2024).

I define firm entry as the first year in which the firm records employment, and I measure a tract's exposure to this entry from the number of the entrant's hires who lived in it or an adjacent tract before entry, weighted by the 30/70 shares above. Within each event, a tract is treated if its exposure is in the top quartile of the event's tracts and serves as a control if it is in the bottom quartile but still has positive exposure. Mean exposure differs between the two groups by 23 workers. I restrict the sample to 40 entries in commuting zones with no other observed entry, so that no control tract is exposed to a second event.

For each event, I build a stack of its top- and bottom-quartile tracts from six years before entry to seven years after. I estimate, for employment and payroll in retail, food services, and other services (NAICS 44, 45, 722, and 81),[11]

$$Y^o_{ejt} = \alpha^o_{ej} + \delta^o_{et} + \sum_{k \in \{-6,\ldots,7\} \setminus \{-2\}} \theta^o_k D_{ej} \mathbb{1}\{t - \tau_e = k\} + u^o_{ejt}, \tag{2}$$

where Y^o_{ejt} is service employment (jobs) or annual service payroll (millions of 2023 dollars) in tract j in year t for event e, D_{ej} indicates a top-quartile tract, and τ_e is the entry year. α^o_{ej} and δ^o_{et} are tract and calendar-year fixed effects, so a tract is compared only with tracts in the same market in the same year. The reference year is two years before entry rather than one, because hiring can begin in the year before the entrant first records employment.[12] Within each event the two groups receive equal weight, and standard errors are clustered by CZ, because tracts in the same market share local shocks.

Under parallel trends and no anticipation, θ^o_k is the effect of entry on service employment in highly exposed tracts relative to less exposed tracts in the same market, k years after entry (Callaway et al., 2026). Reading it as the effect of greater exposure further requires that the most exposed tracts would have responded to low exposure as the least exposed tracts did. Exposure is fixed by where hires lived before being hired, so it cannot reflect moves after hiring or later changes in shopping. The stack-specific year effects remove any shock common to the market, so only differences within the market identify θ^o_k. Additionally, the two groups move together before entry. Every pre-entry coefficient for employment and payroll is small, and every 95 percent confidence interval includes zero (Figure 4), which would be unlikely if service firms responded to the

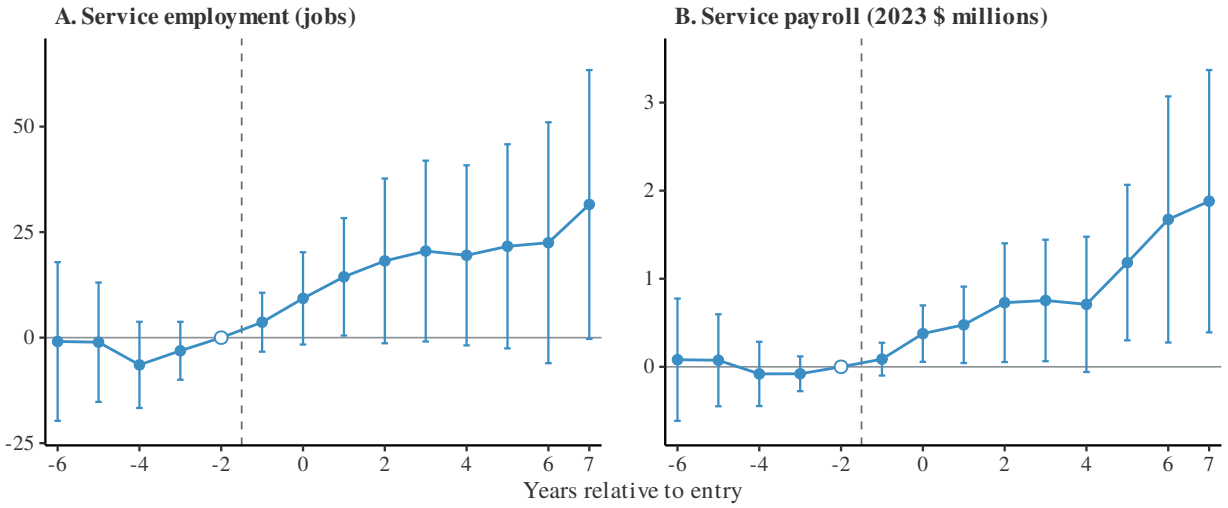
[11] Education and health care (NAICS 61 and 62) are excluded. Their location is tied to schools and hospitals and their funding comes largely from governments and insurers, so their employment responds less directly to nearby household spending.

[12] Employment is measured on March 12, so hiring that begins after March 12 of the preceding year can occur while recorded employment for that year is still zero.

announcement before hiring began. The remaining concern is that entrants hire from neighborhoods whose service sectors were about to grow for other reasons. Appendix A.6 also measures exposure by new income and by residences predicted from public commuting data, and gives the full construction of the stacks.

Seven years after entry, a top-quartile tract has $1.9 million more in annual service payroll, in 2023 dollars, than a bottom-quartile tract in the same market, with a 95 percent confidence interval of $0.39 to $3.4 million (Figure 4). Service employment follows the same path and reaches 32 more jobs by year seven, but it is estimated less precisely, with an interval from just below zero to 63 jobs.

Figure 4: Service employment and payroll around entrant workers' pre-entry residences

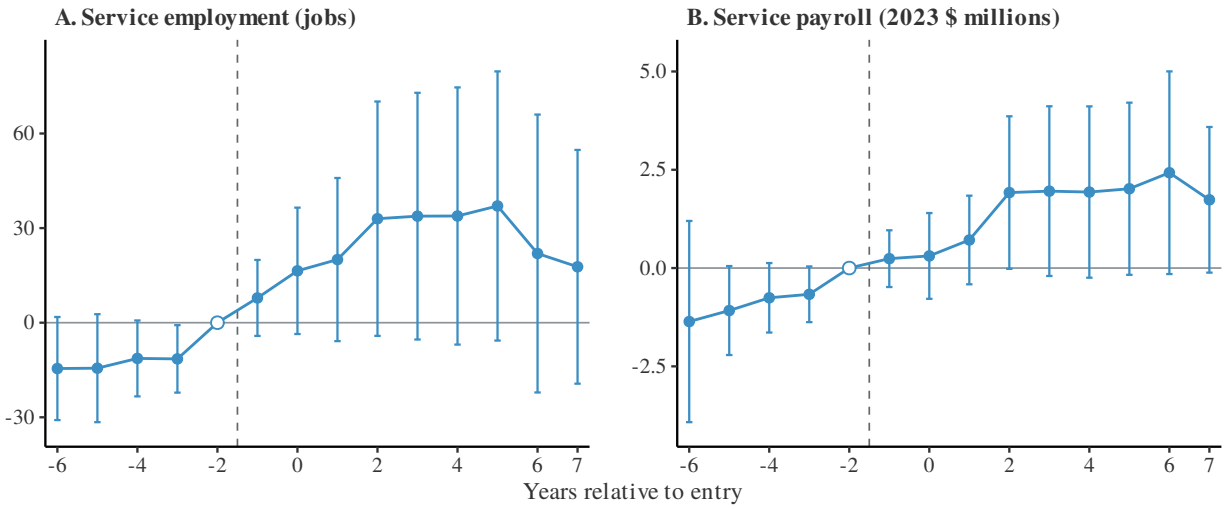


Notes: Stacked event-study coefficients from equation (2), comparing the top and bottom quartiles of pre-entry-resident worker exposure in commuting zones with no other observed events. The difference in mean exposure is 23.5 workers. Each worker's pre-entry home tract receives 30 percent of exposure, and adjacent tracts share 70 percent equally. Outcomes cover retail, food services, and other services (NAICS 44, 45, 722, and 81). Bars are 95 percent confidence intervals; year −2 is normalized to zero. Payroll is displayed in millions of 2023 dollars. The release reports 40 commuting zones.

Figure 5 estimates the same relationship with continuous exposure across all events and exposure tracts, including CZs with more than one observed entry. Exposure from each entry enters as a distributed lag, with endpoint terms that carry exposure from firm entries outside the window, which accommodates several entries of different sizes in the same tract (Schmidheiny and Siegloch, 2023). Distributed lags of this kind are a common way to study policy changes of different sizes (Suárez Serrato and Zidar, 2016; Fuest et al., 2018). This specification uses every entry and the full exposure distribution rather than only those in single-entry CZs, but it assumes that the effect of an entry is proportional to a tract's exposure and that effects add up across entries (Schmidheiny and Siegloch, 2023). Each hire is again placed at their residence before entry, with the same 30/70 split, and the regression includes tract and CZ-by-year fixed effects. The coefficients average 26 service jobs and $2.1 million in annual service payroll per 100 exposed workers over years five

through seven. These are slopes across the full exposure distribution, so their size is not directly comparable to the quartile contrast.

Figure 5: Continuous service responses to pre-entry-residence exposure with 30/70 home-and-adjacent weights



Notes: Continuous event-study coefficients per 100 exposed workers from equation (A.12). Hires are located at their residences before entry. Each home tract receives 30 percent of exposure, and adjacent tracts share 70 percent equally. All events and exposure tracts enter, including CZs with multiple observed events. Outcomes cover retail, food services, and other services (NAICS 44, 45, 722, and 81). The regression includes tract and CZ-by-year fixed effects, is unweighted, and excludes the entrant's workplace tract. Bars are CZ-clustered 95 percent confidence intervals; year −2 is normalized to zero. Payroll is in millions of 2023 dollars. Each panel has its own vertical scale. The release reports 100 commuting zones.

From the facts to the model. The three facts follow entrant payroll to workers' homes, to the shops near those homes, and to the service jobs that grow there. They cannot add up to a net effect for the market. Comparisons across neighborhoods mix new jobs with jobs that move from elsewhere in the CZ, and they miss the wages, rents, and prices through which the entrant also crowds out other firms. The model in Section 4 adds those channels. It is calibrated to the hiring shares in Fact 1 and the shopping gravity in Fact 2, and Section 5 compares its predictions with Fact 3.

4 A Model of Labor and Shopping

I develop a spatial general equilibrium model to measure the service multiplier across neighborhoods and compare it with resident welfare. The economy consists of a single commuting zone (CZ) partitioned into census tracts $i, j \in \mathcal{J}$ differentiated by exogenous amenities and productivity. Individuals choose where to live, work, and shop. Their shopping trips originate from both their home and workplace. The commuting zone is an open economy with workers living just outside its boundary and migration from elsewhere in the country. There are four sectors. Two tradable

sectors, high and low skill intensive, produce with Cobb–Douglas technologies and sell at a national price, and two service sectors, also high and low skill intensive, are monopolistically competitive with free entry. Each county levies a property tax to pay for public services, which it buys from outside the CZ, so the revenue leaves the modeled economy. Residents own residential land, while commercial rents accrue to absentee landlords. Residential and commercial floor space are supplied with separate elasticities.

I model an entering tradable firm as a positive productivity shock in a sector and location. This shock increases demand for labor and raises wages, encouraging individuals to join the labor force, migrate into, or work in the area. A portion of the increase in worker income is spent on local services. Service firms may see revenue gains that translate into increased demand for workers, but in an environment where labor and floor space may be more expensive. The spatial incidence of these net effects is determined by the interaction of workers' choices of where to live, work, and shop, so the gains from job creation accrue not at the firm's entry location but along the commuting and shopping networks that connect it to the rest of the commuting zone.

4.1 Individuals

Individuals differ by skill $k \in \{H, L\}$ and choose where to live, in which sector to work, whether to work, and their residence–workplace pair. They compare living in the CZ, in the accessible outside area, or elsewhere in the country. The order of the region and sector choices differs by skill. High-skill individuals choose their sector first and their region second, so they move between regions more readily than they switch sectors. Low-skill individuals choose their region first and their sector second, so they switch sectors more readily than they move. Within each sector and region, individuals decide whether to work and, if they do, choose a residence–workplace pair. The migration elasticity μ_k differs by skill, while the sector-choice and participation elasticities, ν and η, are common to both.

This nested structure allows the model to distinguish new workers from new consumers. After a positive labor demand shock, substantial shares of new workers migrate, cross the CZ boundary for work, or come from nonemployment (Fact 1). Unemployed residents already consume in the CZ, and those who commute into the CZ spend much of their income outside of it. A semi-open city model with a single migration elasticity could approximate the employment openness of the local market, but it would treat every new worker as a new consumer, and in doing so, overstate the increase in service demand. In this model, these nested employment and location choices determine the labor available to local employers and the locations from which workers originate their shopping trips.

4.1.1 Preferences

Let ℓ index a joint residence and employment choice. For a working individual ω of skill k, this specifies residence i, workplace j, and sector g.[13] The individual consumes housing h, services n, and a tradable good x whose price is normalized to one. For an individual living and working within the CZ, preferences are Cobb–Douglas,

$$U_\ell(\omega) = z_\ell(\omega)\,\frac{B^R_{ik}B^W_{jgk}}{d_{ij}}\,h^{\alpha_H}\left(n^{s_N}x^{1-s_N}\right)^{1-\alpha_H}, \tag{3}$$

where B^R_{ik} and B^W_{jgk} are residential and workplace amenities, with workplace attractiveness allowed to differ by sector. Additionally, $d_{ij} = \exp(\kappa t_{ij})$ is the utility cost of a commute of t_{ij} minutes, and $z_\ell(\omega)$ is an idiosyncratic taste for choice ℓ. The housing expenditure share is α_H, and s_N is the share of remaining income spent on services.[14]

Let y_ℓ denote disposable income, q^R_i the residential rent, and $\mathcal{P}_\ell$ the service price index. Maximizing utility over consumption gives indirect utility $z_\ell(\omega)\Phi_\ell$,[15] where

$$\Phi_\ell = \frac{B^R_{ik}B^W_{jgk}}{d_{ij}}\,\frac{y_\ell}{(q^R_i)^{\alpha_H}\mathcal{P}_\ell^{(1-\alpha_H)s_N}}. \tag{4}$$

Each individual chooses the option ℓ with the highest $z_\ell(\omega)\Phi_\ell$. The value Φ_ℓ is common to everyone of skill k who makes choice ℓ, while $z_\ell(\omega)$ is the individual's idiosyncratic taste for it. Tastes are drawn independently across individuals from a nested Fréchet distribution and are held fixed across equilibria. For an individual of skill k,

$$\Pr\{z_\ell(\omega) \le t_\ell \text{ for every } \ell\} = \exp\left\{-G_k\left((t_\ell^{-1})_\ell\right)\right\}, \qquad t_\ell > 0, \tag{5}$$

where G_k incorporates the nesting structure and the preference parameters (Appendix B.2). The shape parameters μ_k, ν, η, ψ, and χ govern migration, sector choice, labor-force participation, the choice between jobs inside and outside the CZ, and the residence–workplace choice, respectively.

4.1.2 Employment and location choice nests

Whether an individual moves to the CZ depends on the utility it offers, and that utility depends on the jobs in each sector they can reach from places they would like to live. A higher wage at one workplace raises not only the appeal of commuting there, but also the value of working in its sector, and the value of living in the CZ.

[13] For nonworkers, ℓ specifies residence i and a sector with amenities $B^R_{ik}B^U_{ik}$ and no commute. Appendix B.1 gives the corresponding amenities for choices involving the outside area.

[14] For outside residents, the same preferences and demand expressions use the fixed housing share $\overline{\alpha}^o_H$ in place of α_H.

[15] For nonemployment, the amenity component is $B^R_{ik}B^U_{ik}$, income includes nonemployment income y^U_k, and the service price index is P_i, the price of services purchased from the home origin.

The value of working in sector g sums over every residence–workplace pair available in that sector, including workplaces outside the CZ, and the value of not working sums over residences,

$$\mathcal{W}_{gk} = \sum_{\ell \in \mathcal{L}^W_{gk}} \Phi_\ell^\chi, \qquad \mathcal{U}_{gk} = \sum_{\ell \in \mathcal{L}^U_{gk}} \Phi_\ell^\chi. \tag{6}$$

Working and not working then combine into the value of the sector,

$$\mathcal{S}_{gk} = \overline{B}^E_{gk} \mathcal{W}_{gk}^{\eta/\chi} + \overline{B}^U_{gk} \mathcal{U}_{gk}^{\eta/\chi}, \tag{7}$$

where $\overline{B}^E_{gk}$ and $\overline{B}^U_{gk}$ are preference parameters for employment and nonemployment, and the sector values in turn determine the regional and sector choices in equation (8).

Individuals can also live outside the CZ, either in an accessible outside area or elsewhere in the country. The accessible outside area pools the neighboring CZs into a single location whose residents can work in the CZ, work outside it, or remain nonemployed. Moving between the outside area and the CZ is governed by the same parameter μ_k as migration from elsewhere.

For a resident of the outside area, the choice between a job inside the CZ and one outside it is a separate nest with shape parameter ψ, which governs how readily residents of neighboring CZs take CZ jobs when wages rise. CZ residents who work outside it instead treat the outside area as one more workplace in their residence–workplace choice, so ψ affects only outside residents. Commutes across the CZ boundary have no travel time in the model. The costs of travel into the CZ are part of the location-specific exogenous amenity term calibrated to the observed inbound and outbound flows (Appendix C.6.1). The terms $\mathcal{O}_L$ and $\mathcal{O}_{gH}$ in (8) collect the value of living in the outside area or elsewhere in the country. It can change with entry because outside residents benefit from access to CZ jobs, while utility elsewhere in the country is fixed.

Regional location choice. For low-skill individuals, the outermost nest is the choice whether to live in the CZ; for high-skill individuals, it is the choice of sector. Let $\mathcal{N}_k$ be the fixed national population of skill k and $\overline{B}^S_{gk}$ the preference parameter for sector g. For low-skill individuals, $\overline{B}^R_L$ is the preference parameter for CZ residence and $\mathcal{O}_L$ the value of living in the outside area or elsewhere in the country. For high-skill individuals, both are specific to the sector, $\overline{B}^R_{gH}$ and $\mathcal{O}_{gH}$, and $\mathcal{V}_{gH} = \mathcal{O}_{gH} + \overline{B}^R_{gH} \mathcal{S}_{gH}^{\mu_H/\eta}$ is the value of sector g across regions. The number of CZ residents of skill k in sector g is

$$N_{gk} = \begin{cases} \mathcal{N}_L \dfrac{\overline{B}^R_L \, \mathcal{Q}_L^{\mu_L/\nu}}{\mathcal{O}_L + \overline{B}^R_L \, \mathcal{Q}_L^{\mu_L/\nu}} \dfrac{\overline{B}^S_{gL} \mathcal{S}_{gL}^{\nu/\eta}}{\mathcal{Q}_L}, & k = L, \\ \mathcal{N}_H \dfrac{\overline{B}^S_{gH} \mathcal{V}_{gH}^{\nu/\mu_H}}{\sum_{g' \in \mathcal{G}} \overline{B}^S_{g'H} \mathcal{V}_{g'H}^{\nu/\mu_H}} \dfrac{\overline{B}^R_{gH} \mathcal{S}_{gH}^{\mu_H/\eta}}{\mathcal{V}_{gH}}, & k = H, \end{cases} \tag{8}$$

where $\mathcal{Q}_L = \sum_{g \in \mathcal{G}} \overline{B}^S_{gL} \mathcal{S}^{\nu/\eta}_{gL}$. For low-skill individuals, the first factor is the share who live in the CZ and the second is the share of those residents who choose sector g. For high-skill individuals, the order is reversed, so the first factor is the national share who choose sector g and the second is the share of them who live in the CZ. The CZ's population of skill k sums over sectors,

$$N_k = \sum_{g \in \mathcal{G}} N_{gk}. \tag{9}$$

I assume utility elsewhere in the country is fixed. Because the CZ is small relative to the nation, the individuals it draws from do not change wages or prices in the nation. The migration elasticity μ_k is approximately the elasticity of the CZ's population with respect to the value of living there, within each sector for high-skill individuals.[16] When μ_k is low, few individuals move, and a labor demand shock is absorbed by higher wages, prices and rents.

Sector choice. Individuals compare employment opportunities in two tradable sector categories (high and low skill intensive) and two service sector categories (high and low skill intensive), so $\mathcal{G} = \mathcal{G}_T \cup \mathcal{G}_N$ with $\mathcal{G}_T = \{T_H, T_L\}$ and $\mathcal{G}_N = \{N_H, N_L\}$.[17] Equation (8) gives the number choosing each sector. The labor demand shock increases wages in one of the tradable sectors, which draws individuals toward tradable employment, increasing labor competition across sectors for incumbent service firms. The choice of N_{gk} includes the ability to not work. Therefore, the sector choice is best described as an employment opportunity choice.

Labor-force participation. Within sector g, individuals compare working with nonemployment.[18]

The number employed in sector g among CZ residents is

$$E_{gk} = N_{gk} \frac{\overline{B}^E_{gk} \mathcal{W}^{\eta/\chi}_{gk}}{\mathcal{S}_{gk}}. \tag{10}$$

Higher wages can draw nonworkers into employment, expanding the workforce and raising the income available for local services.

Residence–workplace location choice. Conditional on living in the CZ and working in sector g, an individual of skill k chooses residence–workplace pair $\ell = (i, j, g)$ from the feasible set $\mathcal{L}^W_{gk}$

[16] The exact elasticity is $\mu_k(1 - b^{\text{reg}}_k)$, where b^{reg}_k is the share of the national population of skill k living in the CZ or the accessible outside area, computed within sector for high-skill individuals.

[17] Workers of both skill types can work in either the high or low intensity sectors. These sectors simply employ different shares of each worker type.

[18] Nesting participation within sector treats each individual as attached to a sector even when not working, so that a nonworker compares not working with the within-sector wage. This follows the labor literature on sector-specific skills and attachment. I am assuming human capital is partly specific to an industry or occupation. This follows the literature in which displaced workers who switch industries suffer larger earnings losses (Neal, 1995), and Galle et al. (2023, Table 6), who estimate a larger shape parameter for the choice between nonparticipation and work than for the choice across sectors, consistent with $\eta \geq \nu$.

with probability

$$\pi_{\ell k} = \frac{\Phi_\ell^\chi}{\sum_{\ell' \in \mathcal{L}_{gk}^W} \Phi_{\ell'}^\chi}, \qquad F_{ijgk} = E_{gk}\pi_{\ell k}. \tag{11}$$

The numerator captures the attractiveness of the pair through residential and workplace amenities, disposable income, residential rent, service prices, and commuting costs. The denominator, $\mathcal{W}_{gk}$, sums these utility terms across all feasible residence–workplace choices in the sector, so the probabilities sum to one. Multiplying the probability by the number of employed CZ residents in sector g, E_{gk}, gives the corresponding commuting flow. Nonworkers choose among residences with the same parameter χ.

4.1.3 Income and spending

A working individual supplies one unit of labor and earns the wage w_{jk}^g paid to skill k in sector g at workplace tract j (Section 4.3), while a nonworker receives the externally financed payment y_k^U. Current CZ residents also receive a dividend D_k from ownership of residential land.[19] Disposable income is

$$y_\ell = w_{jk}^g + D_k \ \text{ for a worker}, \qquad y_\ell = y_k^U + D_k \ \text{ for a nonworker}. \tag{12}$$

Section 4.4 determines the dividend.[20]

At residential rent q_i^R, Cobb–Douglas preferences imply housing demand h_ℓ and service spending γy_ℓ, with

$$h_\ell = \frac{\alpha_H y_\ell}{q_i^R}, \qquad \gamma = s_N(1-\alpha_H). \tag{13}$$

Therefore, an additional dollar of disposable income raises service spending by γ.

Workers allocate a share ρ of service spending to purchases originating at home and the remainder to purchases originating at work, with $n = n_i^\rho n_j^{1-\rho}$. Let P_i denote the service price index at origin i. The service price index for workers is $\mathcal{P}_\ell = P_i^\rho P_j^{1-\rho}$. Meanwhile, nonworkers purchase all services from home, so their price index is $\mathcal{P}_\ell = P_i$.

Let y_{ijgk} denote the disposable income of a worker of skill k who lives in i and works in sector g at j, and let U_{ik} be the number of nonemployed individuals of skill k at residence i, summed across

[19] In the quantification, housing expenditure includes the rental-equivalent value of owner-occupied housing. This essentially is a rent which residents pay to themselves. The assumption matters because rental income stays in the CZ, so part of it is spent on local services. Because the dividend is shared by all current residents, it spreads the gains from higher rents across the population rather than concentrating them on incumbent owners.

[20] Appendix B.3 gives the corresponding income accounting for outside residents.

sectors. Total service spending originating in tract i is

$$\mathcal{E}_i = \gamma\Big[\underbrace{\rho \sum_{j,g,k} F_{ijgk} y_{ijgk}}_{\text{Workers' purchases from home}} + \underbrace{(1-\rho) \sum_{j,g,k} F_{jigk} y_{jigk}}_{\text{Workers' purchases from work}} + \underbrace{\sum_{k} U_{ik}\big(y_k^U + D_k\big)}_{\text{Nonworkers' purchases}}\Big], \tag{14}$$

where γ uses the outside housing share for inbound commuters.

4.1.4 Shopping destinations

From each shopping origin i, individuals purchase a CES bundle of services available across destinations j, with elasticity of substitution $\sigma_N > 1$. Tract j offers m_j differentiated service varieties, with prices summarized by p_j^N. Each variety is supplied by a firm operating under monopolistic competition with free entry.[21]

Travel from origin i to destination j takes t_{ij} minutes and raises the effective price of purchases by the iceberg factor $\exp(\kappa t_{ij})$, using the same travel-cost parameter κ as commuting. Under CES demand, these preferences and travel costs enter through $T_{ij} = \zeta_{ij} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}$, which determines each destination's contribution to the service price index. The cost of one unit of the service bundle purchased from origin i is[22]

$$P_i = \left[\sum_{j \in \mathcal{J}} T_{ij} m_j (p_j^N)^{1-\sigma_N}\right]^{1/(1-\sigma_N)}. \tag{15}$$

Firm entry expands the range of available services and, holding prices and travel costs fixed, lowers P_i. The expenditure share received by tract j is

$$\pi_{ij}^I = \frac{T_{ij} m_j (p_j^N)^{1-\sigma_N}}{P_i^{1-\sigma_N}}. \tag{16}$$

Individuals direct more spending toward destinations offering more varieties, lower prices, or better travel access. These purchases generate service revenue of $\sum_{i \in \mathcal{J}} \mathcal{E}_i \pi_{ij}^I$ in tract j.

4.2 Service Firms

A service firm in sector $g \in \mathcal{G}_N$ and tract j produces a differentiated variety under monopolistic competition and free entry.[23] With productivity A_{jg}, a firm combines labor $\mathcal{L}$, commercial floor space H, and intermediate inputs X purchased from outside the CZ to produce output Y, using

[21] The model distinguishes two service sectors, N_H and N_L, with separate prices and entry responses. This section presents shopping and revenue aggregated by tract. Appendix B.6 gives the sector-specific equations and derives these aggregates.

[22] The term T_i^O summarizes shopping opportunities outside the CZ. Appendix B.6 describes these opportunities and derives the expenditure share spent outside the CZ.

[23] Appendix B.7 gives the full conditions for each service sector and their aggregation to tracts.

the same Cobb–Douglas form as the tradable sectors (Section 4.3),

$$Y = A_{jg}\,\mathcal{L}^{\alpha_L^g} H^{\alpha_C^g} X^{\alpha_X^g}, \tag{17}$$

where the sector's expenditure shares on labor, commercial space, and intermediates satisfy $\alpha_L^g + \alpha_C^g + \alpha_X^g = 1$.

Under CES demand, firms charge a constant markup over their marginal cost c_{jg},

$$p_{jg} = \frac{\sigma_N}{\sigma_N - 1} c_{jg}. \tag{18}$$

Appendix B.7 derives unit costs and factor demands.

Each firm also incurs an annual fixed cost f_{jg} to maintain its establishment. The fixed cost uses the same inputs as production, so it is proportional to the unit cost c_{jg}, and the input requirement per firm changes with the number of firms m_{jg} in the tract,

$$f_{jg} = \varphi_{jg} c_{jg} m_{jg}^{1/\epsilon_m - 1}, \tag{19}$$

where φ_{jg} scales the requirement. At the CES markup, each firm keeps a share $1/\sigma_N$ of its revenue after variable costs, so free entry drives profits to zero when

$$\frac{R_{jg}}{\sigma_N} = m_{jg} f_{jg}, \qquad\qquad m_{jg} = \left(\frac{R_{jg}}{\sigma_N \varphi_{jg} c_{jg}} \right)^{\epsilon_m}. \tag{20}$$

The entry elasticity ϵ_m is the elasticity of the number of establishments with respect to revenue relative to unit cost. When $\epsilon_m = 1$, each firm's requirement is constant and establishments rise in proportion to revenue relative to cost. A value below one makes each firm's requirement rise as firms enter, so establishments respond less than proportionally.

Since revenue covers variable and fixed costs, total labor expenditure is $\alpha_L^g R_{jg}$. Allocating this expenditure across skills and dividing by the corresponding wage gives employment:

$$L_{jk}^g = \frac{\alpha_L^g \lambda_{jgk} R_{jg}}{w_{jk}^g}. \tag{21}$$

4.3 Tradable Firms and Entry

Tradable producers in sector $g \in \mathcal{G}_T$ (high and low skill intensive) sell a homogeneous good on the national market at price p^T, which the CZ takes as given.[24]

Tradable production in each tract and sector is competitive, with productivity A_{jg}. Production is Cobb–Douglas in labor, commercial floor space, and intermediate inputs. With the price of

[24]The high-skill industry comprises NAICS 51, 52, 54, and 55, and the low-skill industry comprises manufacturing (NAICS 31–33). The labels refer to industries' skill intensity, and individuals of either skill can work in either industry.

intermediate inputs normalized to one, cost minimization gives the unit cost

$$c_{jg} = \frac{(q_j^C)^{\alpha_C^g}}{A_{jg}} \prod_k \left(w_{jk}^g\right)^{\alpha_L^g \lambda_{jgk}}. \tag{22}$$

Production has constant returns and the output price is set nationally, so profits are zero in every location and sector,

$$c_{jg} = p^T, \tag{23}$$

and output Y_{jg} adjusts until the labor and floor-space markets clear. Labor demand is a constant share of revenue,

$$L_{jk}^g = \frac{\alpha_L^g \lambda_{jgk}\, p^T Y_{jg}}{w_{jk}^g}, \tag{24}$$

and demands for commercial space, H_{jg}^C, and intermediate inputs take the same form (Appendix B.8). Equation (23) means that a tradable producer cannot pass higher costs on to its consumers. Given its productivity, each producer can pay only the wages at which it breaks even, and its employment adjusts to the labor supply those wages attract. When the entrant's productivity gain raises the wages workers can earn elsewhere, incumbent tradable producers cannot match them, so they lose workers and shrink.

Entry. I model the entry of a tradable firm as an increase in the productivity $A_{j_E g_E}$ of its sector g_E in its tract j_E, sized so that employment in this tract and sector rises by J^E, the entrant's promised jobs. Higher productivity raises the wages the location-sector can pay, and its employment rises along the upward-sloping labor supply described in Section 4.1. The model does not distinguish the entrant from incumbent producers in the same tract and sector, so J^E measures the location-sector's net growth. Because the shock raises productivity for the whole location-sector, the higher wage it supports is paid to every worker here, not only to new hires. The model has no separate premium for the entrant's workers.

4.4 Land Market Clearing

Each tract has separate markets for residential and commercial floor space. Space cannot be converted between uses, so each tract has two rents. Owners receive r_i^R and r_j^C, and tenants pay these rents plus the property tax rate $t_{a(i)}$ of the county $a(i)$ that contains the tract,

$$q_i^R = (1 + t_{a(i)}) r_i^R, \qquad q_j^C = (1 + t_{a(j)}) r_j^C. \tag{25}$$

Floor space is built by combining land, which is fixed in each tract, with structures, so its supply slopes upward in the rent owners receive. I take this supply to have a constant elasticity,

$$H_i^R = \bar{H}_i^R (r_i^R)^{\epsilon_{R,i}}, \qquad H_j^C = \bar{H}_j^C (r_j^C)^{\epsilon_C}, \tag{26}$$

as in quantitative urban models with a construction sector (Ahlfeldt et al., 2015). The elasticities measure how much floor space expands when rents rise, and $\epsilon = 0$ gives the fixed supply of Monte et al. (2018). The shifters $\bar{H}_i^R$ and $\bar{H}_j^C$ capture each tract's land area and the ease of building on it.

Residents spend a share α_H of disposable income on housing (13). Let $Y_i^R = \sum_k \sum_{\ell:\, i_\ell = i} n_{\ell k} y_\ell$ denote the total disposable income of tract i's residents, workers and nonworkers alike. Commercial space is demanded by tradable producers, including producers in the entrant's tract and sector, and by service firms. Equating supply and demand in each market gives

$$r_i^R = \left[\frac{\alpha_H Y_i^R}{(1 + t_{a(i)}) \bar{H}_i^R} \right]^{\frac{1}{1+\epsilon_{R,i}}}, \qquad H_j^C = \sum_{g \in \mathcal{G}} H_{jg}^C. \tag{27}$$

Because the entrant's location-sector demands floor space in proportion to its output, entry competes with other producers for commercial space.

Residential land is owned by the CZ's residents. Rental income $\mathcal{R} = \sum_{i \in \mathcal{J}} r_i^R H_i^R$ is paid out through a local portfolio according to fixed housing-value shares ω_k^0, so each current resident of skill k receives

$$D_k = \frac{\omega_k^0 \mathcal{R}}{N_k}, \tag{28}$$

where N_k counts all current residents of skill k, including renters, nonworkers, and new arrivals. Commercial rents accrue to absentee landlords.

4.5 Government

Each county provides public services in proportion to the number of residents and developed land within its jurisdiction. Let N_a denote the number of modeled residents in county a and $\mathcal{J}_a$ its modeled tracts. I measure developed property using residential and commercial space valued at baseline rents,

$$\mathcal{H}_a = \sum_{j \in \mathcal{J}_a} \left(r_j^{R,0} H_j^R + r_j^{C,0} H_j^C \right). \tag{29}$$

County spending on public services is

$$G_a = G_a^0 \left[\varsigma \frac{N_a}{N_a^0} + (1 - \varsigma) \frac{\mathcal{H}_a}{\mathcal{H}_a^0} \right], \tag{30}$$

where ς is the resident-service share of baseline spending. The county finances this spending with the property tax in (25), levied on annual pretax rent. Its tax rate t_a adjusts to balance its budget,

$$\mathcal{T}_a = t_a \left(\mathcal{R}_a + \mathcal{R}_a^C \right) = G_a, \tag{31}$$

where $\mathcal{R}_a = \sum_{i \in \mathcal{J}_a} r_i^R H_i^R$ and $\mathcal{R}_a^C = \sum_{j \in \mathcal{J}_a} r_j^C H_j^C$ are the county's residential and commercial rental income. Baseline spending equals baseline property-tax receipts.

The county produces public services with goods bought outside the CZ at a fixed price, so government spending does not raise demand for local service firms and is not paid out to households. Because spending grows with residents and developed land, services per resident and per unit of developed land stay at their baseline levels, and their value to residents is part of the baseline amenities. Appendix B.9 gives the full property and government accounts.

4.6 Equilibrium and the Effects of Entry

Given the parameters in Table 4, the national populations $\mathcal{N}_k$ and utility elsewhere in the country, and the location fundamentals (productivities A_{jg}, amenities and preference parameters, floor-space shifters $\bar{H}_i^R$ and $\bar{H}_j^C$, fixed-cost scales φ_{jg}, travel times t_{ij}, and baseline county spending G_a^0), an equilibrium is a set of wages w_{jk}^g, rents r_i^R and r_j^C, county tax rates t_a, service prices p_{jg}, establishment masses m_{jg}, and allocations of individuals N_k, N_{gk}, E_{gk}, F_{ijgk}, and U_{ik} such that

(i) individuals choose residence, sector, participation, and commute as in (9), (8), (10), and (11);

(ii) tradable producers break even, (23), and employ workers according to (24);

(iii) service firms price at a markup and enter until profits are zero, (18)–(20), and employ workers according to (21);

(iv) labor markets clear at each workplace, sector, and skill, so the commuting flows from all feasible residences, including inbound commuters, equal employment,

$$\sum_i F_{ijgk} = L_{jk}^g, \qquad g \in \mathcal{G}, \quad j \in \mathcal{J}; \tag{32}$$

(v) residential and commercial floor-space markets clear, (27);

(vi) service revenue in each tract and sector equals the spending directed to it, $R_{jg} = \sum_i \mathcal{E}_i \pi_{ijg}^I$;

(vii) residential rental income is paid out through the portfolio, (28); and

(viii) each county's tax rate balances its budget, (31).

The tradable price and the price of intermediates are set nationally and utility elsewhere in the country is fixed, so the CZ is a small open economy. Appendix B.10 writes out the full system and the feasible choices.

Each workplace tract has its own labor market for each sector and skill, and these markets are linked through individuals' choices, as in quantitative urban models with workplace-specific wages (Ahlfeldt et al., 2015; Monte et al., 2018). A higher wage at one workplace draws commuters from other workplaces, workers from other sectors, nonworkers, and migrants, so the employers they leave must respond.

Entry raises the productivity $A_{j_E g_E}$ of the entrant's tract and sector by the factor that raises employment there by J^E. I compare the equilibrium with this higher productivity to the equilibrium without entry, holding fixed all other productivities, amenities, preference parameters, housing expenditure shares, ownership shares, national populations, and the taste distribution. Wages, rents, floor space, prices, individual choices, service entry, and dividends differ between the two equilibria. I solve both in changes relative to the observed pre-entry economy (Section 5).

5 Quantification

I quantify the model separately for each event in its commuting zone (CZ), indexed by c, and its event year. The main parameters are the recruitment elasticities (migration, inbound commuting, and participation), which jointly determine how readily the CZ attracts workers not previously tied to its labor market. Also central are the sector-choice elasticity, which determines the wage pressure service firms feel from the labor demand shock in the tradable sector; the commuting elasticity, which relieves pressure from an inelastic residential land supply; and the shopping elasticity, which relieves pressure from an inelastic commercial land supply and from wage increases. I estimate sector choice and migration from shift-share regressions on public employment, earnings, and population data that vary across commuting zones, and the responses of commuting and shopping from flows that vary across tract pairs. A calibration to the hiring sources in Fact 1 sets the remaining two recruitment margins, participation and inbound commuting. The remaining cost and supply parameters are set from BEA national accounts, shopping trip data, and the literature. Given these parameters (Table 4), I recover the remaining primitives (amenities, productivities, fixed costs, and the service spending share) by inversion so that the model reproduces each commuting zone's economy before entry. None of these steps uses the service responses of Fact 3, which leaves those estimates free to validate the model in Section 5.4.

5.1 Individuals

Sector choice. Individuals' sector choice determines whether the increase in tradable labor demand is met by workers in the same sector or workers who leave other sectors, governed by the elasticity ν. Using the Quarterly Workforce Indicators for 2012–2019, I aggregate private employment and earnings into commuting-zone–sector–skill groups, with sectors defined as tradable and nontradable industries. I estimate the pooled specification

$$\Delta \log E_{csk} = \beta^{\mathrm{QWI}} \Delta \log \bar{w}_{csk} + \alpha_{ck} + \delta_{d(c)s} + \gamma_{sk} + \varepsilon_{csk}, \tag{33}$$

Table 4: Parameters used in the equilibrium calculations

Object	Value	Basis
Behavioral parameters		
Migration, (μ_H, μ_L)	$(2.32, 1.51)$	Estimated; 2010–2018 shift-share 2SLS by skill (Appendix C.2.3)
Sector choice, ν	1.64	Estimated; 2012–2019 QWI shift-share 2SLS (Appendix C.2.1)
Participation, η	3.32	Recruitment calibration to Fact 1 hiring sources
Inbound-to-pair ratios, (q_S, q_M, q_L)	$(0.67, 0.81, 1.00)$	Recruitment calibration; small/medium/large CZs, $\psi_e = q_{\text{size}(e)}\chi_{c(e)}$
Residence–workplace choice, χ	4.93–11.37	CZ-by-year LODES PPML; five-minute within-tract time
Service substitution, $\sigma_{N,c}$	3.95–18.37	Pooled CZ shopping PPML; $\kappa = 0.01$
Establishment entry, ϵ_m	0.40	County CBP regression, 2002–2022; descriptive calibration (Appendix C.5.3)
Accounting and individual calibration		
Service cost shares (L, C, X)	(0.53, 0.13, 0.34)	2012 BEA intermediates; 80/20 labor–space split
High-skill tradable cost shares	(0.49, 0.12, 0.39)	Same; NAICS 51, 52, 54, 55
Low-skill tradable cost shares	(0.27, 0.07, 0.67)	Same; NAICS 31–33
Housing-value shares, ω^0_{ck}	By event	ACS owner values; split equally between partners
Nonemployment income, y^U_k	By event and skill	ACS transfers among nonworkers
Housing share, $\alpha_{H,c}$	0.17–0.28	CZ household-median expenditure share
NT share of nonhousing spending, $s_{N,c}$	0.63–0.89	By event baseline, 10th–90th percentile; service revenue inversion (BEA benchmark 0.48)
Maintained spending, travel, technology, and policy parameters		
Home-origin shopping share, ρ	0.74	74/26 home/work routing share
Travel cost, κ	0.01 per minute	Common to commuting and shopping
Residential supply, $\epsilon_{R,i}$	Tract-specific	Mapped Baum-Snow–Han values; eight-neighbor extension
Commercial supply, ϵ_C	0.60	Maintained commercial-space benchmark
Baseline property tax, t^0_c	0.28	Rate adjusts after entry to finance service needs
Resident-service share of county spending, ς	0.79	2012 Census of Governments
Entrant jobs, industry, site, added space	By event	Baseline capacity rule; 50% tax exemption on new space

Notes: Migration differs by skill; sector choice and participation are common across skills. Border scales vary by event through three CZ-size ratios; commuting and shopping scales also vary by CZ. The recruitment calibration uses all 55 original events (Section 5.1). Cost shares list labor, commercial space, and purchased intermediate inputs (Appendix C.5.2). Appendix C gives the estimands and diagnostics. The separate local-benefit exercise replaces the common 50-percent exemption with event-specific fractions for 54 scenarios (Appendix F.1); its labor parameters remain fixed.

where c indexes commuting zones, s sectors, and k worker skill, distinguishing workers with and without a bachelor's degree. Employment is denoted by E_{csk}, average earnings by $\bar{w}_{csk}$, and Δ denotes the change between 2012 and 2019. The pooled specification imposes a common coefficient across skills. The fixed effects α_{ck}, $\delta_{d(c)s}$, and γ_{sk} absorb CZ-by-skill, Census-division-by-sector, and sector-by-skill differences in growth, respectively, where $d(c)$ is the division containing CZ c.

Because earnings and employment are jointly determined, I estimate equation (33) by weighted two-stage least squares, instrumenting earnings growth with 2007–2011 within-sector industry employment shares interacted with subsequent industry employment growth outside the focal CZ. The pooled estimate is $\widehat{\beta}^{\text{QWI}} = 1.64$, with a CZ-clustered standard error of 0.51 and a first-stage F of 15.03. Under the identifying assumptions, a 1 percent rise in one sector's earnings relative to another's raises the employment ratio by about 1.64 percent over seven years. I use seven years to

match the length of time in my event-study designs. Appendix C.2.1 provides the full specification, the instrument, and the robustness checks.

Migration. The migration elasticity μ_k determines how many workers the CZ draws from the rest of the country when the value of living there rises. I estimate it separately by skill with the same kind of shift-share variation used for sector choice. For each skill, I regress the 2010–2018 change in the log odds of living in a commuting zone rather than elsewhere in the country on the change in the value of living there, measured as log expected earnings (the employment rate times average earnings) less one quarter of log rent, with Census-division effects. I instrument the value change with the CZ's 2010 industry mix of skill-specific employment interacted with national industry employment growth outside the CZ. The estimates are $\mu_H = 2.32$ for high-skill and $\mu_L = 1.51$ for low-skill workers, with CZ-clustered standard errors of 0.75 and 0.51. High-skill workers are therefore more willing to move for better opportunities, consistent with their choosing a sector before a region. The low-skill first stage is weaker, with a Wald statistic of 8.4 against 10.7 for high skill. Appendix C.2.3 gives the specification and inference.

Participation and inbound commuting. I calibrate the remaining two recruitment margins to the hiring sources in Fact 1, holding sector choice and migration at their estimates. The calibration uses all 55 original events, the 25–64 nonworker pool, and the skill-specific nesting orders. It chooses the participation elasticity η and three border ratios (q_S, q_M, q_L), which scale each event's commuting elasticity into its border elasticity, $\psi_e = q_{\text{size}(e)}\chi_{c(e)}$. Participation is disciplined by the share of continuing-resident hires with no observed prior employment, which is 9.5 percent in the source data. Some of these hires held jobs that are missing from the records, so I allow up to five percentage points of missing employment and treat 4.5 to 9.5 percent as the admissible range. The calibrated model predicts 6.8 percent, inside that range. The largest remaining miss is inbound commuting in large CZs, where the model predicts that 21 percent of hires commute in against 23 percent in the data. Across all targets, the root-mean-squared error is 1.3 percentage points. The calibration gives $\eta = 3.32$ and $(q_S, q_M, q_L) = (0.67, 0.81, 1.00)$.

Commuting. Within the commuting zone, residential and workplace choices determine how far a labor demand shock spreads and where additional income accrues. I estimate commuting gravity separately by CZ and year using annual LODES flows from 2007–2022 and a fixed matrix of driving times between tract centroids using driving routes from OpenStreetMap. Fact 2 in Section 3 presents the 2019 comparison with shopping flows. The commuting coefficient satisfies $\beta^C_{ct} = -\chi_{ct}\kappa$, so gravity identifies only the product of the choice scale and the per-minute cost. Following Ahlfeldt et al. (2015) and Tsivanidis (2026), I normalize the per-minute commuting cost to $\kappa = 0.01$, which gives $\chi_{ct} = -\widehat{\beta}^C_{ct}/0.01$. In the 526 CZs included in the 2019 comparison, χ has a median of 6.29 and ranges from 0.87 to 10.80.

Shopping. Where workers shop determines where the entrant's payroll becomes service revenue. With CES demand across service destinations and the same per-minute travel-cost parameter κ as commuting, the coefficient on travel time in a gravity equation for shopping visits is $\beta_c^S = -(\sigma_{N,c}-1)\kappa$, so the rate at which visits decay with travel time reveals how substitutable destinations are. I estimate β_c^S for each commuting zone from 2019 Veraset data restricted to the modeled service sectors (Appendix C.3.2 gives the estimating equation). With $\kappa = 0.01$ per minute, the estimates imply $\sigma_{N,c}$ between 3.95 and 18.37, with a median of 13.28. The same elasticity sets service markups and the share of revenue available to cover fixed costs under free entry.

Individual budgets. I calibrate the housing expenditure share $\alpha_{H,c}$ using household and person records from the American Community Survey (ACS) Public Use Microdata Sample (PUMS).[25] Housing expenditure is annual rent for renters and predicted rent for homeowners, estimated from renter households using dwelling characteristics and location effects. I set $\alpha_{H,c}$ to the weighted median of housing expenditure divided by disposable income within each CZ, where weights combine ACS household survey weights with geographic allocation to the CZ. In the national comparison, the median housing expenditure share ranges from 13 to 35 percent, with a cross-CZ median of 20.2 percent. Figure A.11 compares the housing budget share of a typical household in each CZ with the share of the CZ's total household income devoted to housing. Appendix A.7 provides the measurement details.

After housing, individuals in CZ c spend the fraction $s_{N,c}$ of their remaining resources on services, common to both skills. I set $s_{N,c}$ in the inversion of Section 5.3, so that household spending covers the service revenue implied by each CZ's payroll. Between the 10th and 90th percentile event baselines, the calibrated shares range from 63 to 89 percent. Thus, at fixed prices, an additional dollar of disposable income generates $s_{N,c}(1-\alpha_{H,c})$ dollars of service expenditure. I maintain that workers allocate the fraction $\rho = 0.74$ of this spending to shopping trips originating from home and the remainder to trips originating from work.

I calibrate nonemployment income y_k^U using ACS PUMS person records, taking average public assistance, Supplemental Security Income, and Social Security income among nonworkers of skill k, including those receiving no payments.

I construct the housing ownership shares ω_{ck}^0 from ACS PUMS housing records linked to the individual's education. The model allocates residential rental income across skills in proportion to these ownership-value shares and equally among individuals within each skill group. Appendix C.4.3 gives the rental-income distribution rule.

5.2 Firms and Space

Production cost shares. I measure purchased intermediate inputs, net of purchased real-estate services, in the 2012 BEA supply–use accounts and split the remaining share of gross output 80/20

[25] I use one-year ACS data where available in the input construction. For the full sample, housing shares come from the 2008–2012 five-year household file, including for earlier events, so they are not necessarily measured before entry. Employment baselines use the year before entry.

between labor and commercial space, following the labor–floor-space calibration in Ahlfeldt et al. (2015), Hoelzlein (2023), and Miyauchi et al. (2025) (Appendix C.5.2). Intermediate inputs account for 0.34 of gross output in the nontradable composite, 0.39 in high-skill tradables (NAICS 51, 52, 54, and 55), and 0.67 in manufacturing, which gives labor shares of 0.53, 0.49, and 0.27. The model treats these intermediates as supplied from outside the CZ at a fixed price. Service firms and tradable producers have the same Cobb–Douglas form, and each sector uses its group's shares. With constant returns and Cobb–Douglas production, the average wage a tradable producer can pay at given rents has an elasticity of $1/\alpha_L^g$ with respect to its productivity, so a 10 percent productivity increase raises it by about 21 percent in high-skill tradables and 43 percent in manufacturing.

Skill weights. Each sector combines high- and low-skill workers with Cobb–Douglas weights $\lambda_{jgk} = \lambda_{gk}$, which are common across tracts and fixed in counterfactuals. I set them to each skill's national share of the sector's wage bill in the 2012 Quarterly Workforce Indicators. Within each sector, I multiply full-quarter employment by three months of average monthly earnings for each education group and sum these wage bills across quarters, states, and industries. Workers with a bachelor's degree or more receive 55 percent of the wage bill in high-skill tradables, 26 percent in low-skill tradables, 50 percent in high-skill services, and 27 percent in low-skill services, and low-skill workers receive the rest. The weights fix each skill's share of the wage bill rather than its share of employment, so employment shares still respond to relative wages.[26]

To impose the common weights on the baseline data, I reconstruct employment and wages by detailed industry and skill. The reconstruction keeps employment in each tract and industry unchanged and, within tradables and within services, keeps each skill's total employment, total payroll, and commuting flows by broad sector. Wages and payrolls in detailed industries can change. I then rebuild the household, production, floor-space, and service-revenue accounts from the new allocation.

Establishment entry. Free entry sets the number of service establishments in a tract with elasticity ϵ_m with respect to net revenue, and the fixed-cost shifter φ_j is recovered from the observed establishment count (equation (C.39)). I set $\epsilon_m = 0.40$, in line with a county regression of five-year growth in service establishments on growth in service revenue relative to wages. The regression pools the five service sectors (NAICS 44–45, 61, 62, 72, and 81) over 9,798 county five-year windows between 2002 and 2022, with end-year effects. Establishments and payroll come from County Business Patterns, revenue is payroll divided by the national service labor share of value added in the BEA accounts, and wages come from the Quarterly Workforce Indicators. The OLS coefficient is 0.39 (standard error 0.01), so a 10 percent rise in net revenue adds about 4 percent to the number of establishments (Appendix C.5.3). Because entry also raises revenue by attracting spending, and

[26] QWI education categories cover workers aged 25 and older, so these shares proxy for the wage bills of the model's labor bundles. The wage bills include private and state and local government jobs. Industries keep the high- and low-skill-intensity classification used throughout, including the industries reassigned from other employment to tradables. Industries with missing or ambiguous labels are allocated using the national employment-weighted mix of classified industries within the same broad industry.

the revenue measure moves closely with employment, I treat the value as a descriptive calibration rather than a causal estimate. The value is also close to the response implied by Almagro and Domínguez-Iino (2025) for Amsterdam, where congestion in commercial real estate makes the number of establishments rise with amenity spending with an elasticity of about 0.40. They calibrate the congestion parameter from the range of housing supply elasticities in Saiz (2010).

Housing and commercial space. Residential supply uses tract-specific elasticities $\epsilon_{R,i}$ mapped from Baum-Snow and Han (2024). I retain each valid nonnegative prediction and replace a missing or negative value with the unweighted mean of the eight nearest tracts with valid predictions. Distance is measured between 2020 tract centroids. This extends the existing estimates without using employment or payroll validation outcomes. Across the 36,557 distinct tracts covered by the current event baselines, the unweighted mean is 0.53 and the median is 0.57. I set the commercial space elasticity as $\epsilon_C = 0.60$, at the upper end of the metropolitan office estimates of Marcato and Tong (2025, p. 412), which range from 0 to 0.61, and above the retail estimate of 0.49 in Benjamin et al. (1998, p. 9). This is close to the median residential elasticity across the current tracts. Greaney et al. (2024, pp. 29, 52) take a similar approach. Noting the absence of local estimates for commercial floor space, they set commercial supply elasticities in their model of the San Francisco Bay Area from residential ones.

Property taxes. Users of housing and commercial space pay the property tax $t_c = 0.28$ on top of the rent. Because the model has annual rents but no property values, t_c is the tax bill per dollar of annual rent rather than per dollar of value. A tax of 1 percent of value on a property worth 20 years of rent is 20 percent of a year's rent. Couture et al. (2024, pp. 878–879) measure taxes the same way and find that they are 20 percent of the annual user cost of housing downtown and 30 percent in the suburbs. The 2010–2014 American Community Survey implies about 32 percent of annual rent.[27] I use an average of these estimates and apply the same rate to commercial space.

Government service costs. I set the resident-service share ς to 0.79, the share of resident services in county-government spending on resident and developed-property services in the 2012 Census of Governments. Resident services are health, welfare, police, corrections, libraries, and recreation. Developed-property services are roads, drainage, sewerage, waste collection, fire protection, and transportation infrastructure. The share is the same across counties, and each county's baseline spending G_a^0 equals its baseline property-tax receipts.

5.3 Constructing the Pre-Entry Economy

I recover the remaining fundamentals so that each event's baseline equilibrium reproduces its pre-entry economy. Given observed wages, rents, and commuting costs, residential amenities B_{ik}^R and

[27] Because t_c is levied on top of the rent, taxes are $t_c/(1+t_c) \approx 22$ percent of what users pay, which lies within this range on either reading.

workplace–sector amenities B^W_{jgk} match residence and workplace employment by skill. Tracts that have more residents than their rents and access to jobs predict must offer high amenities. The same logic applies for workplaces that employ more workers than their wages predict. The taste shifters $\overline{B}^R_k$, $\overline{B}^S_{gk}$, $\overline{B}^E_{gk}$, and $\overline{B}^U_{gk}$ match the CZ's population, sector shares, and participation rates by skill, and tradable productivity sets baseline unit costs equal to the national price.

Service productivities are recovered so that the model reproduces where service purchases occur, adapting the procedure of Oh and Seo (2023). I infer each tract's service revenue from its worker payroll, which equals the labor share of costs under free entry. I then set the service share of individuals' nonhousing spending, $s_{N,c}$, so that the spending of the CZ's residents, net of the share of shopping that leaves the CZ (observed in the Veraset data), covers total CZ service revenue. Given this total budget, I invert service productivities so that consumers' demand, which follows the gravity structure of shopping, matches the service revenue observed in each tract. The fixed-cost parameter then makes revenue per establishment consistent with free entry at the observed establishment count.

The calibrated shares, from 0.63 to 0.89 at the 10th and 90th percentiles of event baselines, are well above the 0.48 share of nonhousing consumer spending that goes to local services as measured in the BEA accounts. The residents of the CZ are all workers observed in the LODES data plus the number of working-age nonemployed residents from the ACS data. This is not an accurate reflection of all of the people living there nor does it capture the full demand for local services. Local service establishments also sell to retirees, to visitors from outside the CZ, to businesses, and to the public and private insurers that pay for much of health care and education. The spending share of the model's individuals must support the demand of these other groups, so I scale $s_{N,c}$ to observed service activity rather than households' budgets. This choice materially matters for the multiplier, because each new dollar of resident income raises local service spending by $s_{N,c}(1-\alpha_H)$ in the model. With this calibration, I am assuming the spending that $s_{N,c}$ absorbs moves in the same proportion as the individuals in the model. Employer-provided insurance pays for workers' health care, public insurance and school budgets grow with the resident population, and local businesses buy more services as local employment grows, so I view this as a reasonable assumption.

I recover the amenity, productivity, fixed-cost, and taste fundamentals to rationalize the observed baseline economy as an equilibrium. These fundamentals are held fixed across counterfactuals. I solve for the entry equilibrium in changes relative to the baseline using exact hat algebra. Writing $\widehat{x} = x^1/x^0$, the fundamentals cancel, so the equilibrium conditions depend only on baseline shares and quantities, the parameters described above, and the entry shock. I solve jointly for the changes in wages, rents, service establishment counts, and dividends that clear all markets.

5.4 Validation

The purpose of this model is to explain the spatial distribution of the service multiplier. The model is successful only if it correctly predicts where the entrant's workers live, how much of its payroll becomes new income for CZ residents, and how that income translates into nearby service jobs. I

exploit the fact that estimation makes limited use of the entry events. Only the participation and border elasticities are calibrated using one-fourth of the entrants from the public data to match the hiring shares from the Census events. The shopping elasticity comes from 2019 trip data, after every event, but it fixes only how fast trips decay with travel time, not the model's shopping flows. The model inversion uses data in the year before entry to reproduce the baseline economy of each commuting zone.

Who is hired. First, the out-of-sample fit on the recruitment sources in the model closely mirrors the shares observed in the Census events. On the 161 firm entry events not used in the calibration of the participation or in-bound CZ share of workers, the model's split of hires among continuing residents, migrants, and outside-CZ commuters is 57, 18, and 25 percent against 58, 14, and 28 in the Census. Additionally, the income of the productivity shock matched to the entrant's hiring goes to the same workers in similar proportions as the Census data. Table 5 presents the share of each $100 of entrant payroll that becomes new income for residents. On the held-out events the model's shock generates the same net injection of about a third of entrant payroll, with no entrant wage premium and no moment that fixes the gain of job switchers. The income gains being close is more of a statement about the hiring shares and wages matching. The model pays each hire the wage of the tract and sector it joins, and the hire-weighted average of that wage is $102,000, between the data's median of $75,300 and mean of $125,000 for the same hires.

Table 5: The hired cohort on the events outside the calibration sample: payroll per $100 of entrant payroll (dollars) and residence within the commuting zone

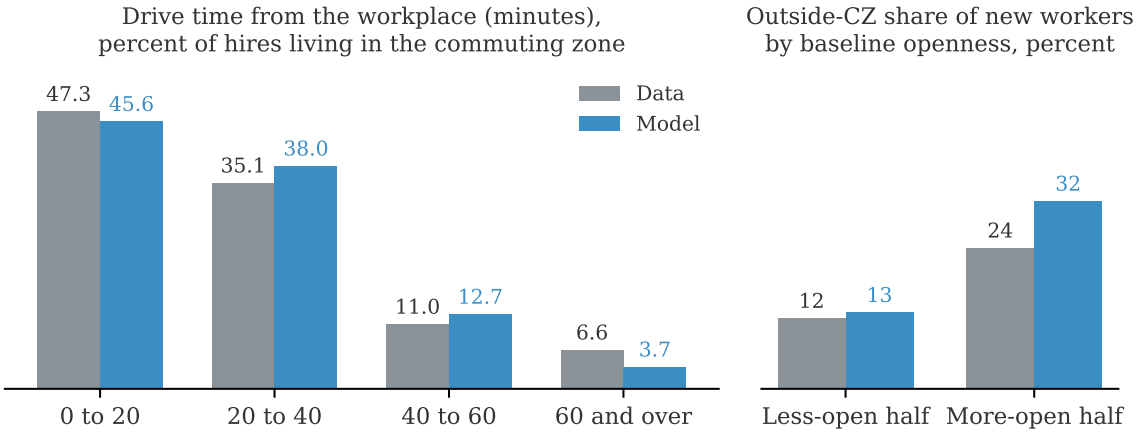
	Census data	Model (not in calibration)
Paid to workers living outside the CZ	28	25.1
Replaces residents' prior earnings	40	39.7
New income of residents	32	35.2
migrants	15	18.8
previously nonemployed	4	3.3
job switchers' earnings gain	14	13.1
Hires living within 20 minutes of the site (percent of resident hires)	47.3	45.6
Events	400	161

Notes: The Census values are the Fact 1 earnings approximations for hires at large tradable entrants, rounded and without confidence intervals. The three components of residents' new income sum to it up to a $1 offset for out-movers, which is omitted. Model values are the workers who switch into the entrant's tract and industry, pooled over events with cohort-payroll weights. The residence row pools all 216 events with net-hire weights, commuting-zone residents only, since the model's outside area has no travel time; its Census value is the Fact 1 tabulation of hires' residences.

The model is able to closely match the distance of its workers' residential neighborhoods as shown in Figure 6. This indicates that the commuting gravity estimated on baseline flows predicts out of sample, without inverting bilateral amenities to match every tract pair's flow exactly (Dingel and Tintelnot, 2026). I then check whether the model pulls workers from outside the commuting zone in proportion to a site's prior openness to outside commuters. The border term is calibrated

to the inward-commuting share of Census hires within three size groups, not to how that share varies with prior openness. I use the public event sample to divide the entrant sites at the median of prior openness and measure the share of new workers. This rises from 12 to 24 percent in the data and from 13 to 32 in the model. This is an important margin that separates a high-multiplier commuting zone from a low one.

Figure 6: Drive time from the workplace and the openness split, data and model (percent)



Notes: Grey bars are the data, blue bars the model. Left panel: hires at the entrant who live in the commuting zone by drive time from the workplace, as a share of all hires. The model reports the same share for the workers in the tract and sector of the shock. Right panel: the share of each subsidized site's new workers who live outside the commuting zone, public LODES at the 216 award sites split at the median of baseline commuting openness, against the model with promised-job weights. The drive-time bins are not targeted; the outside-CZ level is targeted within size groups and its split by openness is not.

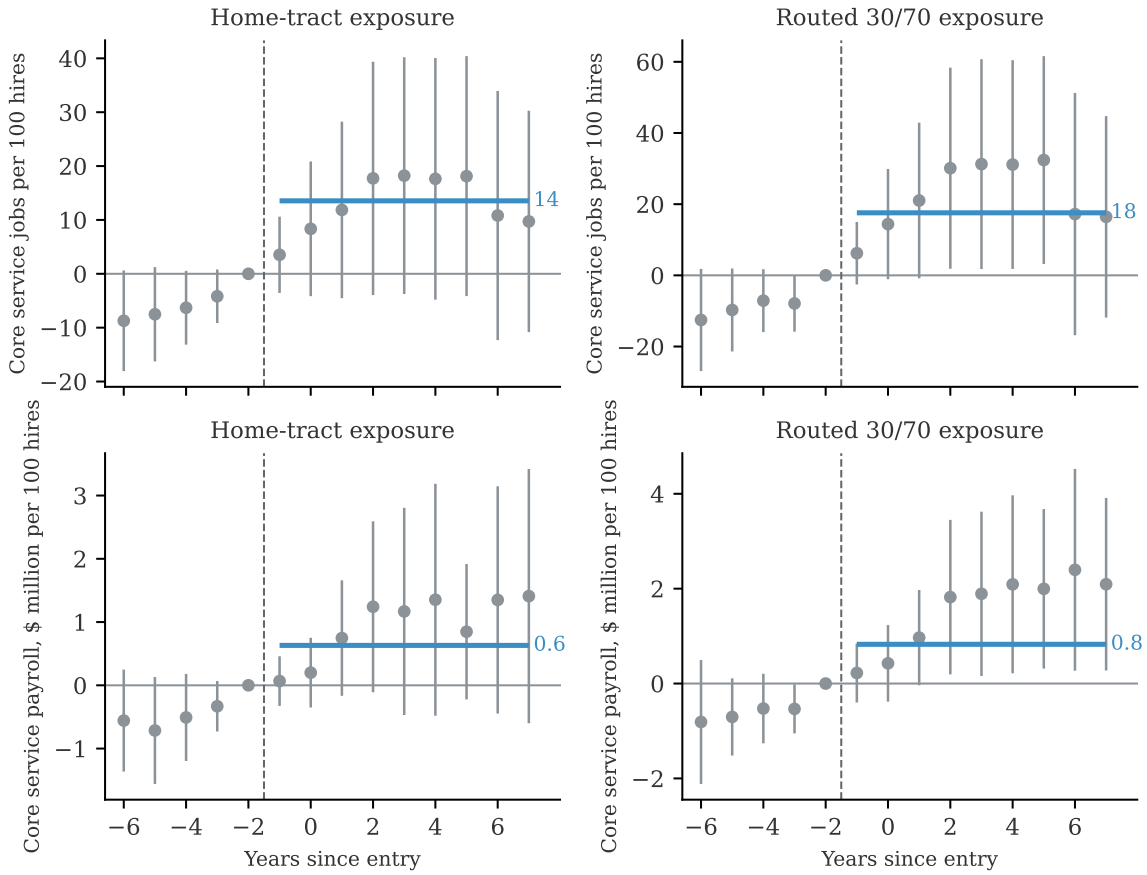
Where service jobs appear. Where service jobs appear in the model is broadly consistent with the patterns observed in the Census data. The central validation is the continuous neighborhood event study of Fact 3 for core services, retail, food and accommodation, and other services (NAICS 44 to 45, 72, and 81), which I replicate on the model for all firm entry events. I measure a tract's exposure as the number of the entrant's hires who live there and convert it into local demand for services either by claiming all of a person's spending occurs in their home tract, or giving 30 percent to their home tract and 70 percent spread over the adjacent tracts.[28] The outcome is the change in the tract's core service employment.[29] Figure 7 places the model beside the Census estimates. Under both exposures the model lies within about one clustered standard error for jobs

[28]The 30/70 split follows the hyperlocal share of service visits in the 2019 Veraset data, the 11 percent of within-CZ core service visits that end in the home tract against the 24 percent that end in an adjacent tract (Table 3). The home tract and its neighbors carry the part of shopping that differs across tracts, while the rest of a resident's visits follow the common hub pattern of the commuting zone (Section 3), so the routed exposure captures the component of shopping that is specific to the shopping footprint of where the hire lives.

[29]Each event is solved against its own no-entry equilibrium with parameters and shocks fixed. The Census data estimates on approximately 100 commuting zones; the model's values are static event-fixed-effect projections on 216 events, so samples and estimators differ. The comparison for all nontradables, which adds education and health, is in Appendix D.

and, for payroll, within one under the home-tract design and between one and one and a half under the routed design. The model does miss the spread more. Routing the hires 70 percent to the adjacent tracts raises the empirical coefficient by a factor of 1.7 and the model's by 1.3, in both jobs and payroll, so the model places too much spending in the home tract and cannot capture the concentration of services.

Figure 7: Core service employment and payroll where the entrant's workers live: empirical event study and the model



Notes: Circles are the continuous event-study coefficients of Fact 3 for core services (NAICS 44 to 45, 72, 81) per 100 exposed hires with 95 percent confidence intervals, with exposure in the hires' home tracts (left) and routed 30 percent to the home tract and 70 percent to the adjacent tracts (right); year −2 is the reference and hiring begins in year −1 (dashed line). The blue line is the model's response from the same design applied to its own hires on all 216 events. The model is static.

The model has two limits here. First, it performs much worse against the data when the service sector includes health and education. With all nontradables the model's response is about half of the empirical one in jobs and two fifths in payroll under the home-tract exposure, and two fifths and one fifth under the hyperlocal exposure (Appendix D). Schools and hospitals are funded in large part not by individuals' disposable income and they are located in specifically zoned places, neither of which the model is equipped to address. Second, the model cannot reproduce the stacked design that compares the top and bottom quartiles of exposed tracts. The gap in exposure between

those quartiles is two to three times larger in the data than in the model, because hires' residences are concentrated in a few tracts near each site in the data and spread more evenly in the model. The two shortfalls compound to a contrast of a few jobs per 100 hires in the model against several dozen in the data.

6 Employment, Welfare, and Spatial Incidence

I use the quantified model to compare the baseline economy with the economy after a productivity increase in the entrant's tract and tradable sector for 216 events in 95 commuting zones. This sample of firm entry events comes from the subsidy awards data described in Section 2 and Appendix A.1. Each firm entry event identifies a subsidy amount, a tract, an industry, and the number of jobs the firm promises to create.[30] Employment changes are per 100 promised jobs.

The firm entry (productivity) shock adds about three quarters of the jobs it promises. For every 100 entrant jobs, the median event adds 74 jobs to the commuting zone. Tradable employment outside the target location-sector falls by approximately 49 jobs. In the model, tradable sectors sell at fixed national prices. Tradable producers compete for workers and floor space but sell at a national price, so they cannot pass higher costs on to consumers and shrink instead. Increased demand for services translates into increased employment of 22 jobs in the service sector. Measured per net tradable job, the median service-job gain is 44 per 100. That is within the range of 30 to 80 in Bartik and Sotherland (2019) and about a quarter of the 160 in Moretti (2010).

Following $100 of additional payroll in the shocked tract and industry, total CZ workplace payroll rises by about $78, after accounting for employment and wage changes elsewhere. Wage earnings rise by about $40 for residents of the CZ. The difference reflects earnings paid to workers living outside the CZ and changes in residents' earnings from those who previously worked outside the CZ. After adding dividends from ownership of residential land, housing costs, spending on tradable goods, and shopping outside the CZ, spending at service firms inside the CZ rises by about $25 (Appendix E.1). At the median, residents' equivalent variation, the annual cash transfer without the firm entry that leaves residents as well off as with the firm entry, is about $7,600 per entrant job. This is approximately 8 percent of the average wage in the entrant's tract and industry, or $10.3 per resident (10th–90th percentile range $2.4 to $59).

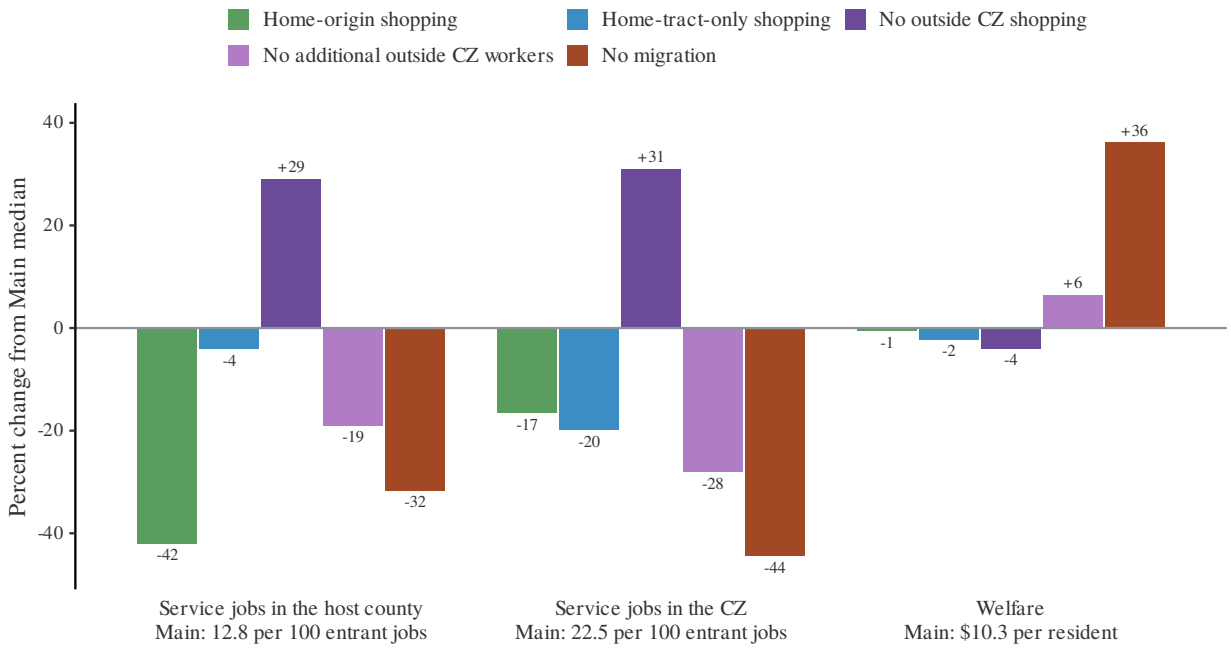
Section 6.1 shows what shopping, commuting, and migration add, and Section 6.2 traces the spatial incidence across neighborhoods. Section 7 asks how much the multiplier varies across markets and across locations within a county, and what each job costs the county that subsidizes it.

[30] I set the productivity increase so that employment in that tract and industry rises by the promised number of jobs before the property-tax exemption. I then hold this productivity increase and added commercial capacity fixed when exempting half of the new property from tax.

6.1 What do shopping, commuting, and migration add?

The model departs from other spatial models by allowing individuals to travel to consume services and by allowing them to migrate, live outside the CZ, and not participate in the workforce. Fact 1 showed that the in-bound, migration, and participation margins are important for matching the sources of workers hired by these entering firms. I included them to distinguish new workers from new service consumers. Fact 2 showed that allowing travel for services was important at such a granular spatial scale because only 10 percent of service trips occur in the tract where individuals live. In Figure 8, I show how much including these model features changes the outcomes of the model in terms of service employment and residential welfare. I keep the same productivity shock, and I shut down one feature of the model at a time. Three of the restrictions change where residents shop – all shopping starts from the home tract (74 percent at baseline), all shopping stays in the home tract, or no shopping can leave the CZ. The other two close a source of the entrant's workers either by not allowing any new commuters from outside the CZ, or by shutting down migration into the CZ.

Figure 8: Host-county service jobs, CZ service jobs, and welfare under five restrictions



Notes: Each bar is the percent change of a restriction's median from the main model median on the same events. Every specification receives the same productivity increase and subtracts its own no-entry baseline. Service jobs are per 100 original promised jobs. The host-county (where the shock occurs) service jobs are those at workplaces in the entrant's county. Welfare is equivalent variation per baseline resident in annual 2023 dollars. Home-origin shopping makes all shopping originate from the home tract; home-tract-only shopping restricts all shopping to the home tract and removes the outside option; no outside CZ shopping removes destinations outside the CZ; no additional outside CZ workers holds the number of workers commuting in from outside the CZ at its no-entry level; no migration holds inside-CZ and adjacent-region populations fixed.

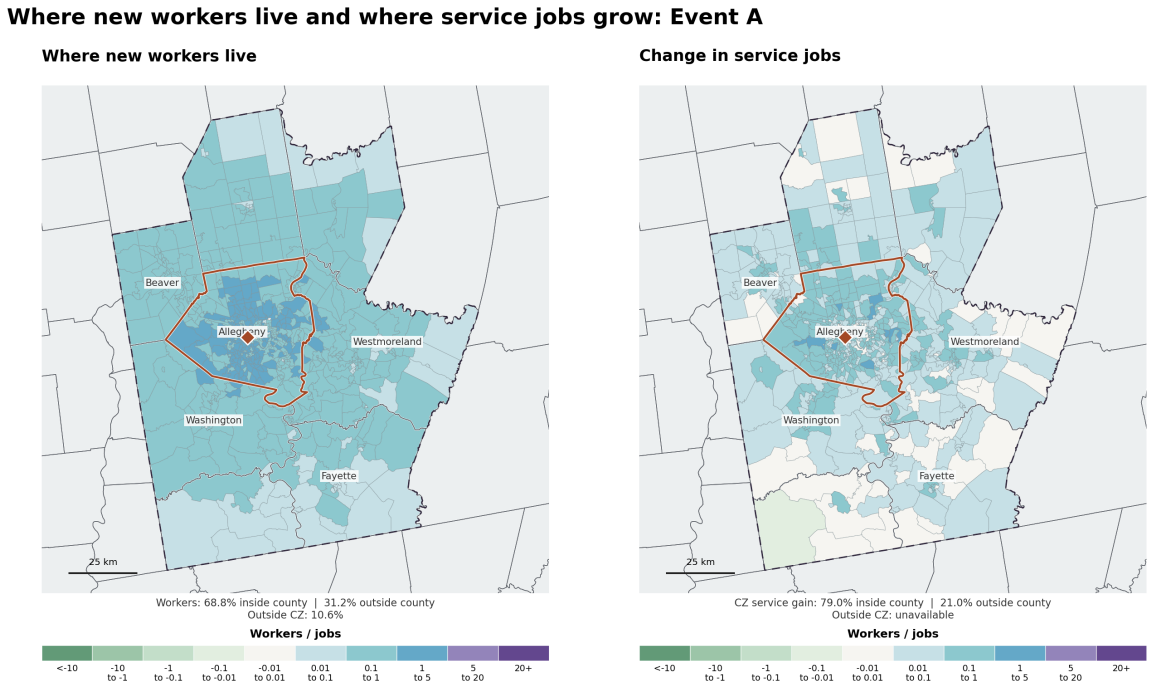
When all shopping must start from the home tract and workers can no longer shop near their

jobs (26 percent of trips in the data), service jobs in the county with the productivity shock fall by 42 percent, more than twice the 17 percent drop for the CZ. The location with the entrant shock loses the most as both workers living in the CZ and commuters no longer shop nearby. Restricting shopping to the home tract does not generate the same spatial incidence because most of the new hires live in the adjacent tracts, so this counteracts the loss from the workplace shopping despite losing more jobs in the CZ as a whole. These restrictions are not nearly as impactful as removing the option for individuals to shop outside the CZ (15 percent of trips in the data). Closing shopping outside the CZ keeps spending that would have left, and service jobs rise by about 30 percent in both the county and the CZ. Under all three shopping restrictions, welfare moves by 4 percent or less because gains and losses in the service sector affect how easy it is for tradable firms in other locations to increase employment. Without additional commuters from outside the CZ, the productivity shock increases wages for local residents and increases migration. Service jobs fall by 19 percent in the county because of increased labor competition, but welfare rises 6 percent as wages rise further. Without migration, service jobs fall by 32 percent in the county. Welfare falls to zero in that run, but mechanically, because the wedge that holds populations fixed counts as a utility cost. A model without travel to shop would put service jobs in the wrong place, and one without commuting and migration would understate the job gains. Appendix Tables E.3 and E.4 report all eleven restrictions run with the model.

6.2 Where do the gains land within the CZ?

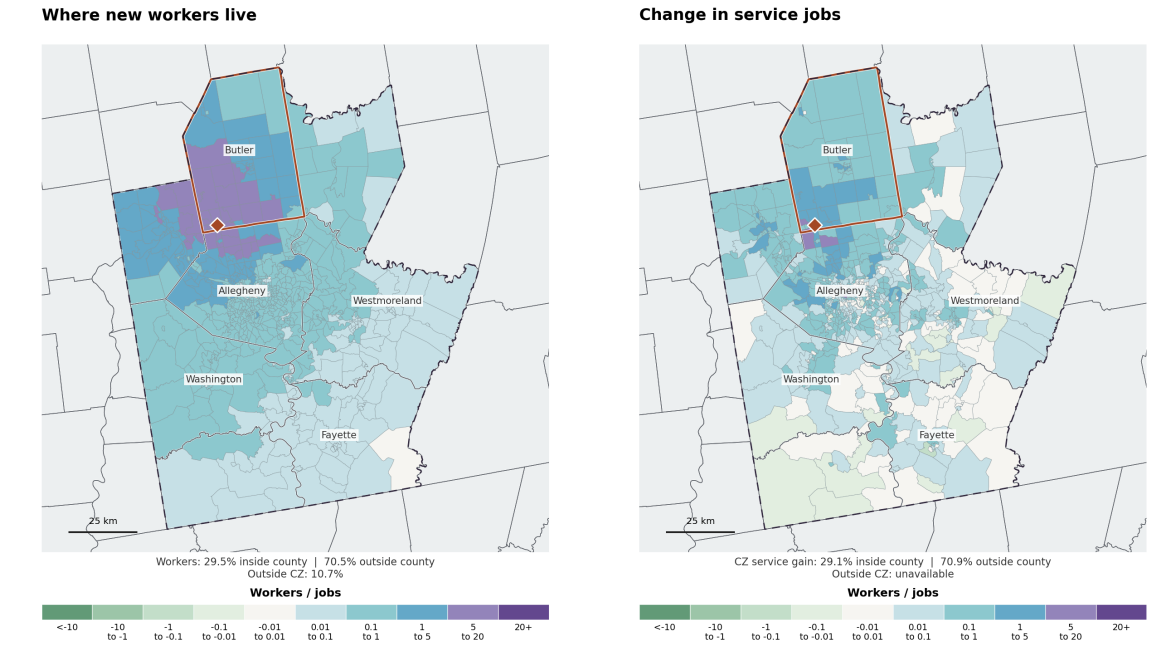
At the median, the county experiencing the productivity shock contains 28 percent of the CZ's residents but receives 60 percent of the resident income gain and 64 percent of the service job gain. Figures 9 and 10 show two entries in the same CZ that each add about 24 service jobs per 100 entrant jobs in the CZ, but their counties see very different gains. Event A enters the CZ's central county, where most of its workers live, and that county keeps 79 percent of the gain. Event B enters at the edge of a neighboring county, where about 70 percent of its workers live outside the county, and it keeps only 29 percent, with most of the gain across the county line.

Figure 9: Where new workers live and where service jobs grow: Event A



Notes: The left panel counts the entrant's workers by their residence tract after entry. The right panel shows the change in service jobs by workplace tract. The diamond marks the entrant's tract and the outlined area in red is the host county.

Figure 10: Where new workers live and where service jobs grow: Event B



To explain why service jobs grow in certain tracts, I regress each tract's service-job gain on

its baseline service employment, wages, and rents, and its exposure to the entrant's payroll either through the home or work origin shopping (Table 6). Both exposures come from the baseline equilibrium, before the productivity shock changes the distribution of individuals, wages and prices. I assign the baseline payroll of the entrant's location and sector to the tracts where its workers live using baseline commuting flows, keep the share residents spend on services after housing, and send that spending to destination tracts using baseline shopping probabilities, starting from workers' homes for home-route exposure and from the entrant's tract for work-route exposure. I find that pre-period exposure explains most of this variation. Comparing tracts within the same event and county, and leaving out the entrant's own tract, baseline service employment, wages, and rents explain only 10.3 percent of the variation in tracts' service-job gains. Home-route exposure alone explains 74.3 percent. Adding work-route exposure raises this to 76.8 percent, and adding the tract controls back moves it only to 77.6 percent. Where an entrant's workers live and shop largely determines which neighborhoods gain service jobs. Conditional on those factors, service jobs are lower in higher-wage places. Appendix E.5 defines the exposures and repeats the exercise leaving out each CZ in turn.

Table 6: Commuting and shopping exposure and tract service-job changes

	(1)	(2)	(3)	(4)
Home-route exposure		0.70***	0.43***	0.41***
		(0.03)	(0.04)	(0.04)
Work-route exposure			0.54***	0.56***
			(0.07)	(0.07)
Log(1 + baseline service jobs)	0.04***			0.01***
	(0.01)			(0.00)
Log baseline service wage	0.01			-0.07***
	(0.01)			(0.01)
Log baseline residential rent	0.00			0.00
	(0.00)			(0.00)
Event × county effects	Yes	Yes	Yes	Yes
Within-county R^2 (%)	10.29	74.32	76.76	77.57
Tracts	114,796	114,796	114,796	114,796
CZ clusters	95	95	95	95

Notes: The outcome is the change in service jobs per 100 promised jobs. The expansion tract is excluded. Exposures are dollars reaching each tract per $100 of baseline payroll in the location-sector, including housing and service expenditure shares and the 0.74/0.26 home/work routing split. Commuting is weighted by skill-specific wages. Parentheses give CZ-clustered standard errors.

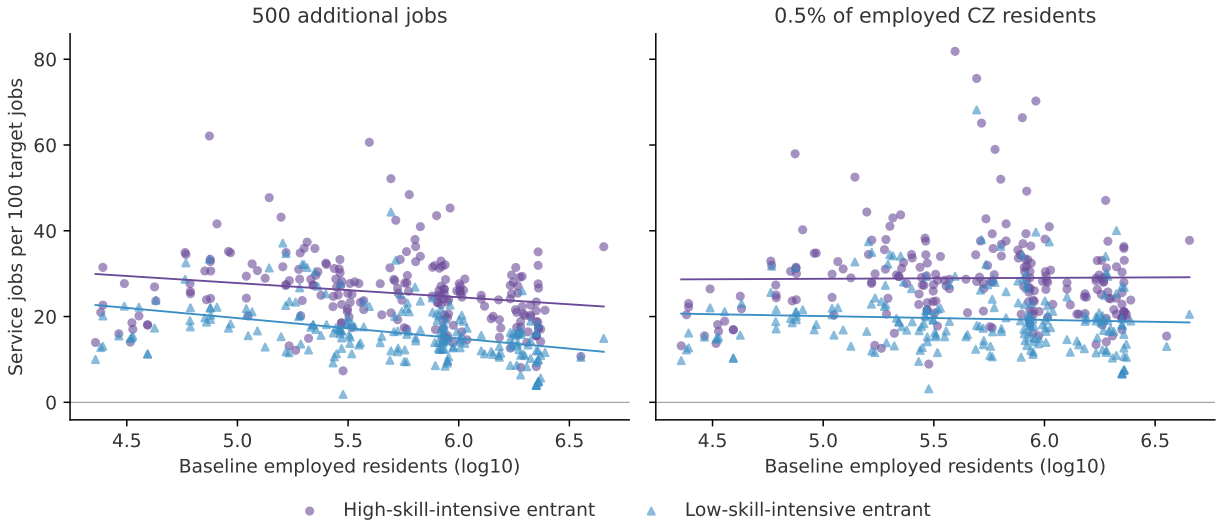
7 Variation in the Multiplier

7.1 How much does the CZ market matter?

So far, I have presented all the results as the median number of service jobs and the median welfare gains. This masks substantial heterogeneity in the response to the productivity shock across CZs. Between the 10th and 90th percentiles of events, the service job gain ranges from about 14 to 33

jobs per 100 entrant jobs. Net jobs range from about 62 to 85, and welfare per resident from about \$2.4 to \$59, a factor of more than ten. A natural first explanation is that the events differ greatly in size, from 126 to 25,000 promised jobs. To control for this, I recalibrate the productivity shock at each of the 216 observed sites, in the entrant's observed sector, so that employment there rises by 500 jobs. I then calibrate it to match a 0.5% employment increase in the CZ. Fixing every shock at 500 jobs reduces the variance of the service gain by only 19.5 percent. Scaling every shock instead to 0.5 percent of the CZ's baseline employed residents actually raises the variance by 29 percent, but removes the impact of the CZ baseline size. This size-proportional scaling gives the largest third of CZs shocks of about 9,000 jobs at the median, against about 800 in the smallest third, while observed firm entries are about 900 to 1,000 jobs in each. The remaining differences reflect local market and site characteristics, together with the entrant's industry. Even together, the market characteristics I measure explain only about 21 percent of the variation within industries (Table 7), leaving substantial variation across sites and interactions outside this aggregate regression.

Figure 11: Market size and service multipliers under standardized entry sizes



Notes: Each point is an entrant shock at the observed site. Lines are separate linear fits against log baseline employed residents. The simulations hold job gains at the entrant's site fixed at the absolute or proportional size with productivity and payroll adjusting to meet that target. The blue line represents low-skill-intensive entrants and the purple line represents high-skill-intensive entrants.

Table 7 relates the 500-job multiplier to market characteristics, controlling for the entrant's industry. Market size alone explains 8.5 percent of the variation within industries, with smaller multipliers in larger markets. Adding inbound commuting, shopping leakage, and the residential supply elasticity raises this to 17.4 percent in column (5). Column (6) adds the shopping elasticity $\sigma_{N,c}$ and the commuting elasticity χ_{ce} used in each event's calibration. These govern substitution across shopping destinations and residence–workplace choices. With both included, the explained share is 20.6 percent, and the market-size coefficient falls to 0.44 and is no longer significant. The remaining 79 percent of the variance is from the networks available within CZs.

Table 7: Market characteristics and service multipliers across commuting zones

	(1)	(2)	(3)	(4)	(5)	(6)
Log baseline employed residents	-2.15***	-2.75***	-0.89	-3.83***	-2.79***	0.44
	(0.79)	(0.74)	(0.92)	(0.76)	(0.99)	(1.42)
Inbound employment share		-1.28**			-0.04	-0.12
		(0.62)			(0.78)	(0.78)
Residential land supply elasticity			1.77**		1.26*	1.12*
			(0.74)		(0.72)	(0.67)
Shopping leakage				-2.63***	-2.37***	-1.31
				(0.63)	(0.79)	(0.84)
Shopping elasticity, $\sigma_{N,c}$						-2.64**
						(1.17)
Commuting elasticity, χ_{ce}						-0.60
						(0.77)
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes
Within-industry R^2 (%)	8.46	10.80	11.35	15.99	17.36	20.64
Observations	432	432	432	432	432	432
CZ clusters	95	95	95	95	95	95

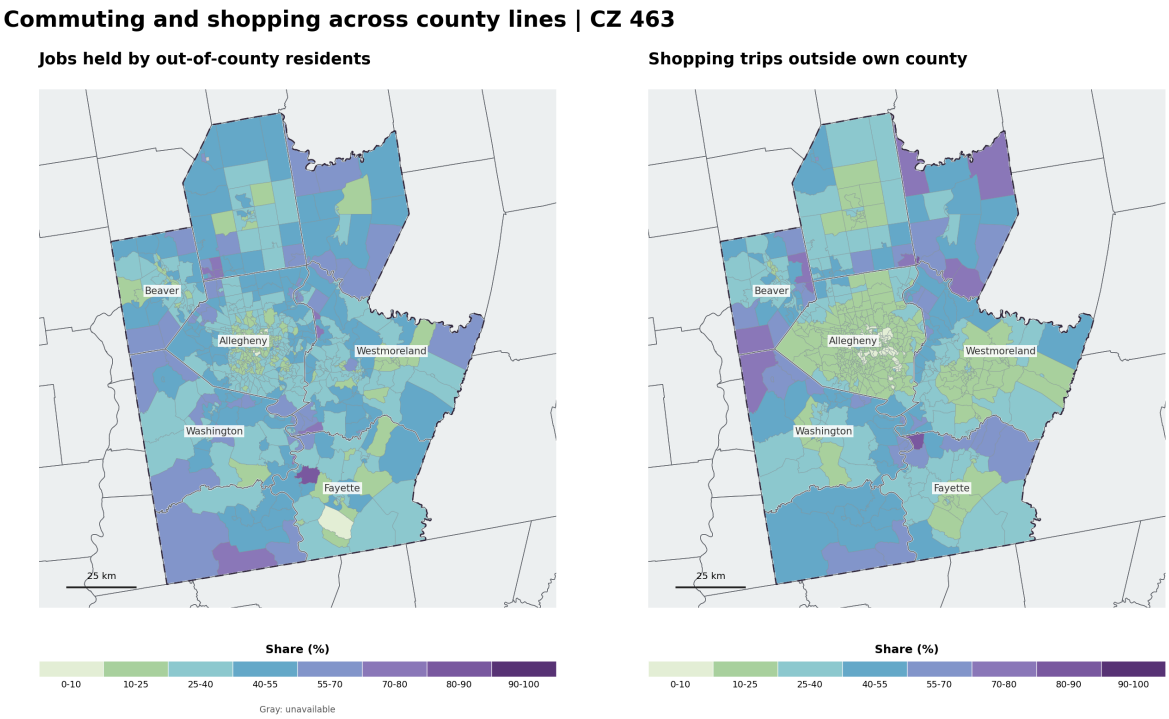
Notes: The outcome is service jobs per 100 entrant jobs under the 500-job shock. Each coefficient is the change associated with one standard deviation of the predictor after removing industry effects. Column (6) adds the estimated shopping and commuting elasticities. Parentheses give CZ-clustered standard errors.

7.2 How much does the location within a county matter?

Where in a county a firm locates matters nearly as much as which market it enters. Within the typical host county, the site at the 90th percentile adds about 1.8 times as many service jobs per 100 entrant jobs as the site at the 10th, a gap of about 14 jobs; across markets, the corresponding ratio is 2.4. Differences across locations within a county account for nearly half of the variance of the multiplier across all entries. Figure 12 shows how much tracts can differ even within one county. Shopping is more local, and six of the CZ's eight counties have tracts where at least 75 percent of shopping trips stay within that county. Workplaces are more mixed, and even in Allegheny County, the center of the CZ, residents of other counties hold at least a quarter of the jobs in about 60 percent of tracts. In the typical tract of each surrounding county, they hold about 30 to 50 percent. This is a key tension in choosing a site. A firm in the central county locates where residents do most of their shopping, but because so many of the county's jobs are held by commuters, much of the new payroll goes to residents of the surrounding counties, who spend much of it near their home.

Ultimately, a location creates more local service jobs when its new payroll translates to more new income for CZ residents in particular. That depends on the labor market around the location rather than on where people shop (Table 8). When the entrant's industry already employs many workers nearby, the jobs can be filled without raising wages much, so the entrant adds little payroll — there is a thicker local labor market of people already working in that industry. When that location has a large share of workers who commute from outside the CZ, little of that payroll reaches residents. These two features account for most of the difference between locations in the

Figure 12: Commuting and shopping across county lines in the CZ of Events A and B



Notes: CZ 463. The left panel shows the share of each tract's jobs held by workers who live in another county. The right panel shows the share of each tract's residents' shopping trips that end outside their own county. Denominators include all primary workplace jobs and all resident-origin shopping trips, including flows that cross the CZ boundary. The dashed line marks the CZ boundary; gray tracts are unavailable.

same county. Shopping decides where in the CZ the new service jobs appear, not how many there are.

About half of the gap between locations reflects the payroll each site needs to create the same jobs. When every site instead receives the payroll the expansion pays at the observed site, the best-to-worst gap in service jobs falls from 22 to 11 per 100 promised jobs and the welfare gap from \$7.6 to \$3.1 per resident (Table 9). The rest of the gap reflects how far a given payroll travels through the local economy around each location.

Observed sites are far from the best available. At the median, the observed site ranks at the 40th percentile of CZ welfare among its alternatives, and the best alternative raises welfare by 48 percent. For a county recruiting a firm, which site the firm uses can matter nearly as much as which market it is in, and what makes a site valuable is mainly the labor market around it.

7.3 What does a job cost, and is the subsidy worth paying?

At the median, the local share of a subsidy costs about \$3,600 a year for each net job the entry adds to the CZ. The local share most frequently takes the form of property taxes forgone in the county where the firm locates, so the host county bears the cost while the gains spread across the CZ. Residents across the CZ gain \$2.8 for each public dollar, against \$0.86 for residents of

Table 8: Decomposing the spread in the service multiplier across locations within a county

Panel A: Within-event decomposition of the log multiplier

Term	Share of within-event variance (%)	Median within-event P90/P10
Payroll per entrant job	49.5	1.38
Resident income per payroll dollar	52.2	1.43
CZ service spending per dollar of resident income	-2.4	1.15
Service jobs per dollar of CZ service spending	0.6	1.05
Service multiplier (product)	100.0	1.83

Panel B: Site characteristics and the within-event log multiplier

Site characteristic (baseline, standardized within event)	(1)	(2)	(3)
Log location-sector jobs	-0.212***		-0.203***
	(0.009)		(0.008)
Log location-sector wage	0.185***		0.183***
	(0.010)		(0.010)
Inbound-commuting share		-0.161***	-0.038*
		(0.035)	(0.020)
Shopping retention from workers' homes		-0.068	0.042**
		(0.043)	(0.021)
Shopping retention from the location		0.068***	0.053***
		(0.016)	(0.005)
Event effects	Yes	Yes	Yes
Within-event R^2 (%)	68.0	19.0	81.9
Locations	2,239	2,239	2,239
Events	213	213	213
CZ clusters	93	93	93

Notes: 213 complete county grids and 2,239 evaluated locations, including the observed locations. The multiplier is service jobs per 100 entrant jobs and equals the product of the four terms in Panel A. Payroll is the change in the tract–industry wage bill; resident income is CZ resident disposable income; service spending is revenue of CZ service firms. Each Panel A share is the covariance of a term's within-event log deviation with the log multiplier's within-event deviation, divided by the variance of the latter, so the shares sum to 100. Panel B regresses the log multiplier on the site characteristics with event effects; each characteristic is demeaned within event and divided by its standard deviation, so coefficients are log points per standard deviation. Parentheses give CZ-clustered standard errors.

Table 9: Relocating the same expansion at equal jobs or equal payroll

	Same jobs at every site	Same payroll at every site
Gap between the best and worst site		
Service jobs per 100 promised jobs	22.0	11.0
Net jobs per 100 promised jobs	15.2	22.0
Welfare per resident ($)	7.6	3.1
Best site compared with the observed site		
Welfare gain (%)	47.8	12.3

Notes: Each design moves the expansion across the observed site and up to ten alternative tracts in the host county. Same jobs holds the job increase at the target site at the promised jobs. Same payroll holds the incremental tract–industry payroll at its value at the observed site and lets jobs adjust, to between 0.44 and 2.16 times the promised jobs. Entries are medians across 213 complete county grids. All outcomes refer to the CZ. Payroll is the change in the target tract–industry wage bill, including wage changes on existing jobs.

the host county (Table 10). Much of the gain goes to residents of the neighboring counties where the entrant's workers live and shop (Section 6.2). Added property-tax receipts do not close the gap, because in the model they pay for the public services that new residents and new floor space require.

Table 10: Annual local-benefit costs and outcomes, conditional on entry

Measure	P10	Median	P90
Commuting zone			
Service jobs per 100 promised jobs	14.2	22.8	34.9
Annual tax expenditure per net job ($)	192.0	3,562.8	8,873.8
Annual welfare per baseline resident ($)	5.7	16.9	60.2
Resident welfare per public dollar	1.0	2.8	44.2
Host county			
Service jobs per 100 promised jobs	2.8	14.9	28.5
Annual tax expenditure per net job ($)	177.3	3,025.2	6,597.4
Annual welfare per baseline resident ($)	3.6	22.7	63.6
Resident welfare per public dollar	0.4	0.9	10.9

Notes: Equal-weight quantiles across 54 entry scenarios built from the 70 records with a verified local subsidy. The cost numerator is annual equilibrium tax expenditure on added commercial space. It covers the local property-tax benefit, not the full state package. The same cost is divided by CZ or host-county net job gains. Welfare is annual residential equivalent variation relative to no entry. Ratios are calculated within each scenario before taking quantiles. All dollars are in 2023 prices.

I represent each local subsidy as an exemption on the property tax owed on the entrant's new commercial space. Where a record gives an annual dollar amount, the exemption equals that amount, up to the full tax on the new space; otherwise, I apply the given percentage. Of the 70 records with a verified local subsidy, 54 can be priced this way, and I solve each one as an entry scenario (Appendix F.1). The cost of a job is the annual tax the exemption forgoes in equilibrium, divided by the net change in workplace employment relative to no entry. Per net CZ job, it ranges from $190 at the 10th percentile to $8,900 at the 90th.

However, measured against no entry, the exemption does not leave the county with less revenue, because the entry increases the number of individuals living in the county, which increases the residential property tax base. Through new residents and new built floor space, the entry with its exemption raises the host county's annual property-tax receipts by a median of about $1.9 million, against a median exemption of about $2.2 million. The county spends these receipts on the public services that the new residents and floor space require, and it still sets a lower tax rate on other property than before entry in 46 of the 54 scenarios, by a median of about 9 basis points. Because residents' welfare already counts the taxes they pay and the services they receive, the gains per public dollar in Table 10 need no further adjustment for these receipts.

The welfare-per-dollar ratios are fiscal accounting in the spirit of the marginal value of public funds (Hendren and Sprung-Keyser, 2020). Policymakers more often quote the cost per job. At the median, the exemptions forgo about $2,800 a year per promised job, and about $3,600 per net CZ job, because an entry adds only about three quarters of its promised jobs to the CZ on net. Over

a ten-year exemption, that comes to about $28,000 per promised job and $36,000 per net CZ job. The megadeals in Slattery and Zidar (2020) averaged $178 million for 1,500 promised jobs between 2002 and 2017, about $120,000 per promised job, so these local exemptions are modest by that standard, and they cover only the local property-tax benefit rather than the full state and local package. Program evaluations instead measure the jobs an incentive causes (Criscuolo et al., 2019; Siegloch et al., 2025; Freedman et al., 2023). For the California Competes Tax Credit, Freedman et al. (2023) estimate about $3,960 per job created in recipient tracts, against $9,900 per pledged job.[31]

An exemption brings the gains of entry only when the firm would not have entered the locality without it. When the firm would have come anyway, the exemption only gives up tax the county would otherwise have collected. Bartik (2018) places the share of incentivized entries that typical incentives actually cause between 2 and 25 percent.[32] At 25 percent, CZ residents can expect to gain about $0.40 per exemption dollar, and they gain in 40 of the 54 scenarios. At 2 percent, they can expect to lose about $0.34 per exemption dollar, and they gain in only 8 (Table 11). In the median scenario, the exemption benefits CZ residents if it causes at least 12 percent of the entries it subsidizes, and host-county residents if it causes at least 15 percent. Both thresholds fall inside Bartik's range, so whether these exemptions pay off for residents depends on how often they actually change where firms locate.

Table 11: Residents' expected gain from offering the exemption, by the share of entries it causes

Share of subsidized entries the exemption causes	100%	25%	2%
CZ residents' expected gain per exemption dollar	2.82	0.40	-0.34
Scenarios where CZ residents gain	54	40	8
Host-county residents' expected gain per exemption dollar	0.86	0.10	-0.18
Scenarios where host-county residents gain	52	33	7
Expected net CZ jobs per 100 promised jobs	74.5	16.7	-1.2

Notes: Medians and counts across the 54 recorded-subsidy scenarios in Table 10. Gains are per dollar of the exemption's annual tax expenditure, and Appendix F.6 gives the calculation. County budgets balance in every scenario, so welfare already includes the taxes residents pay and the services they receive. The 25 and 2 percent columns bracket the range Bartik (2018) considers plausible. Welfare is annual residential-opportunity equivalent variation. Annual 2023 dollars.

8 Conclusion

State and local governments spend about $20 billion a year on firm-specific subsidies, justified by the direct jobs these firms create and the indirect jobs their payroll supports. This paper asks, once a large tradable firm has entered, how many local service jobs its payroll supports and where they

[31] These figures keep each study's price basis and are not converted to 2023 dollars. The ten-year horizon is illustrative and undiscounted; the scenarios record annual amounts. Differences in geography, duration, the scope of the incentive, and additionality also limit direct comparison.

[32] The lower end reflects the median of the estimates without an obvious bias, 3.4 percent, and the upper end is close to the 23.5 percent median of estimates biased upward (Bartik, 2018, pp. 1–2).

appear. It follows the payroll from the firm to its workers, from the workers to the neighborhoods where they live and shop, and from those neighborhoods to local service firms. It uses linked worker histories to see who is hired and a tract-level model of an open commuting zone to add up the pieces in a local multiplier.

The data show why the indirect jobs are fewer than the payroll suggests. About half of an entrant's new employees were not previously attached to the local labor market, and many of the rest leave other local jobs, so only about $32 of every $100 of entrant payroll becomes new income for residents. Residents spend that income close to home, and service employment grows in the neighborhoods where the entrant's workers lived before entry.

The model turns these pieces into a net effect. An entry adds about three quarters of the jobs it promises on net, because other tradable producers lose workers and floor space to it, and 100 entrant jobs support about 22 service jobs at the median, or about 44 per 100 net tradable jobs, within the 30 to 80 range in Bartik and Sotherland (2019) and about a quarter of the 160 in Moretti (2010). Residents' welfare rises by about $7,600 a year per entrant job, small relative to the wage because each hire gains only the new wage net of their outside option. The gains spread across the market. The host county holds 28 percent of the commuting zone's residents but receives 64 percent of its service-job gain. Across markets, the multiplier is about 2.4 times as large at the 90th percentile event as at the 10th. Fixing every entry at 500 jobs removes only about a fifth of that variance, and the aggregate characteristics of the local economy explain only about 21 percent of the variation within industries. Where in a county the firm locates matters nearly as much: differences across locations within a county account for nearly half of the variance of the multiplier, mainly through the labor market around the site.

For the 54 entry with an observed local subsidy award, annual local tax expenditure is a median of $3,600 per CZ job. Median resident welfare per public dollar is $2.8 for the CZ and $0.86 for the host county. These figures distinguish regional benefits from benefits to the jurisdiction bearing the local cost.

The model takes the firm's entry as given, so it cannot say whether an award changed the location decision. If an award causes 25 percent of the entries it subsidizes, CZ residents can expect to gain about $0.40 per tax exempt dollar; at 2 percent, they can expect to lose about $0.34. The exemption pays off for CZ residents only if it causes at least 12 percent of the entries it subsidizes, and for host-county residents at least 15 percent, both inside the range in Bartik (2018). Within that limit, the lesson is that what an entry is worth to a locality depends less on how many jobs the firm promises than on who fills them, where they live, and where they shop.

References

John M. Abowd and Lars Vilhuber. National estimates of gross employment and job flows from the quarterly workforce indicators with demographic and industry detail. *Journal of Econometrics*, 161(1):82–99, 2011. ISSN 0304-4076. doi: 10.1016/j.jeconom.2010.09.008. URL https://www.sciencedirect.com/science/article/pii/S0304407610001880.

Daron Acemoglu and Pascual Restrepo. Robots and jobs: Evidence from us labor markets. *Journal of Political Economy*, 128(6):2188–2244, 2020. doi: 10.1086/705716. URL https://www.journals.uchicago.edu/doi/full/10.1086/705716.

Rodrigo Adão, Michal Kolesár, and Eduardo Morales. Shift-share designs: Theory and inference. *Quarterly Journal of Economics*, 134(4):1949–2010, 2019. doi: 10.1093/qje/qjz025.

Sumit Agarwal, J. Bradford Jensen, and Ferdinando Monte. Consumer mobility and the local structure of consumption industries. Working Paper 23616, National Bureau of Economic Research, 2020. URL https://www.nber.org/papers/w23616.

Gabriel M. Ahlfeldt, Stephen J. Redding, Daniel M. Sturm, and Nikolaus Wolf. The economics of density: Evidence from the berlin wall. *Econometrica*, 83(6):2127–2189, 2015. ISSN 1468-0262. doi: 10.3982/ECTA10876. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA10876.

Milena Almagro and Tomás Domínguez-Iino. Location sorting and endogenous amenities: Evidence from amsterdam. *Econometrica*, 93(3):1031–1071, 2025.

Erhan Artuç, Shubham Chaudhuri, and John McLaren. Trade shocks and labor adjustment: A structural empirical approach. *American Economic Review*, 100(3):1008–1045, 2010. ISSN 0002-8282. doi: 10.1257/aer.100.3.1008. URL https://www.aeaweb.org/articles?id=10.1257/aer.100.3.1008.

David Autor, David Dorn, and Gordon H. Hanson. On the persistence of the china shock. National Bureau of Economic Research, 2021. URL https://www.nber.org/papers/w29401.

David H. Autor, David Dorn, and Gordon H. Hanson. The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, 103(6):2121–2168, 2013. ISSN 0002-8282. doi: 10.1257/aer.103.6.2121. URL https://www.aeaweb.org/articles?id=10.1257%2Faer.103.6.2121.

Timothy J Bartik. *Who Benefits from State and Local Economic Development Policies?* WE Upjohn Institute for Employment Research, 1991. URL https://research.upjohn.org/up_press/77/?ref.

Timothy J. Bartik. "But For" percentages for economic development incentives: What percentage estimates are plausible based on the research literature? Upjohn Institute Working Paper 18-289,

W.E. Upjohn Institute for Employment Research, 2018. URL https://research.upjohn.org/up_workingpapers/289/.

Timothy J. Bartik and Nathan Sotherland. Local job multipliers in the united states: Variation with local characteristics and with high-tech shocks. *Working Paper*, 2019. doi: 10.17848/wp19-301. URL http://research.upjohn.org/up_workingpapers/301/.

Nathaniel Baum-Snow and Lu Han. The microgeography of housing supply. *Journal of Political Economy*, 132(6):1897–1946, 2024. ISSN 0022-3808, 1537-534X. doi: 10.1086/728110. URL https://www.journals.uchicago.edu/doi/10.1086/728110.

John D. Benjamin, G. Donald Jud, and Daniel T. Winkler. A simultaneous model and empirical test of the demand and supply of retail space. *Journal of Real Estate Research*, 16(1):1–14, 1998. doi: 10.1080/10835547.1998.12090938.

Kirill Borusyak, Peter Hull, and Xavier Jaravel. Quasi-experimental shift-share research designs. *The Review of Economic Studies*, 89(1):181–213, 2022. ISSN 0034-6527. doi: 10.1093/restud/rdab030. URL https://doi.org/10.1093/restud/rdab030.

Kirill Borusyak, Peter Hull, and Xavier Jaravel. A Practical Guide to Shift-Share Instruments. *Journal of Economic Perspectives*, 39(1):181–204, 2025. doi: 10.1257/jep.20231370. URL https://www.aeaweb.org/articles?id=10.1257/jep.20231370.

Matias Busso, Jesse Gregory, and Patrick Kline. Assessing the incidence and efficiency of a prominent place based policy. *American Economic Review*, 103(2):897–947, 2013. ISSN 0002-8282. doi: 10.1257/aer.103.2.897. URL https://www.aeaweb.org/articles?id=10.1257/aer.103.2.897.

Lorenzo Caliendo, Maximiliano Dvorkin, and Fernando Parro. Trade and labor market dynamics: General equilibrium analysis of the china trade shock. *Econometrica*, 87(3):741–835, 2019. ISSN 1468-0262. doi: 10.3982/ECTA13758. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA13758.

Brantly Callaway, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. Difference-in-differences with a continuous treatment. *American Economic Review*, 2026. doi: 10.1257/aer.20240137. URL https://www.aeaweb.org/articles?id=10.1257/aer.20240137. Forthcoming.

Doruk Cengiz, Arindrajit Dube, Attila Lindner, and Ben Zipperer. The effect of minimum wages on low-wage jobs. *The Quarterly Journal of Economics*, 134(3):1405–1454, 2019. ISSN 0033-5533. doi: 10.1093/qje/qjz014. URL https://doi.org/10.1093/qje/qjz014.

Victor Couture, Cecile Gaubert, Jessie Handbury, and Erik Hurst. Income growth and the distributional effects of urban spatial sorting. *Review of Economic Studies*, 91(2):858–898, 2024.

Chiara Criscuolo, Ralf Martin, Henry G. Overman, and John Van Reenen. Some causal effects of an industrial policy. *American Economic Review*, 109(1):48–85, 2019.

Donald R Davis, Jonathan I Dingel, Joan Monras, and Eduardo Morales. How segregated is urban consumption? *Journal of Political Economy*, 127(4):1684–1738, 2019.

Manasi Deshpande and Yue Li. Who is screened out? Application costs and the targeting of disability programs. *American Economic Journal: Economic Policy*, 11(4):213–248, 2019. doi: 10.1257/pol.20180076. URL https://www.aeaweb.org/articles?id=10.1257/pol.20180076.

Jonathan I. Dingel and Felix Tintelnot. Spatial economics for granular settings. *Econometrica*, 94 (2):407–464, 2026. ISSN 1468-0262. doi: 10.3982/ECTA19350. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA19350.

Pablo D Fajgelbaum and Cecile Gaubert. Optimal spatial policies, geography, and sorting. *The Quarterly Journal of Economics*, 135(2):959–1036, 2020. ISSN 0033-5533. doi: 10.1093/qje/qjaa001. URL https://doi.org/10.1093/qje/qjaa001.

Matthew Freedman. Targeted business incentives and local labor markets. *Journal of Human Resources*, 48(2):311–344, 2013. ISSN 0022-166X, 1548-8004. doi: 10.3368/jhr.48.2.311. URL https://jhr.uwpress.org/content/48/2/311.

Matthew Freedman, Shantanu Khanna, and David Neumark. Combining rules and discretion in economic development policy: Evidence on the impacts of the California Competes Tax Credit. *Journal of Public Economics*, 217, 2023.

Clemens Fuest, Andreas Peichl, and Sebastian Siegloch. Do higher corporate taxes reduce wages? micro evidence from germany. *American Economic Review*, 108(2):393–418, 2018. ISSN 0002-8282. doi: 10.1257/aer.20130570. URL https://www.aeaweb.org/articles?id=10.1257/aer.20130570.

Simon Galle, Andrés Rodríguez-Clare, and Moises Yi. Slicing the pie: Quantifying the aggregate and distributional effects of trade. *The Review of Economic Studies*, 90(1):331–375, 2023. ISSN 0034-6527, 1467-937X. doi: 10.1093/restud/rdac020. URL https://academic.oup.com/restud/article/90/1/331/6569095.

John Gardner. Two-stage differences in differences. Working paper, arXiv:2207.05943, 2021. URL https://arxiv.org/abs/2207.05943.

Andrew Garin. Do place-based industrial interventions help "left-behind" workers? lessons from wwii and beyond. National Bureau of Economic Research, 2025. URL https://www.nber.org/papers/w33418.

Genevieve Giuliano and Kenneth A. Small. Subcenters in the Los Angeles region. *Regional Science and Urban Economics*, 21(2):163–182, 1991. doi: 10.1016/0166-0462(91)90032-I.

Paul Goldsmith-Pinkham, Isaac Sorkin, and Henry Swift. Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624, 2020. ISSN 0002-8282. doi: 10.1257/aer.20181047. URL https://pubs.aeaweb.org/doi/10.1257/aer.20181047.

Brian Greaney, Andrii Parkhomenko, and Stijn Van Nieuwerburgh. Dynamic urban economics. Social Science Research Network, 2024. URL https://papers.ssrn.com/abstract=4897077.

Michael Greenstone, Richard Hornbeck, and Enrico Moretti. Identifying agglomeration spillovers: Evidence from winners and losers of large plant openings. *Journal of Political Economy*, 118(3): 536–598, 2010. ISSN 0022-3808, 1537-534X. doi: 10.1086/653714. URL https://www.journals.uchicago.edu/doi/10.1086/653714.

Andrew Hanson and Shawn Rohlin. Local employment multipliers for large publicly subsidized firms: Evidence from a synthetic control approach. *Journal of Regional Science*, 64(2):491–526, 2024. doi: 10.1111/jors.12680.

Gordon H. Hanson, Dani Rodrik, and Rohan Sandhu. The u.s. place-based policy supply chain. *National Bureau of Economic Research*, 2025. doi: 10.3386/w33511. URL https://www.nber.org/papers/w33511.

Nathaniel Hendren and Ben Sprung-Keyser. A unified welfare analysis of government policies. *Quarterly Journal of Economics*, 135(3):1209–1318, 2020.

Brad Hershbein and Bryan A. Stuart. The evolution of local labor markets after recessions. *American Economic Journal: Applied Economics*, 16(3):399–435, 2024. ISSN 1945-7782. doi: 10.1257/app.20220132. URL https://www.aeaweb.org/articles?id=10.1257/app.20220132.

Matthias Hoelzlein. Two-sided sorting and spatial inequality in cities. Dropbox, 2023. URL https://www.dropbox.com/scl/fi/c4mmkriwszcq4yw9sdm63/Two_Sided_Sorting_and_Spatial_Inequality_in_Cities_Jun23.pdf?dl=0&rlkey=3kc7eo0qll2m34hhnjg49w6t4.

Richard Hornbeck and Enrico Moretti. Estimating who benefits from productivity growth: Local and distant effects of city productivity growth on wages, rents, and inequality. *The Review of Economics and Statistics*, 106(3):587–607, 2024. ISSN 0034-6535. doi: 10.1162/rest_a_01208. URL https://doi.org/10.1162/rest_a_01208.

Patrick Kline and Enrico Moretti. People, places, and public policy: Some simple welfare economics of local economic development programs. *Annual Review of Economics*, 6(1):629–662, 2014a. ISSN 1941-1383, 1941-1391. doi: 10.1146/annurev-economics-080213-041024. URL https://www.annualreviews.org/doi/10.1146/annurev-economics-080213-041024.

Patrick Kline and Enrico Moretti. Local economic development, agglomeration economies, and the big push: 100 years of evidence from the tennessee valley authority. *Quarterly Journal of Economics*, 129(1):275–331, 2014b. doi: 10.1093/qje/qjt034.

Gianluca Marcato and Lok Man Michelle Tong. Supply constraints and search equilibrium in office markets. *Journal of Real Estate Finance and Economics*, 70(2):378–428, 2025. doi: 10.1007/s11146-023-09955-y.

Daniel P. McMillen. Nonparametric employment subcenter identification. *Journal of Urban Economics*, 50(3):448–473, 2001. doi: 10.1006/juec.2001.2228.

Atif Mian and Amir Sufi. What explains the 2007–2009 drop in employment? *Econometrica*, 82 (6):2197–2223, 2014. ISSN 1468-0262. doi: 10.3982/ECTA10451. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA10451.

Yuhei Miyauchi, Kentaro Nakajima, and Stephen Redding. The economics of spatial mobility: Theory and evidence using smartphone data. *Working Paper*, page w28497, 2021. doi: 10.3386/w28497. URL http://www.nber.org/papers/w28497.pdf.

Yuhei Miyauchi, Kentaro Nakajima, and Stephen J. Redding. The economics of spatial mobility: Theory and evidence using smartphone data. *Quarterly Journal of Economics*, 140(4):2507–2570, 2025. doi: 10.1093/qje/qjaf038.

Ferdinando Monte, Stephen J. Redding, and Esteban Rossi-Hansberg. Commuting, migration, and local employment elasticities. *American Economic Review*, 108(12):3855–3890, 2018. ISSN 0002-8282. doi: 10.1257/aer.20151507. URL https://pubs.aeaweb.org/doi/10.1257/aer.20151507.

Enrico Moretti. Local multipliers. *American Economic Review*, 100(2):373–377, 2010. ISSN 0002-8282. doi: 10.1257/aer.100.2.373. URL https://pubs.aeaweb.org/doi/10.1257/aer.100.2.373.

Emi Nakamura and Jón Steinsson. Fiscal stimulus in a monetary union: Evidence from u.s. regions. *American Economic Review*, 104(3):753–792, 2014. doi: 10.1257/aer.104.3.753.

Derek Neal. Industry-specific human capital: Evidence from displaced workers. *Journal of Labor Economics*, 13(4):653–677, 1995. doi: 10.1086/298388.

David Neumark and Jed Kolko. Do enterprise zones create jobs? evidence from california's enterprise zone program. *Journal of Urban Economics*, 68(1):1–19, 2010. ISSN 0094-1190. doi: 10.1016/j.jue.2010.01.002. URL https://www.sciencedirect.com/science/article/pii/S0094119010000033.

Ryungha Oh and Jaeeun Seo. What causes agglomeration of services? theory and evidence from seoul. *Working Paper*, 2023.

Carlianne Patrick. Identifying the local economic development effects of million dollar facilities. *Economic Inquiry*, 54(4):1737–1762, 2016. ISSN 0095-2583, 1465-7295. doi: 10.1111/ecin.12339. URL https://onlinelibrary.wiley.com/doi/10.1111/ecin.12339.

Carlianne Patrick and Mark Partridge. Identifying agglomeration spillovers: New evidence from large plant openings. 2022. URL https://www.dropbox.com/scl/fo/xyy27tvgi16glt4v3td2t/ABKKfGkFh9-Za3sd1voCot8?dl=0&rlkey=du2xk5sewqzzyxh5ek6bc8fi5.

Albert Saiz. The geographic determinants of housing supply*. *The Quarterly Journal of Economics*, 125(3):1253–1296, 2010. ISSN 0033-5533. doi: 10.1162/qjec.2010.125.3.1253. URL https://doi.org/10.1162/qjec.2010.125.3.1253.

Kurt Schmidheiny and Sebastian Siegloch. On event studies and distributed-lags in two-way fixed effects models: Identification, equivalence, and generalization. *Journal of Applied Econometrics*, 38(5):695–713, 2023. doi: 10.1002/jae.2971. URL https://onlinelibrary.wiley.com/doi/10.1002/jae.2971.

Sebastian Siegloch, Nils Wehrhöfer, and Tobias Etzel. Spillover, efficiency, and equity effects of regional firm subsidies. *American Economic Journal: Economic Policy*, 17(1):144–180, 2025. doi: 10.1257/pol.20220667.

J. M. C. Santos Silva and Silvana Tenreyro. The log of gravity. *The Review of Economics and Statistics*, 88(4):641–658, 2006. ISSN 0034-6535. URL https://www.jstor.org/stable/40043025.

Cailin Slattery and Owen Zidar. Evaluating state and local business incentives. *Journal of Economic Perspectives*, 34(2):90–118, 2020. ISSN 0895-3309. doi: 10.1257/jep.34.2.90. URL https://pubs.aeaweb.org/doi/10.1257/jep.34.2.90.

Caitlin Slattery. Bidding for firms: Subsidy competition in the us. *Working Paper*, 2024. URL https://static1.squarespace.com/static/5acd55db36099bfff90bf5d6/t/67055a7061d6386f1971959b/1728404085624/Slattery+%282024%29+Bidding+for+Firms.pdf.

Juan Carlos Suárez Serrato and Owen Zidar. Who benefits from state corporate tax cuts? a local labor markets approach with heterogeneous firms. *American Economic Review*, 106(9):2582–2624, 2016. ISSN 0002-8282. doi: 10.1257/aer.20141702. URL https://www.aeaweb.org/articles?id=10.1257/aer.20141702.

Juan Carlos Suárez Serrato and Owen Zidar. Who benefits from state corporate tax cuts? a local labor market approach with heterogeneous firms: Reply. *American Economic Review*, 113(12):3401–3410, 2023. ISSN 0002-8282. doi: 10.1257/aer.20230208. URL https://www.aeaweb.org/articles?id=10.1257/aer.20230208.

Kasia Tarczynska. 2022: A mega-year for megadeals! - good jobs first. Good Jobs First, 2022. URL https://goodjobsfirst.org/2022-a-mega-year-for-megadeals/.

Charles M. Tolbert and Molly Sizer. U.s. commuting zones and labor market areas: A 1990 update. Staff Report AGES-9614, U.S. Department of Agriculture, Economic Research Service, 1996.

Nick Tsivanidis. Evaluating the Impact of Urban Transit Infrastructure: Evidence from Bogotá's TransMilenio. *American Economic Review*, 116(2):418–463, February 2026. ISSN 0002-8282. doi: 10.1257/aer.20190874. URL https://www.aeaweb.org/articles?id=10.1257/aer.20190874.

U.S. Census Bureau. Introduction. *LEHD Snapshot Documentation*, 2025. URL https://lehd.ces.census.gov/data/lehd-snapshot-doc/latest/sections/introduction.html. Release S2024_R2025Q4. Accessed 2026-05-21.

Jasper Jacob Van Dijk. Local employment multipliers in u.s. cities. *Journal of Economic Geography*, page lbw010, 2016. ISSN 1468-2702, 1468-2710. doi: 10.1093/jeg/lbw010. URL https://academic.oup.com/joeg/article-lookup/doi/10.1093/jeg/lbw010.

Veraset. Visits. Dataset, 2022. URL https://doi.org/10.82551/TP7E-VA02.

Coady Wing, Seth M. Freedman, and Alex Hollingsworth. Stacked difference-in-differences. Working Paper 32054, National Bureau of Economic Research, 2024. URL https://www.nber.org/papers/w32054.

Danny Yagan. Employment hysteresis from the great recession. *Journal of Political Economy*, 127 (5):2505–2558, 2019. doi: 10.1086/701809. URL https://www.journals.uchicago.edu/doi/full/10.1086/701809.

Appendix

Appendix roadmap. Appendix A presents the data construction and additional empirical results. Appendix B derives the model from individual preferences, including employment and location choices, consumption, production, and equilibrium accounts. Appendix C describes quantification and equilibrium computation. Appendix E supports the results in Section 6, and Appendix F supports the counterfactuals in Section 7.

A Data and Additional Empirics

A.1 Award data and event samples

Award records and site verification. The award data begin with three public sources. State economic development agencies publish registers of the awards they approve, Good Jobs First's Subsidy Tracker compiles company-specific awards from state and local disclosures (Tarczynska, 2022), and the Slattery series records the announced packages of large incentive competitions, which often bundle state and local incentives (Slattery, 2024). Where supporting documents exist, I reconcile each recorded amount with agency reports, executed agreements, legislative materials, and local authorizing documents, matching on the recipient, the program, the project location, the award or agreement date, and the contract identifier when one is reported. Because an award names a firm rather than an establishment, I also match each award to its establishment in Data Axle and check the site against a document, preferring primary incentive documents to agency announcements and news reports, and against 2020 tract boundaries and available parcel records. I drop 36 awards that do not attach to a single employer's site, such as grants to prepare industrial land, film credits, and tax increment financing for retail developers, and 11 more whose address cannot be placed in a 2020 tract (Table A.1).

From award records to events. One project often appears in several award records. A state credit, a training grant, and a local abatement may each finance it, and Good Jobs First and a state register may list the same award. I therefore consolidate records into firm-site events. After normalizing firm names, which merges four sites where one firm appears under two spellings, I group the awards to the same firm in the same tract and start a new episode when at least five years separate two awards. Only one site has awards that far apart. When every award at a site promises the same number of jobs, the awards form one event, the jobs count once, and the subsidy is the sum across instruments. When the promises differ, I split the awards by award year and combine the instruments within each year, which turns six sites into nine events. The one exception is a project whose three annual awards report slightly different totals for the same plant, which I keep as a single event. Each event is dated to its first award year, and its promised jobs are the largest promise among its awards. Overlapping valuations of the same award count once, so when a package from the Slattery series already contains a state program's award, I keep the package

figure and do not add the program. These steps turn 299 award records into 226 events involving 212 firms at 201 tracts in 32 states, and 179 of the events rest on a single award.

Realized jobs and payments. Promised jobs come from the award itself, but realized jobs and payments are observed only where a program requires reports and the agency publishes them. Realized employment is observed for 95 events, most of them in Texas, Virginia, North Carolina, and New Jersey, which publish performance reports for their largest programs. The administering agency verified or certified 53 of these counts. The rest are recipients' self-reports republished by the agency, figures carried in the award record, or press reports. When an event has more than one count, I prefer verified or certified counts, then a count attached to a documented post-performance review, then the most recent count, and finally the larger one. The 64 events with a count at least four years after the award, when most of the promised hiring should have occurred, form the descriptive attainment sample in Figure A.1, which is not used in the regressions or the model.

An approved amount is not the amount paid, so I collect payment and clawback records separately where they are public. In Indiana, I match awards to the 2017 Economic Incentives and Compliance Report by recipient, funding program, and exact contract amount, and record the paid or certified amount with its reporting date. In Virginia, I match the Commonwealth's Development Opportunity Fund awards to the state's post-performance reports by grant amount, locality, and performance-agreement date, keep the latest project status among the editions examined, and separate clawback requests from repayments received where the report allows. This yields paid or certified amounts for 22 events and clawback records for five. These records give cumulative payments or recoveries as of a reporting date rather than spending over the completed life of an award.

State and local support. Separating the local share starts with whether a site received any local property-tax relief. Among the 226 events, I establish the local relief for 86, either as a dollar amount (43) or as a documented absence of any abatement (43). For another 13, an abatement is documented but its amount is not published, and the remaining 127 are unresolved. For 33 events, I can document both the state and the local share of the award. Fifteen receive state support only through state programs, fourteen are large packages whose state side I rebuild from the individual programs named in the granting documents, and four have a local deduction whose dollar value I compute from the exempted assessed value. I build the state amount from documented state instruments, rather than as the residual left after subtracting an announced local contribution from the package total. Each component is classified by the revenue source it reduces, so a local property-tax exemption that the state administers remains a local cost. I also set apart spending that does not accrue to the firm, such as public infrastructure, transfers to third parties, and bond face values, and I separate generally available tax preferences from negotiated, firm-specific incentives.

Property-tax relief is researched facility by facility. I first identify the parcel, its owner of record, and the taxing jurisdictions, including the municipality and school district, allowing for

subsidiaries, successor firms, landlords, and industrial development authorities that hold title to the property the recipient occupies. The sources are assessor and auditor records, state abatement registers, industrial development board filings, executed agreements, ordinances, council materials, and project disclosures in Annual Comprehensive Financial Reports. Government-wide abatement totals are not assigned to a firm without project-level evidence. When a source reports an abated assessed value rather than a dollar saving, annual relief equals the jurisdiction's tax rate per dollar of assessed value times the exempted value, and market values are converted only when the assessment basis is known. Under a payment in lieu of taxes, relief is the tax that would have been owed less the required payment. Relief repeated across several state award records counts once, while concessions from separate taxing authorities are added.

Observed annual savings, approval-stage estimates, and projected full-term values are kept separately. When the observation window is incomplete, the full-term value follows the contract's term and schedule, including any recorded cancellation, and otherwise extends the most recent observed annual saving. Cases that require an assumed horizon or a declining deduction schedule are flagged, and all amounts are in 2023 dollars. A search that finds nothing is recorded as missing rather than as zero support, and a documented absence of support is distinguished from an instrument whose amount is unknown. Public-ownership exemptions and substitute cash grants are recorded separately from conventional abatements. Because disclosure and historical coverage vary across jurisdictions, and realized abatements depend on the investment the assessor eventually values, these comparisons describe only the projects whose components can be documented, and their full-term values are estimates rather than verified spending.

Table A.1: From all-state award records to firm-site events

Stage	Change	Count
Raw all-state award records	—	202,070
Package > $1 million or promised jobs > 100	−184,903	17,167
Exactly one usable announced location	−2,587	14,580
Exclude loans	−974	13,606
Append Slattery awards	+385	13,991
Geographic and establishment matching	−5,857	8,134
Award years 2004–2017	−3,503	4,631
Tradable industry	−2,687	1,944
Promised jobs > 500 or > 0.50% of baseline CZ employment	−1,598	346
Restrict to single-employer, site-specific awards	−36	310
Require substantiated address and 2020 tract	−11	299
Consolidate award records into firm-site events	—	**226**

Notes: Counts refer to award records until the final row. CZ employment is measured in the year before the award. Multiple awards for the same firm-site project are combined into one event, dated to the first award year; distinct project commitments and episodes separated by at least five years are kept separate.

The award universe. Table A.2 describes the awards of at least 100 promised jobs in the 29 states with usable records, 2002–2024. Tax credits and rebates are half of the awards but a fifth of the value. Property tax abatements are 7 percent of the awards but a quarter of the value, and they

are local rather than state revenue. The 2 percent of awards that bundle several instruments carry 46 percent of all value, so mean-based statistics are driven by the tail. Most programs require an application before the siting decision, the largest awards are discretionary, and local property-tax deals need a second approval from the county, school board, or industrial development board.

Table A.2: Instrument mix of awards with at least 100 promised jobs, 2002–2024

Instrument	Share of awards	Value ($bn)	Share of value
Tax credit or rebate	52.40%	15.50	21.40%
Grant (incl. loan hybrids)	37.90%	5.70	7.80%
Property tax abatement	6.70%	18.00	24.80%
Megadeal (mixed package)	2.30%	33.20	45.80%
Other (infrastructure, reimbursement, assistance)	0.80%	0.10	0.20%
Total	100%	72.50	100%

Notes: Author's award file, 29 states, 2002–2024, all awards of at least 100 promised jobs. Values are reported package values.

Table A.3 codes the tax base of the 17 programs that carry 99 percent of the credit and rebate dollars. Three fifths are payroll or withholding based, so the firm keeps part of the income tax withheld from its own employees' wages. Only 6 percent are conventional corporate income tax credits. Most "corporate tax credits" in this universe are therefore wage subsidies financed by employees' own withholding. Awards are nominally conditional on job counts, wage floors, and investment minimums, but enforcement ranges from post-performance payment on verified jobs to self-reported counts with weak audit, and reported jobs are usually projections.

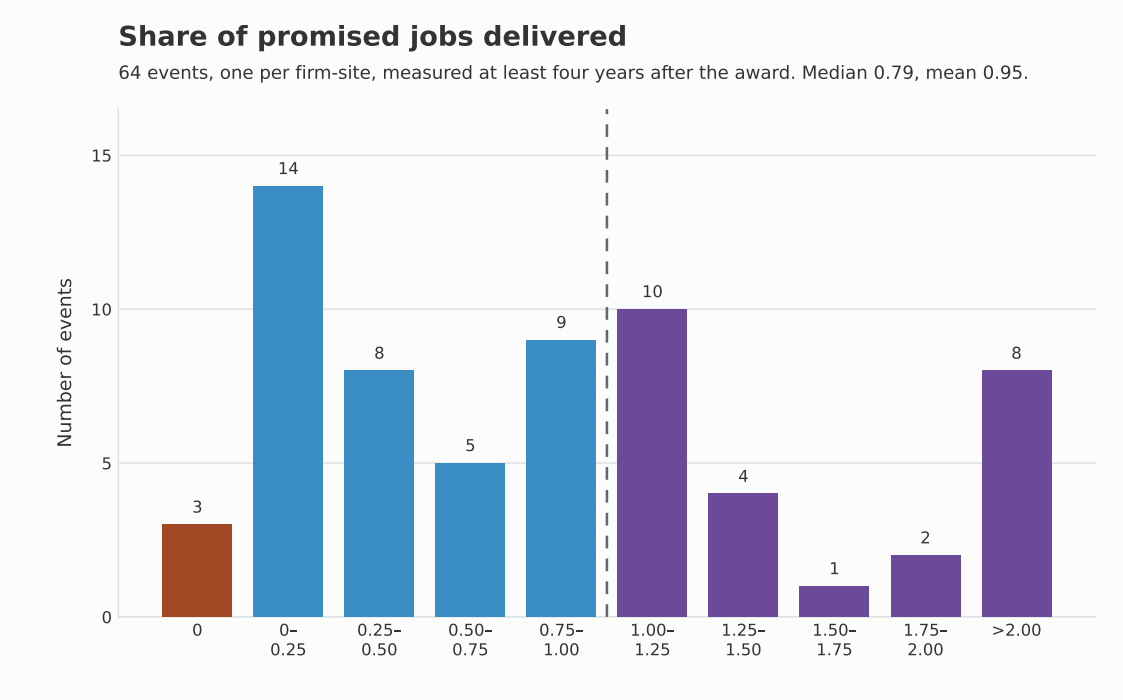
Table A.3: Tax base of credit and rebate awards

Base	Share of credit and rebate value
Payroll or withholding	61.30%
Sales or property, in a state without an income tax	8.10%
Corporate income, conventional	5.90%
Mixed income, sales, and withholding	3.00%
Local or unclassified	3.20%

Notes: Shares of the $15.50 billion coded as tax credit or rebate in Table A.2, after hand-coding the 17 programs that carry 99 percent of it. Two New Jersey programs (25.20 percent of the total) are split between the first and third rows.

Promised and delivered jobs. Figure A.1 uses the 64 firm-site events with reported employment at least four years after award. The median delivered 0.79 of the jobs promised and the mean 0.95. 39 fell short, 17 delivered under a quarter of the promise, 3 recorded zero or negative job gains, and 8 more than doubled the promise. Total realized jobs divided by total promised jobs is 1.06. Because disclosure is selective, these are descriptive statistics rather than estimates of the effect of subsidies.

Figure A.1: Share of promised jobs delivered, firm-site events observed at least four years after award



Notes: 64 events from the consolidated award registry, with employment observed at least four years after award. Each bar counts firm-site events by realized-to-promised employment; the first includes zero and negative gains. The dashed line marks delivery of exactly the promised count. Median 0.79, mean 0.95, pooled ratio 1.06.

The samples used in the paper. The paper draws on several samples of entries, and Table A.4 relates them. The award registry provides the subsidy events. The LEHD entrant sample contains all large tradable openings and expansions in the 29 states with restricted data, whether or not they were subsidized, and supports Facts 1 and 3. The public registry links awards to an entry year, a 2020 tract, and a commuting zone; a candidate enters it only if LODES employment in the entry tract and industry rises by more than 500 jobs relative to baseline in at least one of event years 1 through 5, and when two events occupy the same tract within two years the earlier is kept. The model is quantified for the 55 registry events with complete model inputs. Because the public registry conditions on a realized employment increase, it describes what follows a large entry that occurred and does not identify whether the award caused the firm to locate where it did.

A.2 Public data sources

Employment and commuting. Workplace employment by tract comes from LODES Workplace Area Characteristics, resident workers from Residence Area Characteristics, and tract-to-tract commuting flows from the Origin–Destination files. All tract identifiers are harmonized to 2020 geography. The public service-employment outcome is raw annual workplace employment in NAICS 44–45, 53, 61, 62, 72, and 81, available for 97.48% of tract-years. In the public exposure designs,

Table A.4: Event samples

Sample	Screen	Events	CZs	Used in
Award registry (Table A.1)	Tradable; $>$ 500 jobs or $>$ 0.5% of CZ employment; 2004–2017	226	—	Section 2
with realized jobs $\geq$ 4 years after award	Reported employment	64	—	Figure A.1
LEHD entrant sample	Same size and industry screen; 29 states	400	—	Table 1, Fact 1
pre-entry-residence stack	CZs with one observed event	—	40	Fact 3, Figure 4
new-income stack	CZs with one observed event	—	30	Figure A.8
continuous exposure	All events and tracts	—	100	Figure 5
incumbent-industry event study	Matched control CZs	—	80	Appendix A.4
Public registry	LODES tract-industry employment +500 in years 1–5	124	69	Appendix A.6
boundary comparison	All verified award sites, +5 years	180	86	Section 2, Table A.11
Model sample	Registry events with complete inputs	55	38	Sections 5–7

Notes: Entrant and event counts from restricted data are disclosure-rounded. A dash means the count is not released or not applicable. The LEHD entrant sample includes the subsidized events that meet its screen. The award registry's CZ count is not separately tabulated.

entry tracts and their adjacent tracts receive no exposure, so that exposure is never assigned to employment at the plant itself.

Shopping visits. Shopping flows come from 2019 Veraset smartphone visit records (Veraset, 2022). Explicit home and work records are not counted as visits. Remaining visits to service establishments are aggregated from 2010 block groups to 2020 tracts with the official crosswalk. Each device receives total weight one within a month and industry group, so the annual measure is a device-weighted distribution of visits from each home tract to each destination tract. Visits are counts of trips, not dollars. The visit data cover retail, selected real estate, education excluding colleges, food services, and other services; they contain no health care (NAICS 62) bucket, so that sector enters the employment outcome but not the shopping allocations. Because the visits are observed in 2019, after the events, they describe where service demand is located rather than a predetermined treatment map.

Travel times. Driving times between tract centroids use the 2020 OpenStreetMap road network (OSRM). Same-tract travel is set to five minutes. Gravity comparisons use tract pairs with finite driving time of at most 180 minutes.

Wages. Public workplace wages combine QWI county earnings by education with LODES tract employment counts by education. Because LODES publishes education and industry as separate margins, education categories are allocated to the tradable sector using the tract-year tradable

employment share. This is a proxy for workplace wages by tract, not a joint tabulation.

Property. CoreLogic deed and assessor records over 2004–2020 provide residential and commercial repeat-sales and hedonic price indexes, building square footage, land area, and property counts by 2020 tract. The repeat-sales estimator follows a three-stage Case–Shiller design, and the hedonic regressions include physical characteristics, property type, quality, condition, and missingness indicators. Commercial transactions are sparse in some tracts, and tract-years without support are left missing. Property values are converted to annual rent equivalents in the quantification through the price-to-rent assumptions of Appendix C.

Restricted records. Worker histories and establishment records come from the restricted LEHD and business register (Abowd and Vilhuber, 2011; U.S. Census Bureau, 2025). Entrant hires are observed from 12 quarters before to 12 quarters after hire. A worker's primary employer in a quarter is the employer paying the highest earnings. All restricted statistics in the paper have passed Census Bureau disclosure review.

A.3 Fact 1: Hires, residence, and earnings

Residence and prior workplace. Table A.5 expands Table 2 into ten joint residence–workplace categories for the 551,850 hires in the released residence tabulation, pooling entrant types and CZ sizes. Residence is measured one year before and one year after hire. Let C_e be the event CZ, r_i^- and r_i^{+1} residence before and after hire, and w_i^- the prior workplace. Continuing residents satisfy $r_i^- = r_i^{+1} = C_e$; movers satisfy $r_i^- \neq C_e$ and $r_i^{+1} = C_e$, including those who commuted into the CZ before moving; outside residents satisfy $r_i^{+1} \neq C_e$. Residence and prior workplace carry different information. Continuing residents who previously worked outside the CZ are 9.31 percent of hires, and movers who previously commuted to a job in the CZ are 2.17 percent. Among continuing residents, 9.56 percent have no observed prior job. For these workers I treat no observed job as nonemployment, which includes nonparticipation. For workers who lived outside the CZ before hire, a missing prior job may instead be a job in a state outside the data, so their prior earnings are treated as missing rather than zero.

Market size. Table A.6 splits hires by the released CZ-size class. Continuing residents are 62.27 percent of hires in small CZs, 60.03 percent in medium CZs, and 54.64 percent in large CZs, while the share living outside the CZ after hire rises to 31.26 percent in large markets from about 25 percent in small and medium ones. Movers are 13.09, 15.42, and 14.10 percent of hires, so migration does not rise with market size. These comparisons pool entrant types and do not separate market size from entrant composition.

Prior industry. Table A.7 reports hires' prior industries. Prior jobs in tradables account for 35.41 percent of hires and prior jobs in nontradables for 17.92 percent, of which 10.60 percent of

Table A.5: Residence transitions and prior workplace of entrant hires

Residence before hire	Residence after hire	Prior workplace	Hires	Share of all hires (%)
Outside CZ	Outside CZ	Outside CZ	83,400	15.11
Outside CZ	Outside CZ	Inside CZ	27,700	5.02
Outside CZ	Outside CZ	No observed job	35,600	6.45
Outside CZ	Inside CZ	Outside CZ	37,700	6.83
Outside CZ	Inside CZ	Inside CZ	12,000	2.17
Outside CZ	Inside CZ	No observed job	29,300	5.31
Inside CZ	Outside CZ	Unresolved	7,250	1.31
Inside CZ	Inside CZ	Outside CZ	51,400	9.31
Inside CZ	Inside CZ	Inside CZ	237,000	42.95
Inside CZ	Inside CZ	No observed job	30,500	5.53
Total			551,850	100.00
Among continuing CZ residents (318,900 hires)				
Share with no observed prior job (%)				9.56

Notes: Low- and high-skill entrants and all CZ-size classes are pooled. Residence is measured one year before and one year after hire; workplace refers to the prior job. The ten joint categories are mutually exclusive and exhaustive. Counts are disclosure-rounded, and shares are calculated from those counts, so they may not sum to 100. The final line divides the 30,500 hires with no observed prior job by the 318,900 continuing CZ residents. No observed prior job does not distinguish unemployment from nonparticipation.

Table A.6: Recruitment and residence by CZ size

CZ size	Hires	Continuing CZ residents (%)	Moved into CZ (%)	Lives outside CZ after hire (%)
Small	106,150	62.27	13.09	24.63
Medium	171,900	60.03	15.42	24.55
Large	273,800	54.64	14.10	31.26

Notes: Entrant types are pooled within each CZ-size class. Each percentage uses all hires in its row as the denominator. Continuing residents live in the event CZ one year before and one year after hire; movers live outside it before and inside it after; the final category lives outside it after hire. Counts are disclosure-rounded. Small, medium, and large are the released size labels; population cutoffs are not released.

all hires came from nontradable employers inside the event CZ. A further 17.45 percent have no observed prior job.

Earnings. For released group g with n_g hires, let q_g^+ and q_g^- be the after-hire and prior annual earnings quasi-medians, and let $\mathcal{O}(\mathcal{G})$ be the groups in a set $\mathcal{G}$ with observed prior earnings. The

Table A.7: Prior industry and workplace of entrant hires

Prior industry / job status	All hires (%)	Prior job inside CZ (%)	Prior job outside CZ (%)
Low-skill nontradables	13.06	7.32	5.74
High-skill nontradables	4.87	3.28	1.59
Low-skill tradables	15.42	10.40	5.01
High-skill tradables	19.99	12.78	7.21
Other observed industries	29.22	17.31	11.91
No observed prior job	17.45	—	—

Notes: Entrant types and worker skills are pooled. Each percentage uses all 550,700 hires in the prior-industry tabulation as its denominator; this sample is separate from the 551,850 hires in the residence tabulation. Low and high skill refer to the prior employer's industry. The inside- and outside-CZ columns split each observed-industry row geographically. Counts are disclosure-rounded.

reported after-hire earnings and earnings increase are

$$Q_{\mathcal{G}}^{+} = \frac{\sum_{g\in\mathcal{G}} n_g q_g^{+}}{\sum_{g\in\mathcal{G}} n_g}, \qquad D_{\mathcal{G}} = \frac{\sum_{g\in\mathcal{O}(\mathcal{G})} n_g (q_g^{+} - q_g^{-})}{\sum_{g\in\mathcal{O}(\mathcal{G})} n_g}. \tag{A.1}$$

These are count-weighted averages of group quasi-medians, not medians of individual earnings changes. Among the 454,600 hires with observed prior earnings, after-hire earnings average $85,500 and the increase $22,500. The increase is $18,400 for hires whose prior job was in the CZ and $29,300 for hires whose prior job was outside it (Table A.8). The 96,100 hires with no observed prior job earn $72,200 after hire and have no reported increase. The increases can reflect hours as well as pay rates.

Table A.8: Earnings after hire and observed earnings increases

Hire group	Hires	2023 $ thousands After hire	Increase
All hires with observed prior earnings	454,600	85.5	22.5
Prior workplace inside CZ	281,350	84.9	18.4
Prior workplace outside CZ	173,250	86.4	29.3
Low-skill workers, observed prior earnings	299,600	61.7	18.3
High-skill workers, observed prior earnings	155,000	131.4	30.7
No observed prior job	96,100	72.2	—

Notes: Entrant types are pooled. Earnings are annual, in thousands of 2023 dollars, and are count-weighted averages of released group quasi-medians; increases are the corresponding averages of after-minus-before quasi-medians. The prior-workplace rows and the worker-skill rows are alternative splits of the same 454,600 hires. Missing prior earnings are not treated as zero. Counts are disclosure-rounded; the earnings sample contains 550,700 hires.

Payroll by residence. Table A.9 splits the payroll shares in Table 2 by worker skill. Of the $69.035 billion in entrant payroll, 71.69 percent is paid to workers living inside the event CZ one year after hire. The inside-CZ total combines continuing residents and movers.

Table A.9: Entrant payroll by post-hire residence and worker skill

	Annual payroll (2023 $ billions)			
Residence one year after hire	Low-skill workers	High-skill workers	Total payroll	Share of total payroll (%)
Inside the CZ	23.852	25.639	49.491	71.69
Outside the CZ	9.284	10.260	19.544	28.31
Total	**33.136**	**35.899**	**69.035**	**100.00**

Notes: Low and high skill refer to individual workers; entrant types are pooled. Residence is measured one year after hire. Payroll is annual, in billions of 2023 dollars, from released earnings sums. These are gross payments by post-hire residence, not additional resident income.

The resident-income accounting. Fact 1 allocates each $100 of entrant payroll among workers living outside the CZ, the prior earnings of continuing residents, and new resident income. This paragraph states the object being approximated and the assumptions the approximation uses. For entrant e, let H be its hire cohort and $P^H_{e,0}$ the entrant's payroll to that cohort in a reference year. For horizon h, let $\mathcal{W}^-_h$ and $\mathcal{W}^+_h$ be annual earnings windows before and after hire, r^-_h and r^+_h the residence reference quarters, $Y^{\text{cov}}_{i,\mathcal{W}}$ worker i's covered earnings from all employers in window $\mathcal{W}$, and $Y^e_{i,\mathcal{W}}$ earnings paid by the entrant. The change in resident earnings and the entrant payroll paid to outside residents are

$$\Delta Y^{H,R}_{e,h} \equiv \sum_{i \in H} \left[\mathbf{1}\{R_{i,r^+_h} = C_e\} Y^{\text{cov}}_{i,\mathcal{W}^+_h} - \mathbf{1}\{R_{i,r^-_h} = C_e\} Y^{\text{cov}}_{i,\mathcal{W}^-_h} \right], \tag{A.2}$$

$$P^{H,C}_{e,h} \equiv \sum_{i \in H} \mathbf{1}\{R_{i,r^+_h} \neq C_e\}\, Y^e_{i,\mathcal{W}^+_h}, \tag{A.3}$$

and the resident-earnings ratio is $\iota^{\text{earn}}_{e,h} \equiv \Delta Y^{H,R}_{e,h} / P^H_{e,0}$. Equation (A.2) requires earnings classified jointly by residence before and after hire. The released tabulations classify residence (Table A.5) and earnings (Table A.8) separately, so the $100 allocation in Fact 1 approximates it under three assumptions. First, each residence–workplace category's earnings equal the released quasi-median for its prior workplace, pooling prior industries and worker skills within entrant type. Second, continuing residents with no observed prior job had zero prior earnings. Third, prior and after-hire allocations are each scaled to match the released payroll totals by entrant type and post-hire residence. The approximation omits replacement hiring at the employers the hires left and any wage and employment response elsewhere in the CZ, which is why Fact 1 presents its $32 as a lower bound on the resident income an entry creates.

No double counting. The accounting above enters the model exactly once. In Section 4, entrant jobs enter tradable labor demand, the resulting wages accrue to the workers who fill them, and their earnings pass once through the housing and service shares and the 70/30 home/work shopping split, with inbound commuters' home-origin spending and outbound commuters' workplace-origin spending leaving the local account. Adding the Fact 1 allocation as a second spending shock would count the same payroll twice. The model's recruitment of entrant workers is an equilibrium outcome rather than an imposed share.

Education and reasons for moving. The higher migrant share at high-skill entrants in Section 2 is consistent with public evidence on why workers move. The CPS ASEC for 2007–2019 records each respondent's main reason for moving. For respondents aged 25–64, I estimate survey-weighted linear probability models of five mobility outcomes on an indicator for a bachelor's degree, controlling for survey year, five-year age group, sex, race and ethnicity, and marital status, and report average standardized predictions by education group with confidence intervals from the 160 CPS replicate weights (Table A.10). College graduates are no more likely to move in a given year (11.44 versus 11.35 percent), but they are 2.07 times as likely to move for a new job or job transfer (1.87 versus 0.90 percent), and among movers 16.41 percent of graduates cite a new job against 8.01 percent of others. Education is associated not with whether people move but with whether the move is tied to a job.

Table A.10: Educational attainment and reasons for residential mobility

Outcome	Below BA (%)	BA+ (%)	Difference pp [95% CI]	Rate ratio [95% CI]
Any residential move	11.35	11.44	0.10 [−0.07, 0.26]	1.01 [0.99, 1.02]
Employment-related move	2.05	3.24	1.18 [1.09, 1.28]	1.58 [1.53, 1.63]
New job or job transfer	0.90	1.87	0.97 [0.90, 1.03]	2.07 [1.98, 2.17]
Employment reason, among movers	18.20	28.38	10.18 [9.41, 10.94]	1.56 [1.51, 1.61]
New job or transfer, among movers	8.01	16.41	8.40 [7.82, 8.98]	2.05 [1.96, 2.14]

Notes: Public-use CPS ASEC, 2007–2019, respondents aged 25–64. BA+ denotes a bachelor's degree or higher; associate degrees are in the below-BA group. The first three rows use all respondents as the denominator; the last two condition on having moved. Rates are average standardized predictions from survey-weighted linear probability models controlling for year, age group, sex, race and ethnicity, and marital status. Confidence intervals use CPS successive-difference replicate weights.

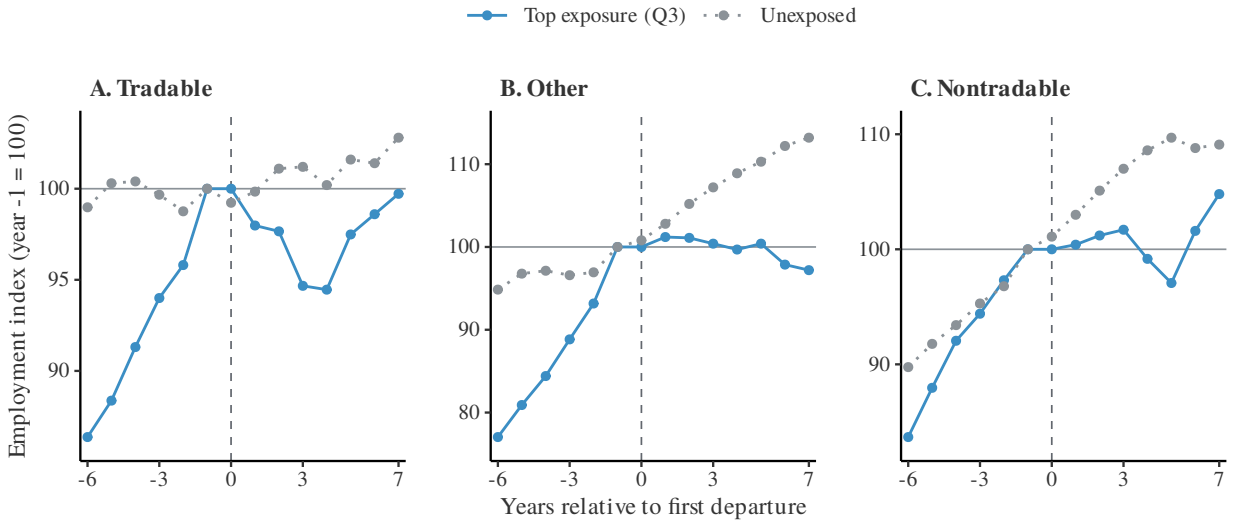
A.4 Incumbent adjustment

Source establishments. For each hire who previously worked in the CZ, the source establishment is the hire's most recent main job. Exposure is the share of a source establishment's payroll that the entrant hired away, measured the year before its first worker left. Within each size bin, the figures compare the most exposed third of source establishments with establishments that

lost no workers to an entrant.[33] Each index equals 100 in the year before the first departure. The all-establishment employment index counts establishments that later close as zero, while the active-establishment employment and pay indices follow only establishments that remain open.

The most exposed establishments stop growing once their workers leave. Among tradable establishments that remain open, employment at the most exposed falls about 6.5 jobs per 100 baseline jobs behind unexposed establishments three years after the first departure, and the gap narrows to about 3 jobs per 100 by year seven as they refill the vacated jobs (Figure A.2). The year-seven gap is about 4 jobs per 100 for nontradable establishments and 16 for establishments in other sectors.[34] Because every establishment in the sample exists at the first departure, the all-establishment index rises before that year and falls after it for exposed and unexposed establishments alike, so the comparison that matters is between the two groups at the same point in time. Once closures are counted, the most exposed tradable establishments end seven years after the first departure about 2 jobs per 100 baseline jobs below unexposed ones (Figure A.3), and those still operating raised average pay by 14 percent, compared with 9 percent at unexposed establishments (Figure A.4). These are descriptive paths without confidence intervals.

Figure A.2: Employment at active source establishments by exposure

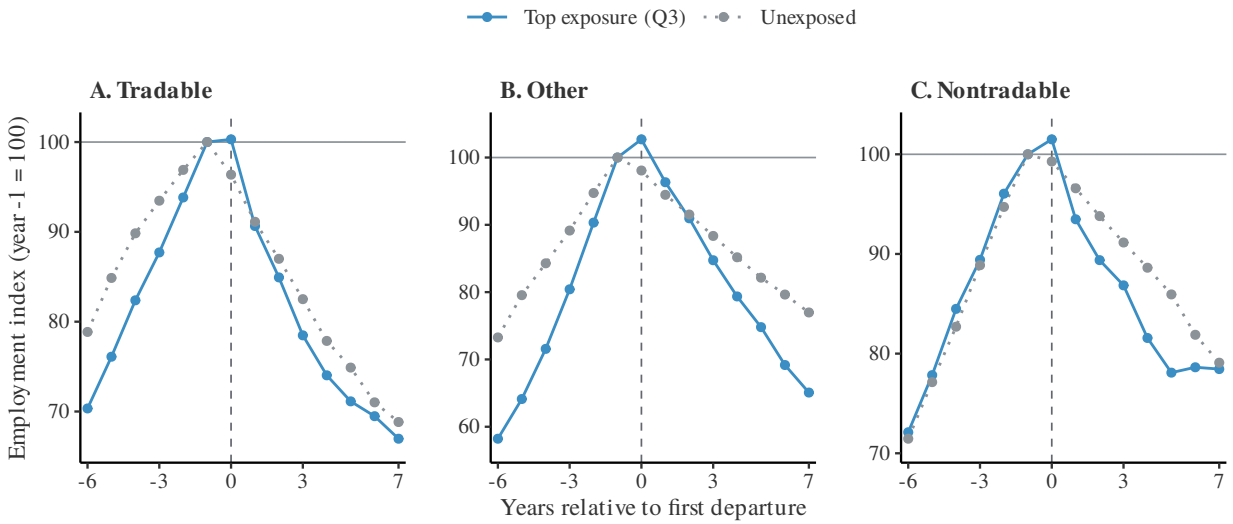


Notes: Released employment index for source establishments that remain active, for the highest-exposure (Q3) and unexposed establishments, by sector of the source establishment. Exposure is measured as the share of an establishment's baseline payroll lost to the entrant. Year 0 is the first departure for exposed establishments; all indices equal 100 in year −1. Missing values at years −1 and 0 are displayed at 100. Each panel has its own scale. Figures A.3 and A.4 show employment across all establishments and average pay at active establishments.

[33]The four employment size bins are small (6–19 workers), medium-small (20–49), medium-large (50–249), and large (250 or more). Exposure groups are formed within each size bin, so the index compares the most exposed firms of each size.

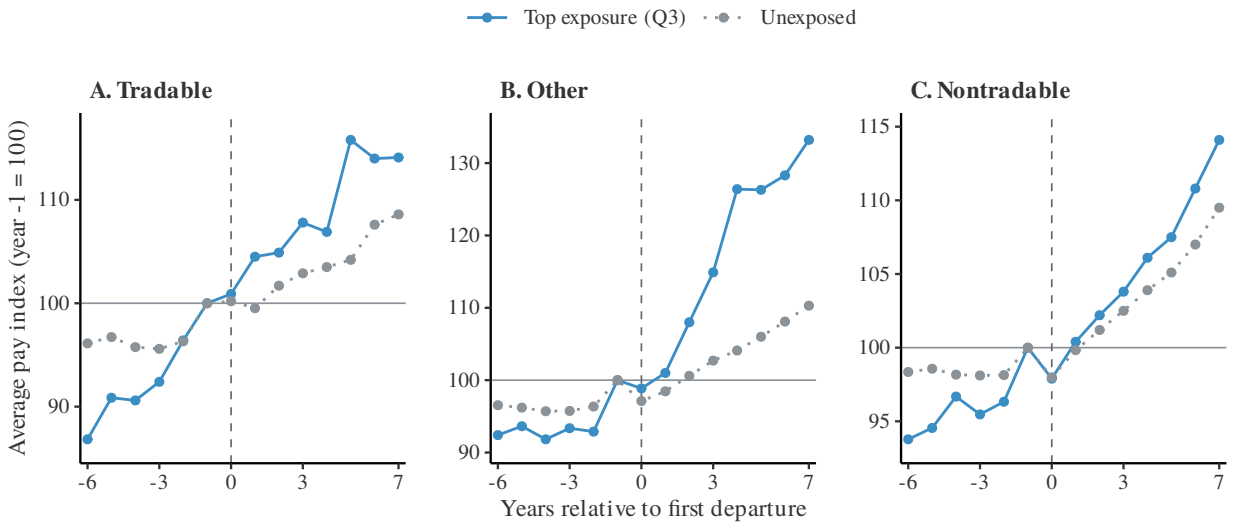
[34]Sectors are defined by the source establishment's four-digit NAICS industry. Tradable industries follow the classification of Mian and Sufi (2014), excluding anything in NAICS 61, 72, and 81 and including all of manufacturing (NAICS 31–33). Nontradable industries are the local services defined in Section 2, and other sectors are all remaining industries.

Figure A.3: Employment at all source establishments by exposure



Notes: Released employment index for all source establishments, which counts establishments that later close as zero, for the highest-exposure (Q3) and unexposed source establishments, by sector of the source establishment. Groups and display conventions follow Figure A.2.

Figure A.4: Average pay at active source establishments by exposure



Notes: Released average-pay index for source establishments that remain active, for the same groups and sectors. Groups and display conventions follow Figure A.2.

Labor-market relatedness and incumbent employment. To investigate whether large tradable entrants displace employment among incumbent firms through competition for workers, I examine whether incumbent industries that are more closely connected to the entrant in the labor market experience larger employment declines. I construct a predetermined measure of labor-market relatedness using national worker flows I construct from the LEHD between three-digit NAICS industries. Industries that historically supply a larger share of workers to the entrant's

industry are expected to face greater recruitment competition following entry. For each entry event e and incumbent industry k, I define labor-market exposure as

$$L_{ek} = \frac{J_e \, s_{k \to h(e)}}{E^0_{ek}}, \tag{A.4}$$

where J_e denotes entrant employment, $s_{k \to h(e)}$ is the predetermined national worker-flow relatedness between incumbent industry k and the entrant's industry $h(e)$, and E^0_{ek} is baseline incumbent employment in industry k in the treated commuting zone. I normalize this measure by its employment-weighted interquartile range, so the coefficients correspond to a one-IQR increase in labor-market exposure. I estimate the event-study specification

$$y_{eckt} = \alpha_{eck} + \delta_{ect} + \gamma_{ekt} + \sum_{\tau \neq -2} \beta_\tau \, \mathbf{1}\{t - T_e = \tau\} \, D_{ec} \, \widetilde{L}_{ek} + \varepsilon_{eckt}, \tag{A.5}$$

where y_{eckt} is log incumbent employment, or log average pay in industry k, commuting zone c, and year t, for event e. The indicator D_{ec} identifies the treated commuting zone, and $\widetilde{L}_{ek}$ is normalized labor-market exposure. The fixed effects absorb time-invariant event–CZ–industry differences, event–CZ–year shocks, and event–industry–year shocks. The omitted period is two years before entry.

For each large-firm entry event, I select five comparison commuting zones using propensity-score matching. Candidate commuting zones are restricted to the same Census division and must not experience another identified large-firm entry within five years of the event. Matching covariates, measured two years before entry, include baseline employment, number of establishments, population, demographic composition, residential rents, and the entrant industry's local employment concentration.

The sample follows a fixed cohort of incumbent establishments identified before entry, retaining establishments that subsequently close with zero employment. Regressions are weighted by baseline incumbent-industry employment, with matched control locations collectively receiving the same weight as the treated location for each event-industry cell. Standard errors are clustered by event commuting zone and incumbent industry. The coefficients β_τ capture differential changes in incumbent outcomes associated with greater labor-market exposure in treated locations relative to matched controls. I summarize the post-entry response using the average coefficient over years two through five and examine the joint significance of pre-entry and post-entry coefficients.

Figure 2 plots the labor coefficients from the joint specification. Tradable industries more connected to the entrant through worker flows lose employment in years four through six, by at most 0.06 percent per interquartile range of exposure, and the year-seven interval includes zero. Average pay in those tradable industries is also slightly lower after entry, by about 0.03 percent per interquartile range, while nontradable industries show no clear change. Several pre-entry intervals also exclude zero, so these paths are suggestive rather than clean event studies.

A.5 Fact 2: Shopping geography and openness

Service expenditure shares. The 2024 BEA PCE Bridge maps each NIPA consumption category to input–output commodities and reports producers' values, transportation costs, and wholesale and retail margins. I define service expenditure N as household consumption of educational services (NAICS 61), health care and social assistance (62), accommodation and food services (72), and other services (81), each at producers' value, plus the retail margin on goods sold by retailers (44–45). Using the margin rather than the full purchase price keeps the manufactured value of goods out of retail. This gives $N = \$9.04$ trillion in 2024, or $N/Y^d = 0.41$ of disposable personal income Y^d, which was \$21.92 trillion (NIPA Table 2.1). Housing $H = \$3.15$ trillion is the PCE housing-services category, including imputed rent on owner-occupied housing; a \$3.29 billion overlap with group housing is removed. The share of after-housing income spent on services is

$$s_N^{AH} = \frac{N - 0.00329}{Y^d - H} = \frac{9.04}{21.92 - 3.15} = 0.48,$$

or 49.34% if utilities are also treated as housing. These are average national shares of total household consumption, including services purchased on households' behalf by employers and governments, not marginal propensities, and the sectors are locally oriented rather than perfectly nontraded.

Openness at county and CZ boundaries. Figure A.5 compares, for each county and CZ, the share of residents' primary jobs located inside the area with the share of residents' service visits that end inside it. The commuting share uses 2019 LODES resident workers; the visit share uses raw 2019 Veraset visits in retail, selected real estate, education excluding colleges, food services, and other services. Neither is a dollar share.

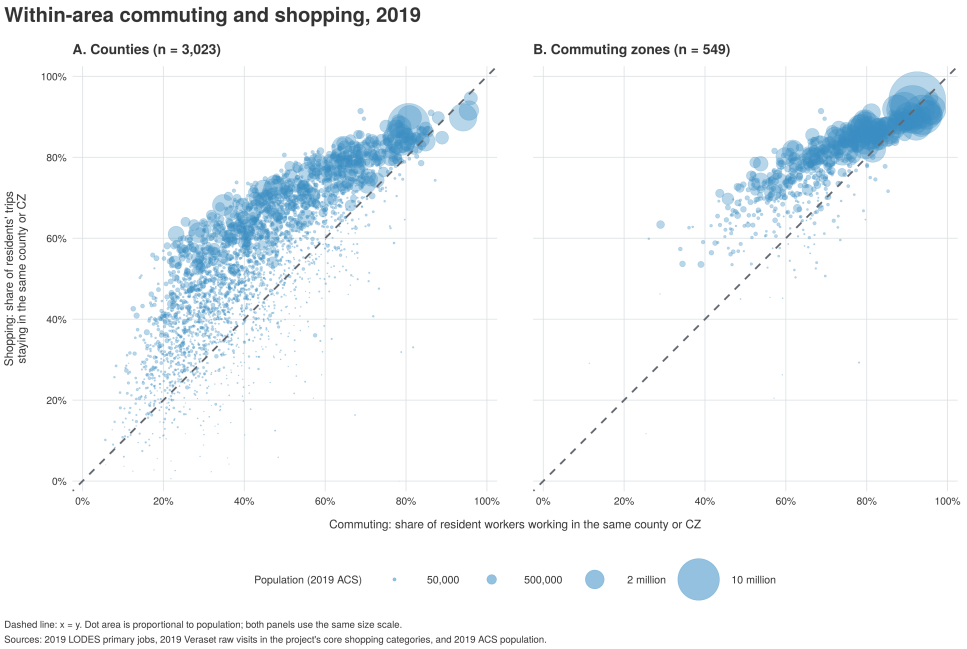
The entrant-level comparison in Section 2 uses the 180 verified award episodes at 167 workplace tracts in 86 CZs, with 4 episodes more than 100 km from their CZ centroid excluded. Let F_{ie,t_e+5} be the number of primary-job workers living in tract i and employed in the event tract e five years after the award. The share living inside boundary B is

$$S_e(B) = \frac{\sum_{i \in B} F_{ie,t_e+5}}{\sum_i F_{ie,t_e+5}}, \tag{A.6}$$

for the event county, the CZ, and the tracts within 60 driving minutes of the event tract. These are post-award levels for all workers in the event tract, not the entrant's own workforce. Entrant industries are classified as high skill when their four-digit industry's college share in the 2000 Census is at or above the national median of 20.66%, giving 132 high-skill and 46 low-skill episodes. To separate accessible population from administrative geography, I also compute the 2019 population within 40 driving minutes of the event tract,

$$P_e^{40} = \sum_i \text{Pop}_{i,2019}\ \mathbb{1}\{\tau_{ie} \leq 40 \text{ minutes}\}, \tag{A.7}$$

Figure A.5: Within-area commuting and shopping, 2019



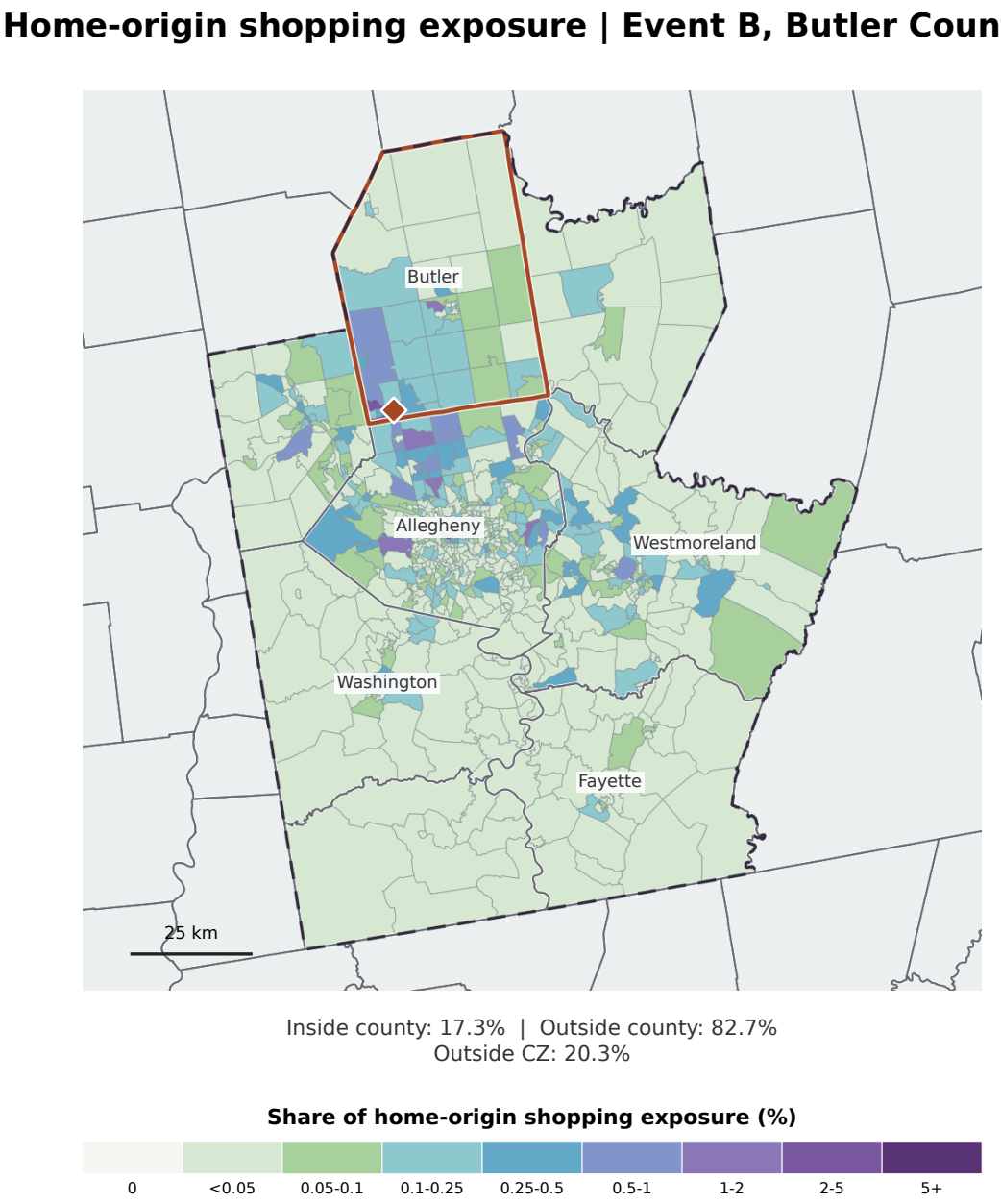
Notes: Each dot is a county (3,023) or CZ (549). The horizontal axis is the share of resident primary-job workers in LODES 2019 who work inside the area; the vertical axis is the share of residents' raw Veraset visits in NAICS 44, 45, selected 53, 61 (excluding 6113), 722, and 81 that end inside it. Cross-county flows within a CZ count as internal. Dot area is proportional to 2019 population; the line is equality.

and split episodes at common terciles (448,246 and 1,735,422 residents). Table A.11 reports the shares. In the lower and middle terciles, high-skill sites keep a larger share of their workers inside the county than low-skill sites, the reverse of the unconditional comparison, so the aggregate skill contrast partly reflects where each type of entrant locates. The upper tercile has only 2 low-skill events.

Figure A.6 maps where the workers at the entrant's tract in Figure 1 shop five years after entry, assigning each worker the 2019 visit pattern of residents of their home tract.

Where within-CZ visits go. Table 3 in Section 3 splits within-CZ visits into three mutually exclusive destinations. A same-tract visit ends in the device's home tract; an adjacent-tract visit ends in a different 2020 tract sharing a boundary point with the home tract; a nonlocal hub visit ends in a shopping-hub footprint outside the home tract and its neighbors. For the core sectors (NAICS 44–45, 722, 81), 10.60 percent of within-CZ visits stay in the home tract, 23.81 percent go to an adjacent tract, and 29.25 percent to a nonlocal hub. Hubs of all kinds receive 43.40% of within-CZ visits; 14.10% of within-CZ visits are to a hub that is also in the home tract or an adjacent tract, so 67.50% of hub visits are away from home. Across all destinations, 84.80% of core-sector visits stay in the home CZ. The core combines NAICS 44, 45, selected NAICS 53, restricted NAICS 61, NAICS 722, and NAICS 81. Selected NAICS 53 consists of 5311 (Lessors of Real Estate), 5312 (Offices of Real Estate Agents and Brokers), 5313 (Activities Related to Real

Figure A.6: Where the workers at the entrant's tract shop



Notes: Share of the shopping visits of the workers at the entrant's tract five years after entry that end in each destination tract, in percent. Each worker is assigned the 2019 visit pattern of residents of their home tract across retail (NAICS 44–45), food services (722), other services (81), and non-college education (61), weighted by public LODES counts of primary jobs in the broad All Other Services group (SI03) at the tract by residence. Workers' own purchases are not observed. Shares under the map include destinations outside the CZ. The diamond marks the entrant's tract and the outlined area its host county.

Table A.11: Share of workers living inside the boundary, by nearby population and entrant industry

Entrant industry	Events	County	CZ	60-minute area
		Mean share in percent (SD in percentage points)		
All retained events				
High-skill	132	44.60 (18.50)	78.50 (11.40)	83.50 (7.50)
Low-skill	46	48.50 (15.20)	75.10 (12.20)	77.70 (9.90)
Lower tercile: at most 448,246 residents				
High-skill	25	57.60 (12.90)	79.60 (8.80)	81.40 (7.70)
Low-skill	34	50.50 (15.70)	77.00 (10.10)	76.10 (10.10)
Middle tercile: 448,246–1,735,422 residents				
High-skill	49	45.60 (19.00)	81.20 (9.50)	81.80 (8.50)
Low-skill	10	42.20 (13.00)	70.80 (15.20)	81.00 (8.60)
Upper tercile: more than 1,735,422 residents				
High-skill	58	38.30 (17.20)	75.70 (13.20)	85.90 (5.70)
Low-skill	2	45.90 (16.70)	63.70 (25.50)	87.80 (4.20)

Notes: Each event receives equal weight. The denominator is all LODES primary jobs in the event workplace tract in award year +5, not the entrant workforce. Columns change only the residential boundary. Parentheses contain cross-event standard deviations, not standard errors.
Nearby population is 2019 ACS population summed across 2020 tracts whose reference points are at most 40 driving minutes *to* the event tract, irrespective of county or CZ membership. Whole-tract populations are included by reference-point membership. This is an accessible-population measure, not population density or a pre-entry covariate.
Common tercile cutoffs use all 180 retained award episodes, including 2 with unresolved industry skill classifications. Those two episodes are omitted from the displayed high/low rows. Ties stay together; displayed cutoffs are rounded to the nearest resident. The upper-tercile low-skill group has only 2 events.
High skill is at or above the nationwide median four-digit-industry college share in Census 2000; pooled industry matches are flagged in the source data. See Appendix A.5 for the sample and coverage checks.

Estate), 5321 (Automotive Equipment Rental and Leasing), 5322 (Consumer Goods Rental), and 5323 (General Rental Centers). Restricted NAICS 61 includes Educational Services other than 6113 (Colleges, Universities, and Professional Schools). Mobility records for NAICS 62 (Health Care and Social Assistance), including ambulatory health-care providers, hospitals, nursing and residential-care facilities, and social-assistance providers, are excluded for privacy reasons. NAICS 721 is not included in the core or reported separately. Table A.12 reports the cumulative driving-time distribution of all visits, including those that cross CZ boundaries.

Shopping hubs. Hubs are built from destination activity, adapting the employment-subcenter method of Giuliano and Small (1991) and McMillen (2001) to visits. Let V_j be all 2019 device-weighted core-sector visits received by tract j, and $\widetilde{V}_j = V_j + \sum_{k \in \mathcal{N}(j),\, c(k)=c(j)} V_k$ its sum with its same-CZ neighbors. A tract is a candidate peak when $\widetilde{V}_j$ is at least as large as that of every adjacent tract in its CZ. Peaks are processed from strongest to weakest, and a weaker peak is

Table A.12: Cumulative driving time of U.S. service visits, 2019

Sector	≤ 5	≤ 10	≤ 15	≤ 20	≤ 30	≤ 40	≤ 60	> 60
NAICS 44	18.99	38.11	52.51	62.61	74.86	81.16	86.82	13.18
NAICS 45	14.86	34.60	50.95	62.16	75.23	81.72	87.43	12.57
NAICS 722	16.25	34.75	49.25	59.60	72.23	78.79	84.80	15.20
NAICS 81	19.20	36.28	50.92	61.99	75.72	82.59	88.45	11.55
Core	*16.86*	*35.52*	*50.48*	*61.11*	*73.97*	*80.52*	*86.38*	*13.62*
Selected NAICS 53	9.26	22.44	35.35	45.79	59.50	66.84	73.76	26.24
Restricted NAICS 61	11.28	26.24	41.35	53.74	69.77	78.22	85.99	14.01

Notes: Percentages of annual device-weighted visits, including visits that cross CZ boundaries. Columns are cumulative minutes of tract-center driving time on the 2020 road network; same-tract visits are assigned five minutes. Pairs more than 100 straight-line miles apart, routed times above 60 minutes, and unreachable pairs are in the > 60 column. The core combines NAICS 44, 45, selected NAICS 53, restricted NAICS 61, NAICS 722, and NAICS 81. Selected NAICS 53 consists of 5311 (Lessors of Real Estate), 5312 (Offices of Real Estate Agents and Brokers), 5313 (Activities Related to Real Estate), 5321 (Automotive Equipment Rental and Leasing), 5322 (Consumer Goods Rental), and 5323 (General Rental Centers). Restricted NAICS 61 includes Educational Services other than 6113 (Colleges, Universities, and Professional Schools). Mobility records for NAICS 62 (Health Care and Social Assistance), including ambulatory health-care providers, hospitals, nursing and residential-care facilities, and social-assistance providers, are excluded for privacy reasons. NAICS 721 is not included in the core or reported separately.

merged into a stronger one when the mean driving time between them is at most ten minutes. A hub's footprint $\mathcal{F}_g$ is the union of its peak tracts and their adjacent tracts, with shared tracts assigned to the stronger hub, and its share of destination activity is $H_g = \sum_{j \in \mathcal{F}_g} V_j / \sum_{j:c(j)=c(g)} V_j$. A hub qualifies under the 2 percent definition when $H_g \geq 0.02$; the 1 and 5 percent definitions use 0.01 and 0.05. The same footprints are used for every sector in Table 3.

Hubs and market size. Across the 446 CZs with 2019 population of at least 50,000, the number of hubs rises with population. A log-link count model of the hub count on log population gives an elasticity of 0.59 under the 1 percent definition, 0.43 under the 2 percent definition, and 0.10 under the 5 percent definition, which mechanically caps how many hubs can qualify (Table A.13).

Table A.13: Scaling of within-CZ shopping hubs with CZ population

Minimum hub share	Preferred model	Intercept	Robust SE	Elasticity	Robust SE	95% CI
1%	Negative binomial	-5.95	(0.30)	0.59	(0.02)	[0.54, 0.63]
2%	Negative binomial	-3.99	(0.32)	0.43	(0.03)	[0.38, 0.48]
5%	Poisson	-0.20	(0.30)	0.10	(0.02)	[0.06, 0.15]

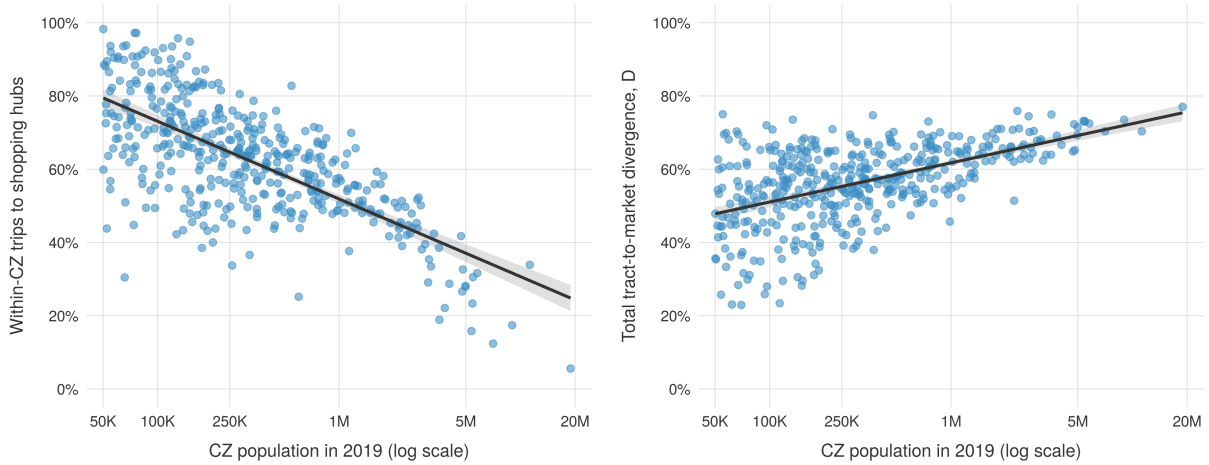
Notes: Each row reports the valid AIC-preferred count model for one hub-mass threshold. The sample contains 446 CZs with 2019 population of at least 50,000. The dependent variable is the number of within-CZ hubs. Standard errors are heteroskedasticity-robust HC1 estimates. At the 5% threshold, the negative-binomial dispersion collapses to the Poisson boundary and is excluded from model comparison.

A tract's shopping versus its market's. Fact 3 assigns exposure to the home tract and its neighbors rather than through the observed shopping network, on the grounds that the home-and-adjacent component is the part of shopping that differs across tracts. This paragraph measures that. For home tract i in CZ c, let p_{ij} be the share of its device-weighted within-CZ visits ending in tract j, and let

$$q_{-i,j} = \frac{\sum_{k \in c,\, k \neq i} v_{kj}}{\sum_{k \in c,\, k \neq i} \sum_{\ell \in c} v_{k\ell}} \tag{A.8}$$

be the destination shares of all other residents of the CZ. The total-variation distance $D_i = \sum_j \max\{p_{ij} - q_{-i,j}, 0\}$ is the share of tract i's visits that would have to be reassigned to make its pattern match its market's. Across 438 CZs with population of at least 50,000, 34.40% of within-CZ visits end in the home tract or an adjacent tract, against 3.20% under the other residents' pattern, and the home-and-adjacent set accounts for 48 percent of the pooled positive deviation $\sum_i V_i D_i$ (weighting origins by their visits V_i). Figure A.7 shows how the two parts of this vary with market size. The share of within-CZ visits going to 2 percent hubs falls from three quarters in the smallest tract-count decile to two fifths in the largest (correlation -0.70 with log population), while the visit-weighted mean of D_i rises from about one half to two thirds (correlation 0.52). Small markets send most visits to a few shared hubs, so tracts look like their markets; large markets spread visits over many destinations, and tracts differentiate.

Figure A.7: Shopping hubs and total tract-to-market divergence by market size



Notes: Each dot is one of the 438 CZs with 2019 population of at least 50,000 and positive within-CZ shopping support. Left: share of residents' within-CZ device-weighted visits received by 2 percent hub footprints. Right: visit-weighted mean of D_i across the CZ's home tracts. Lines are OLS fits in log population with pointwise 95 percent HC1 bands.

Shopping and commuting gravity. This paragraph documents the comparison of travel-time decay behind Figure 3 and equation (1). It is distinct from the sector-by-CZ shopping gravity used to quantify the model in Section 5, which adds same-tract and same-county terms. For shopping, F^S_{ijc} is the number of 2019 device-weighted visits from residents of i to establishments in j, pooling

NAICS 44–45, 5311–5313, 5321–5323, 61 (excluding 6113), 722, and 81. For commuting, F^C_{ijc} is the 2019 LODES count of workers living in i and working in j. For each mode m and CZ c, Poisson pseudo-maximum likelihood solves

$$(\widehat{\alpha}^m_c, \widehat{\delta}^m_c, \widehat{\beta}^m_c) = \arg\max_{\alpha,\delta,\beta} \sum_{(i,j)\in\mathcal{J}_c} \left[F^m_{ijc} \log \mu^m_{ijc} - \mu^m_{ijc}\right], \qquad \mu^m_{ijc} = \exp\left(\alpha^m_{ic} + \delta^m_{jc} + \beta^m_c t_{ijc}\right), \quad \text{(A.9)}$$

with standard errors two-way clustered by origin and destination tract. PPML accommodates the many tract pairs with zero flows, which carry information about decay, and avoids the retransformation problems of log-linear OLS (Silva and Tenreyro, 2006). Both modes use the same tract-pair support: all non-water pairs within a CZ with finite driving time of at most 180 minutes, with same-tract pairs at five minutes and absent flows entered as zero. A CZ needs at least ten candidate pairs and ten positive-flow pairs. The shopping estimator converges in 542 CZs and the commuting estimator in 530; the comparison uses the 526 CZs with both. Because travel time enters in levels, I report the exact percentage decline per minute, $d^m_c = 100[1 - \exp(\widehat{\beta}^m_c)]$. Table A.14 reports the coefficient moments; the mean declines are 8.26% for shopping and 5.92% for commuting, and the cross-CZ correlation of the two coefficients is 0.86.

Table A.14: CZ-level shopping and commuting decay estimates

Flow	Converged CZ fits	Mean $\widehat{\beta}_c$	Median $\widehat{\beta}_c$	Median decline per minute
Shopping	542	−0.09	−0.08	8.08%
Commuting	530	−0.06	−0.06	6.10%

Notes: Coefficient moments and exact declines are computed on the 526-CZ matched comparison sample; the converged-fit column counts estimates available before matching the two modes.

The estimates are in line with the literature. Ahlfeldt et al. (2015) estimate a commuting semi-elasticity of about 0.07 per minute in Berlin, against the median 6.10% here, and Monte et al. (2018) estimate a county-to-county distance elasticity of −4.43. For consumption trips, Miyauchi et al. (2021) estimate travel-time elasticities of −1.82 from home and −1.81 from work in Tokyo, and Oh and Seo (2023) find that ten additional minutes reduce service-trip flows by about 80 percent in Seoul, against 57.00% here. The comparison is descriptive. Identification of β^m_c requires that travel time be uncorrelated with omitted pair-specific determinants of flows conditional on the fixed effects, and the visit data count trips rather than dollars. In the model the two coefficients map to different structural objects, $\beta^S_c = (1 - \sigma_N)\kappa$ and $\beta^C_c = -\chi\kappa$, which is why the comparison is made on the reduced-form declines rather than on the parameters.

A.6 Fact 3: Residential exposure designs

Fact 3 compares service employment in the tracts most and least exposed to entrant workers' spending. This section gives the alternative exposure measures and specifications estimated on the restricted data, the public analog built from predicted residences, and the reasons the observed

shopping network is not used as exposure.

The home-and-adjacent kernel. Every exposure measure in this section places each hire's spending at a home tract i and spreads it over $\mathcal{N}(i)$, the home tract and all 2020 tracts sharing a boundary point with it, with weights K_{ij} that assign 30 percent to the home tract and share 70 percent equally among the neighbors,

$$K_{ij} \geq 0, \qquad \sum_{j \in \mathcal{N}(i)} K_{ij} = 1, \qquad K_{ij} = 0 \text{ if } j \notin \mathcal{N}(i), \tag{A.10}$$

so each hire is counted once across tracts. The 30/70 split is the share of visits to the home tract among visits that end in the home tract or an adjacent tract (Appendix A.6). The entrant's workplace tract is excluded from every outcome sample.

Stacked new-income exposure. The first alternative ranks tracts by new income rather than by workers. New income is the earnings increase over the previous job for previously employed local hires and the full entrant wage for migrants and hires without a prior job. It is assigned to the hire's residence four quarters after hire, so that G_{ei} is new income received by hires living in tract i, and exposure is

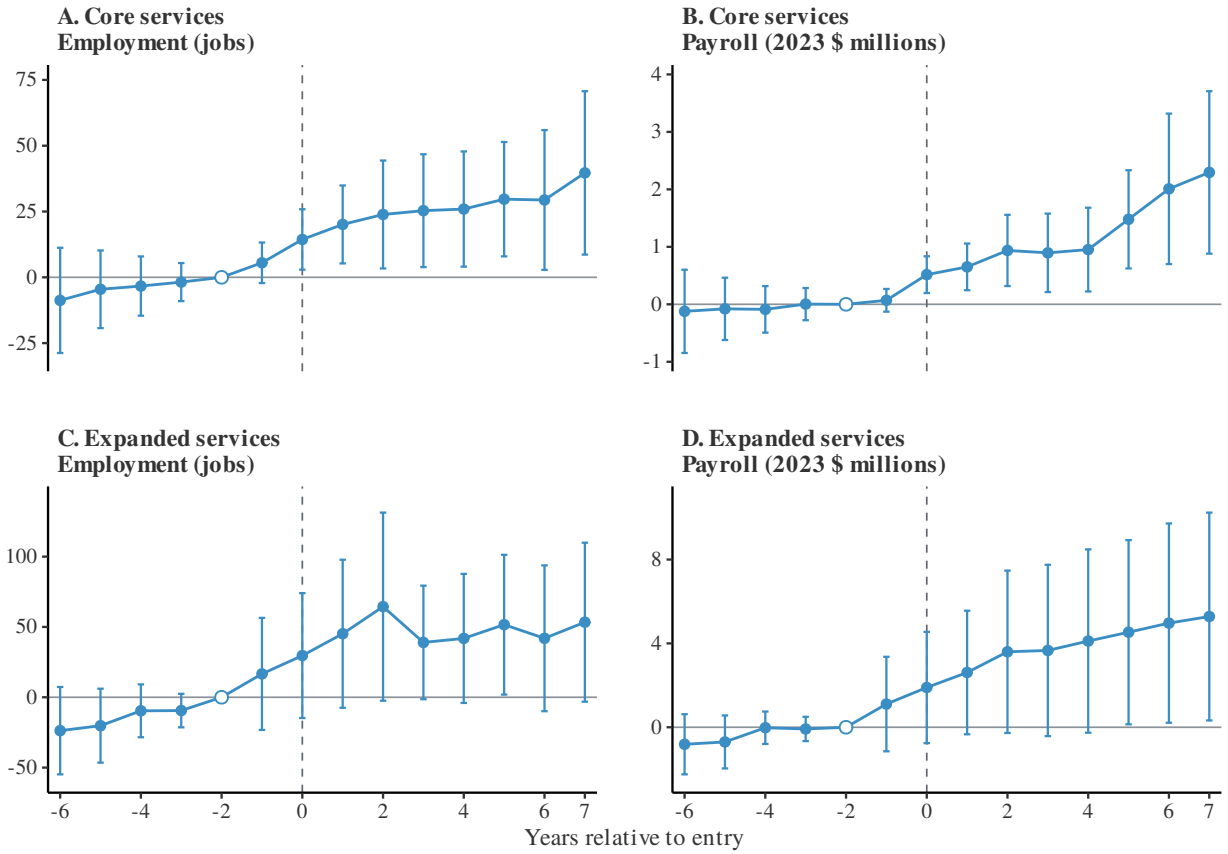
$$A^I_{ej} = \sum_i G_{ei} K_{ij}. \tag{A.11}$$

As in Fact 3, the sample is events in CZs with no other observed event, and I compare the top and bottom quartiles of A^I_{ej} within each event with equation (2). Mean exposure differs between the quartiles by \$1.166 million for the core-service outcomes (NAICS 44, 45, 722, 81) and \$1.164 million for the expanded outcomes that add 61 and 62; the release covers 30 CZs. By year seven, the core-service contrasts are 39.67 jobs and \$2.295 million in annual payroll, with both intervals above zero (Figure A.8). The expanded outcomes give 53.37 jobs and \$5.283 million, with the employment interval including zero. Pre-entry intervals include zero in all four panels. Because this measure uses post-hire residences, it can reflect moves after hiring as well as income gains at existing addresses.

Continuous exposure. The continuous specification uses all events and all tracts, including CZs with several events and the middle of the exposure distribution. For event e, let H_{ei} be the number of hires who lived in tract i of the event CZ before entry, with $A^W_{ej} = \sum_i H_{ei} K_{ij}$ the neighborhood allocation. Calendar-year exposure is $X^W_{jt} = 100^{-1} \sum_{e:\tau_e = t} A^W_{ej}$ for workers and $X^I_{jt} = 10^{-6} \sum_{e:\tau_e = t} A^I_{ej}$ for income. For either measure X and outcome o,

$$Y^o_{jt} = \alpha^o_j + \delta^o_{c(j)t} + \sum_{k \in \{-6,\ldots,7\} \setminus \{-2\}} \beta^o_k X_{j,t-k} + \gamma^o_- X^{\text{old}}_{jt} + \gamma^o_+ X^{\text{future}}_{jt} + \varepsilon^o_{jt}, \tag{A.12}$$

Figure A.8: Stacked service responses to neighborhood new-income exposure



Notes: Top-minus-bottom-quartile contrasts in new-income exposure from equation (2), in CZs with no other observed events. Panels A–B cover NAICS 44, 45, 722, and 81; panels C–D add 61 and 62. Income is assigned to residences four quarters after hire and spread with the 30/70 kernel. Bars are 95 percent confidence intervals clustered by CZ; year −2 is normalized to zero. Employment is in jobs and payroll in millions of 2023 dollars. 30 CZs.

with tract and CZ-by-year fixed effects, no weights, CZ-clustered standard errors, and stocks of exposure from events outside the window carried in X^{old} and X^{future}. Outcomes cover NAICS 44, 45, 722, and 81. Figure 5 in Section 3 reports the worker-exposure estimates. Stacked and continuous coefficients answer different questions. The stacked contrast is the change in the gap between the top and bottom quartiles in single-event CZs, while the continuous coefficient is a slope per 100 workers across the full distribution in all events, so their magnitudes are not directly comparable.

The public analog. The public design replaces identified hires with predicted residences and LEHD outcomes with LODES employment. It covers the 124 registry events dated 2011–2017 in 69 CZs.

Residences are predicted in two steps. First, for each event, the pre-entry LODES origin–destination counts $C_{hw_e,t}$ from residence tract h to the workplace tract w_e are averaged over event

years -4 through -2 and normalized within the CZ,

$$\overline{C}_{eh} = \frac{1}{3}\sum_{\ell=2}^{4} C_{hw_e,\tau_e-\ell}, \qquad p^C_{eh} = \frac{\overline{C}_{eh}}{\sum_{r\in c_e}\overline{C}_{er}}. \tag{A.13}$$

These are the residences of all workers in the tract, not of the entrant's hires. Second, because movers may settle outside the existing commuting catchment, a mover distribution p^M_{eh} is the share of the CZ's recent in-migrants aged 18–64 living in tract h, from the ACS five-year table B07001 ending in $\tau_e - 2$,

$$p^M_{eh} = \frac{M_{h,\tau_e-2}}{\sum_{r\in c_e} M_{r,\tau_e-2}}. \tag{A.14}$$

The predicted residence distribution mixes the two with a fixed weight,

$$p_{eh} = \omega\, p^C_{eh} + (1-\omega)\, p^M_{eh}, \qquad \omega = 0.84, \tag{A.15}$$

where ω is the share of CZ-resident hires who were previously employed or nonemployed in the CZ, and $1-\omega$ the share who migrated in, from the source shares behind Fact 1. The weights were fixed before any outcome was examined. Predicted residences are then scaled to the event's exposure mass N_e, the promised jobs allocated to CZ tracts by pre-entry commuting shares, on the support S_e that excludes the workplace tract and its neighbors,

$$R_{eh} = N_e\, \frac{p_{eh}\, \mathbf{1}\{h \in S_e\}}{\sum_{r\in S_e} p_{er}}. \tag{A.16}$$

Exposure then follows the home-and-adjacent rule. Let $\mathcal{A}_e(h)$ be the tracts in c_e adjacent to h. The 2019 visit data assign 0.09 of all service visits to the home tract and 0.20 to adjacent tracts; conditional on these local destinations, the home share is 0.31 and the adjacent share 0.69, which are the 30/70 weights used throughout. Destination exposure is

$$q^{HT}_{ehj} = 0.31\, \mathbf{1}\{j = h\} + \frac{0.69}{|\mathcal{A}_e(h)|}\, \mathbf{1}\{j \in \mathcal{A}_e(h)\}, \qquad E_{ej} = N_e\, \frac{\sum_h R_{eh}\, q^{HT}_{ehj}}{\sum_m \sum_h R_{eh}\, q^{HT}_{ehm}}, \tag{A.17}$$

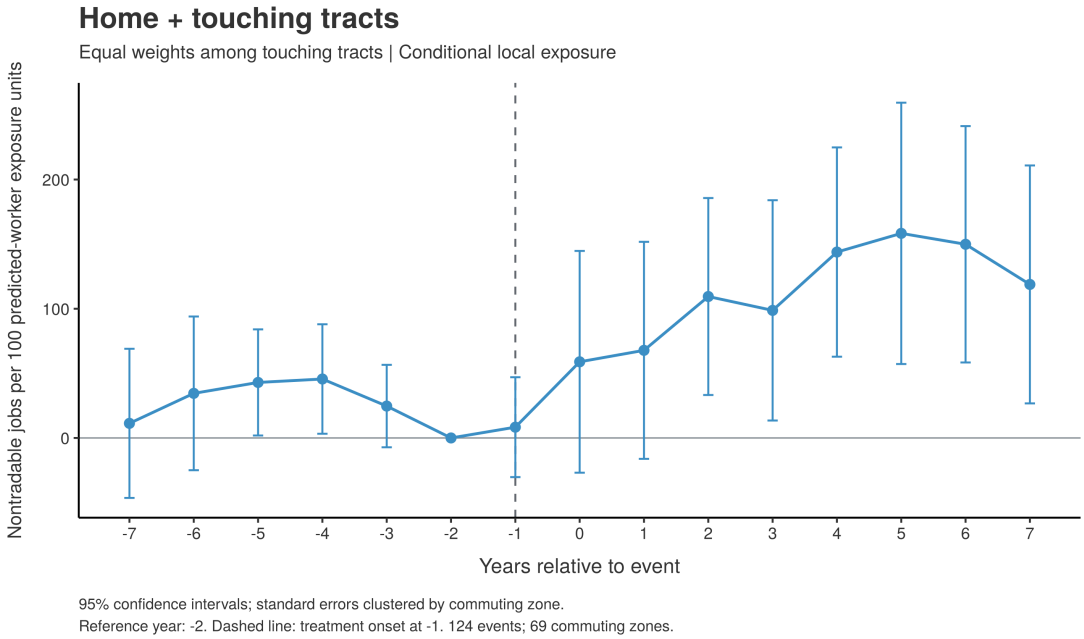
so each event's exposure mass is preserved. With $X^k_{jt} = \sum_e (E_{ej}/100)\, \mathbf{1}\{t-\tau_e = k\}$, the specification is

$$Y_{jt} = \alpha_j + \lambda_{c(j)t} + \sum_{k=-7,\, k\neq -2}^{7} \beta_k X^k_{jt} + \gamma_O O_{jt} + \gamma_F F_{jt} + \varepsilon_{jt}, \tag{A.18}$$

where Y_{jt} is LODES private service employment, α_j and $\lambda_{c(j)t}$ are tract and CZ-by-year fixed effects, and O_{jt} and F_{jt} carry exposure to events more than seven years in the past and future. The regression is unweighted and clustered by CZ, with 772,144 tract-years on 36,791 tracts over 2002–2022. The coefficient β_k is the differential employment response per 100 predicted-worker exposure units relative to year -2. Figure A.9 plots the estimates. The mean over years $+2$ through $+7$ is 129.90 jobs per 100 units (SE 41.00), and the joint test that the leads from -7 through -3 are zero

has $p = 0.06$, so the pre-entry evidence warrants caution. Like Fact 3, this is a within-CZ spatial response, not net job creation.

Figure A.9: Public home-plus-adjacent exposure and service employment



Notes: Coefficients β_k from equation (A.18), in private service jobs per 100 predicted-worker exposure units, with 95 percent confidence intervals clustered by CZ. Year -2 is normalized to zero; the dashed line marks treatment onset at -1. 124 events in 69 CZs.

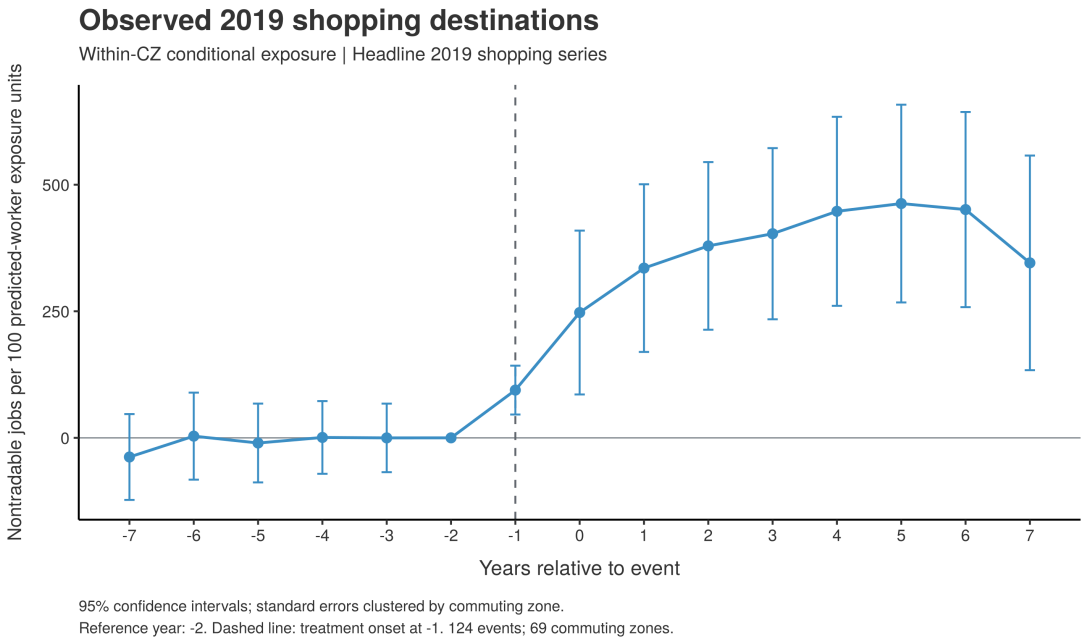
Why the observed shopping network is not used as exposure. The same predicted residences can instead be routed to destinations through the observed 2019 within-CZ visit shares, $S_{ej} = \sum_h R_{eh} s_{ehj}$, or through a gravity network built from pre-entry destination employment and travel time. Both alternatives have the same outcome, specification, and sample as the home-and-adjacent design. Two features rule them out as identifying variation. First, residents of different neighborhoods in a CZ shop at the same hubs, so exposure routed through the network lands on nearly the same destination tracts for every event in a CZ. Among the 85 entries in CZs with more than one entry, destination fixed effects alone explain 76.60% of the variation in network exposure. Second, the 2019 network is measured after the 2011–2017 events and may reflect their effects.

Figure A.10 shows what the observed network produces anyway. The mean coefficient over years +2 through +7 is 414.90 jobs per 100 predicted-worker exposure units (SE 89.20), several times the home-and-adjacent estimate, with a joint pretrend p-value of 0.37. The log and stacked versions of the same design reject the pretrend test ($p = 0.00$ and $p = 0.01$). Placebo networks that shuffle destination exposure within event, size-decile, and travel-time groups give a mean post-entry coefficient of 68.70 against the observed 414.90.[35] The coefficient measures employment growth at

[35]The placebo distribution uses 199 shuffles that preserve each event's exposure mass and destination support; a

the shared shopping destinations of each market in the years after entries, which is where entry-linked spending lands, but it cannot attribute that growth to a particular entrant. The residential design remains the identifying one.

Figure A.10: Observed 2019 shopping-destination exposure and service employment



Notes: Coefficients from equation (A.18) with exposure assigned through observed 2019 within-CZ shopping destination shares, in private service jobs per 100 predicted-worker exposure units, with 95 percent confidence intervals clustered by CZ. Year -2 is normalized to zero; the dashed line marks treatment onset at -1. 124 events in 69 CZs. Because the 2019 network postdates the events, the series is descriptive.

A.7 Aggregate and median housing shares across commuting zones

The aggregate housing share measures the fraction of a CZ's total household resources devoted to housing. The median household share describes the housing expenditure burden of the middle household. Figure A.11 compares these moments using a common sample and measurement procedure across 583 CZs in the 50 states and the District of Columbia. This national comparison uses a common 2008–2017 period; the event calibration in equation (C.14) targets the household median in each CZ's pre-entry sample.

Sample and geographic assignment. The data are the Census Bureau's national ACS five-year Public Use Microdata Samples for 2008–2012 and 2013–2017, linking housing records to their adult person records. Each release receives one half of its published survey weight. Pooling these nonoverlapping releases produces a ten-year household distribution and average population stocks. The housing sample contains 12,113,453 occupied households with at least one adult aged 18 or

restricted wild-cluster bootstrap with 999 CZ-level draws gives $p = 0.00$.

older and usable housing-service measures. Rents and incomes use the ACS adjustment factors followed by the CPI conversion to annual 2023 dollars.

Public Use Microdata Areas (PUMAs) are allocated fractionally to CZs using 2010 census population in their intersections. The allocation uses 2000 PUMAs for 2008–2011 interviews and 2010 PUMAs for 2012–2017 interviews. The former crosswalk is constructed from the Missouri Census Data Center's Geocorr PUMA–county intersections; the latter uses Census tract–PUMA relationships and tract populations. Historical county codes and merged PUMAs are reconciled before aggregation. For household m in PUMA p, let a_{pc} be the fraction allocated to CZ c. Its estimation weight is $w_{mc} = \frac{1}{2}\text{WGTP}_m a_{pc}$.

Housing services and household resources. Paying renters contribute their annual contract rent. Owner-occupied and rent-free dwellings contribute imputed rental services from a national, survey-weighted regression estimated on 3,322,577 paying renter households:

$$\log E_m^{\text{rent}} = X_m'\beta + \lambda_{p(m),t(m)} + u_m. \tag{A.19}$$

Here X_m contains log rooms and its square, bedrooms and its square, structure-type indicators, and structure-type interactions with log rooms. The fixed effects identify state, PUMA vintage, PUMA, and interview year. Imputed annual service expenditure is $E_m = \exp(X_m'\widehat{\beta} + \widehat{\lambda}_{p(m),t(m)})\widehat{s}_{b(m)}$, where $\widehat{s}_b$ is the survey-weighted mean of $\exp(\widehat{u}_m)$ among renters of structure type b. This retransformation incorporates the residual distribution when converting predicted log rents to levels.

Household resources sum income and housing-portfolio dividends over adults. An employed adult receives 95 percent of observed wage and salary income. A nonworker receives the CZ–skill mean of public assistance, Supplemental Security Income, and Social Security income among nonworkers, estimated with person weights. High skill denotes a bachelor's degree or higher. The CZ housing portfolio equals total housing-service expenditure divided by 1.28. Its skill shares are based on reported owner-occupied housing values, split equally between the adult householder and spouse or unmarried partner when both are present. Each skill's share of the portfolio is distributed equally among its represented adults in the eligible sample. Thus

$$Y_{mc} = Y_{mc}^{\text{inc}} + \sum_{k\in\{H,L\}} n_{mk} D_{ck}, \qquad D_{ck} = \frac{\omega_{ck}}{N_{ck}} \frac{\sum_m w_{mc} E_m}{1.28}, \tag{A.20}$$

where n_{mk} is the household's adult count of skill k, $N_{ck} = \sum_m w_{mc} n_{mk}$, and ω_{ck} is that skill's ownership-value share. Property tax revenue leaves the modeled economy, so residents receive no rebate. These resources implement the model's individual budget convention.

The two expenditure shares. Both statistics use the same eligible households, service bills, and resources. With $s_{mc} = E_m/Y_{mc}$, define

$$A_c = \frac{\sum_m w_{mc} E_m}{\sum_m w_{mc} Y_{mc}} = \sum_m \frac{w_{mc} Y_{mc}}{\sum_a w_{ac} Y_{ac}} s_{mc}, \tag{A.21}$$

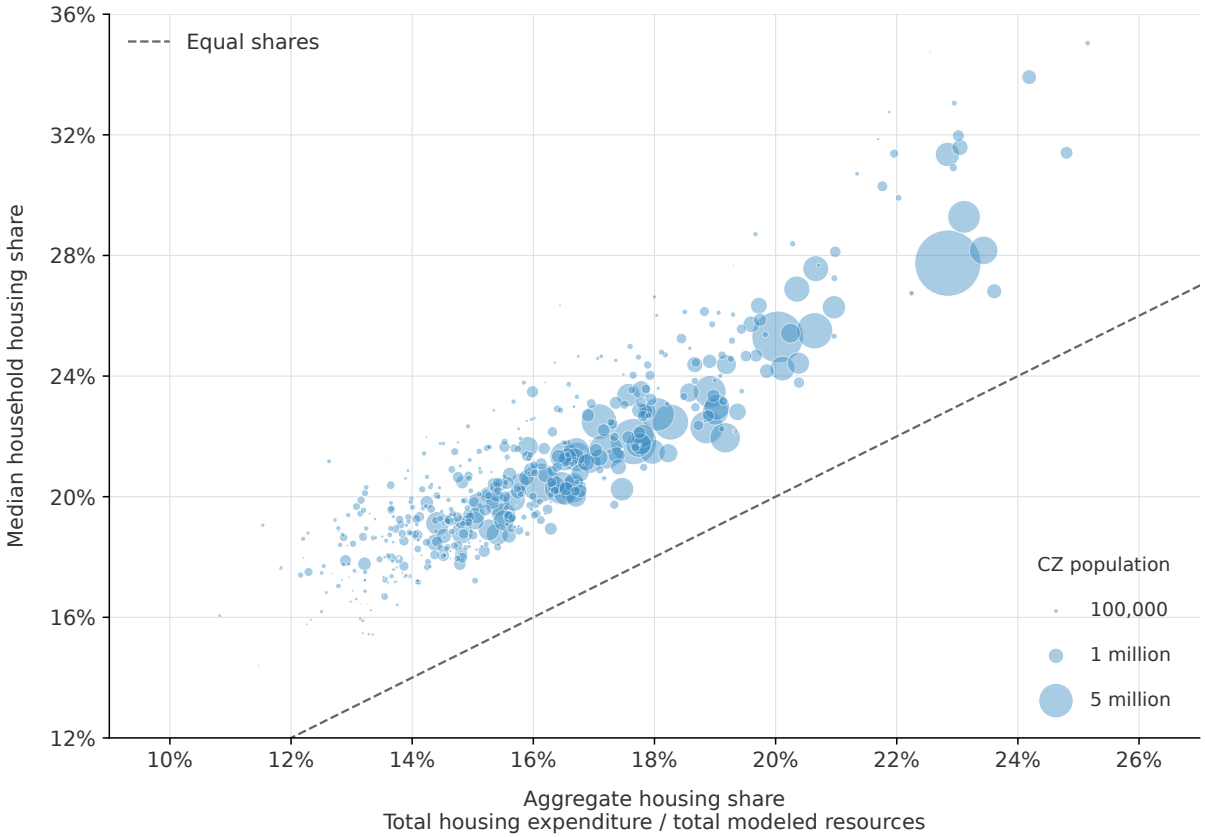
$$M_c = \inf \left\{ s : \frac{\sum_m w_{mc} \mathbf{1}\{s_{mc} \le s\}}{\sum_m w_{mc}} \ge \frac{1}{2} \right\}. \tag{A.22}$$

The aggregate A_c weights household shares by resources as well as survey weights; M_c is the household-weighted median of the pooled observations. The aggregate share describes housing's claim on total resources, while the median describes a typical household's expenditure burden. Greater resource weight on households with lower housing shares can therefore produce a lower aggregate share.

Across CZs, the median of A_c is 15.40 percent and the median of M_c is 20.20 percent. The median within-CZ gap is 4.70 percentage points, and the household median exceeds the aggregate in all 583 CZs. Each dot's area is proportional to average 2008–2017 population, obtained from all-age person records, including group quarters, with the same half-weight pooling and geographic allocation. The cross-CZ summaries give each CZ equal weight.

The comparison isolates the choice of expenditure-share statistic under common measurement. Both moments depend on the rental-service imputation and resource convention. The model targets the median household share; its baseline portfolio and housing quantities are determined jointly by the budget and market-clearing equations. A common Cobb–Douglas share generally cannot reproduce both the observed median and aggregate share. Fractional geographic assignment assumes that households in a PUMA represent its constituent CZ portions; small CZs sharing a PUMA can consequently have similar estimates. The plotted values are point estimates conditional on these imputations and geographic allocations.

Figure A.11: Aggregate and median household housing shares across U.S. commuting zones



Notes: ACS 2008–2012 and 2013–2017 five-year microdata, pooled with one half of each release's survey weights. The horizontal axis is total housing-service expenditure divided by total modeled household resources; the vertical axis is the household-weighted median expenditure share. Both statistics use identical eligible households and measurement rules. Housing services are observed contract rents or imputed rental equivalents. Resources include modeled income and housing-portfolio dividends. Dot area represents average all-age CZ population during 2008–2017. The dashed line denotes equal shares. All 583 CZs are shown; the axes have different ranges. PUMA-to-CZ locations are assigned using population weights.

B Model

This appendix derives individual demand, employment and location choices, firms' input demands, and the equilibrium accounts from the preferences and technologies in Section 4. A superscript 0 denotes the no-entry equilibrium. Appendix C describes the quantification and solution of these conditions.

B.1 Individual preferences

An individual ω of skill k chooses consumption and a joint residence and employment choice ℓ. For workers, $\ell = (i, j, g)$ specifies residence, workplace, and sector. For nonworkers it specifies residence and sector. The housing share is $\alpha_\ell = \alpha_H$ for $i \in \mathcal{J}$ and $\alpha_\ell = \overline{\alpha}_H^o$ for $i = o$. Write $\gamma_\ell = (1 - \alpha_\ell)s_N$

and $\beta_\ell = (1-\alpha_\ell)(1-s_N)$, so that $\alpha_\ell + \gamma_\ell + \beta_\ell = 1$. The utility in (3) can be written

$$U_\ell(\omega) = z_\ell(\omega)\mathcal{B}_\ell h^{\alpha_\ell} n^{\gamma_\ell} x^{\beta_\ell}, \qquad \mathcal{B}_\ell = \begin{cases} B^R_{ik} B^W_{jgk}/d_{ij}, & i,j \in \mathcal{J},\ e = W, \\ \overline{B}^{\text{out}}_{igk}, & i \in \mathcal{J},\ j = o,\ e = W, \\ \overline{B}^{\text{in}}_{jgk}, & i = o,\ j \in \mathcal{J},\ e = W, \\ \overline{B}^{\text{oo}}_{gk}, & i = j = o,\ e = W, \\ B^R_{ik} B^U_{ik}, & e = U. \end{cases} \tag{B.1}$$

Here $e = W$ denotes work and $e = U$ nonemployment. For internal commuters, $d_{ij} = \exp(\kappa t_{ij})$. The boundary amenity composites include fixed commuting costs and the attractiveness of access to outside jobs or residences. Each has at most one location index within the CZ; they preserve the separately calibrated inbound and outbound flows and the outside worker stock. Their construction is given in Appendix C.6.1. Internal commuting retains separate residence and workplace factors, with workplace amenities varying by sector. Housing and the two nonhousing goods have constant expenditure shares. After optimizing consumption and absorbing fixed expenditure-share constants into participation preference parameters, the systematic value is $\Phi_\ell = \mathcal{B}_\ell y_\ell / [(q^R_i)^{\alpha_\ell} \mathcal{P}^{\gamma_\ell}_\ell]$. The next subsection derives the employment and location probabilities; Appendix B.3 derives income, consumption demands, and this normalization.

B.2 Nested tastes and employment and location choices

Individuals choose jointly among residence and employment choices. The nested Fréchet distribution allows their tastes to be more similar among location pairs than across participation choices, sectors, and areas. Draws are independent across individuals and are held fixed across equilibria. For a nest v containing choices $a \in \mathcal{C}(v)$, let $\overline{B}_{va,k}$ be preference parameters and θ_v its shape parameter. The value index and conditional choice probability are

$$V_{vk} = \left[\sum_{a \in \mathcal{C}(v)} \overline{B}_{va,k} V^{\theta_v}_{ak} \right]^{1/\theta_v}, \tag{B.2}$$

$$\pi_{a|v,k} = \frac{\overline{B}_{va,k} V^{\theta_v}_{ak}}{V^{\theta_v}_{vk}}. \tag{B.3}$$

For a specific residence and employment choice, $V_{ak} = \Phi_\ell$ and $\overline{B}_{va,k} = 1$: fixed location attractiveness is contained in the amenities in Φ_ℓ. The $\overline{B}$'s describe preferences over broader choices, whose values summarize the residence and employment opportunities they include. A nest with one potential choice has a preference parameter equal to one. The order of the region and sector nests differs by skill. Low-skill individuals choose their region before their sector, and high-skill individuals choose their sector before their region, so the shape parameters satisfy

$$0 < \mu_L \leq \nu \leq \eta \leq \psi \leq \chi, \qquad 0 < \nu \leq \mu_H \leq \eta \leq \psi \leq \chi. \tag{B.4}$$

The migration parameter μ_k differs by skill, while the sector and participation parameters ν and η are common across skills and CZs. In the calibration, $\mu_L = 1.51$, $\nu = 1.64$, $\mu_H = 2.32$, and $\eta = 3.32$, so both orderings hold. The residence–workplace parameter χ is estimated from commuting gravity for each CZ, and ψ varies across events, so the ordering holds event by event. Their calibration is reported in Appendix C.1.

B.2.1 Value indices and conditional shares

The region consists of two areas, the CZ and the accessible outside area. Let $r \in \{I, O\}$ denote the area of residence, so that $r = I$ for CZ residents and $r = O$ for outside-area residents. Let $g \in \mathcal{G} = \mathcal{G}_T \cup \mathcal{G}_N$ be a sector, with $\mathcal{G}_T = \{T_H, T_L\}$ and $\mathcal{G}_N = \{N_H, N_L\}$ (for a nonworker, the sector in which they would work), and $e \in \{W, U\}$ work or nonemployment. A location is a CZ tract or the outside area, and a location pair ℓ is a residence and workplace location, or a residence location for a nonworker. CZ residents choose directly among location pairs with workplaces inside and outside the CZ. Outside-area residents who work first choose a job location $s \in \{I, O\}$, inside or outside the CZ, and then a location pair within it. Each nest's index equals the value index in (B.2) raised to the nest's shape parameter. The working, nonworking, sector, and area indices are defined in (B.8) and (B.9) below, and applying (B.3) at each margin gives the conditional shares that follow them. For low-skill individuals, residence in the region and in the rest of the country form the top nest,

$$(V_L^{\text{all}})^{\mu_L} = \overline{B}_L^{\text{nat}} (V_L^{\text{nat}})^{\mu_L} + \overline{B}_L^{\text{reg}} (V_L^{\text{reg}})^{\mu_L}, \qquad (V_L^{\text{reg}})^{\mu_L} = \sum_{r \in \{I,O\}} \overline{B}_{rL}^{R} \mathcal{Q}_{rL}^{\mu_L/\nu}, \tag{B.5}$$

where the national value V_L^{nat} is fixed for national population $\mathcal{N}_L$. Because migration and the choice between the two areas share μ_L, the choice to live in the region and the choice of area combine into one upper nest. High-skill individuals instead choose a sector at the top and then a region within it; the paragraph on the high-skill ordering in Appendix B.2.2 gives their indices and allocations.

For a group of location pairs $\mathcal{D}$ with shape χ,

$$\pi_{\ell|\mathcal{D},k} = \frac{\Phi_\ell^\chi}{\sum_{a \in \mathcal{D}} \Phi_a^\chi}. \tag{B.6}$$

For outside-area workers, multiplying this location-pair share by the job-location share yields (B.21). CZ residents and nonworkers have no job-location choice, and nonworkers choose among residence locations with shape χ.

B.2.2 Expanded utilities and population allocations

This subsection writes the indices and allocations explicitly for both areas and then derives the CZ resident notation used in the main text. The preference parameters are $\overline{B}_{rk}^R$ for area r, $\overline{B}_{rgk}^S$ for sector g, $\overline{B}_{rgk}^E$ and $\overline{B}_{rgk}^U$ for work and nonemployment, and $\overline{B}_{gsk}^O$ for job location s, a choice only

outside-area residents make. Write $\mathcal{L}^W_{Igk}$ for the working location pairs available to CZ residents, $\mathcal{L}^W_{Ogsk}$ for those available to outside-area residents with job location s, with $\mathcal{L}^W_{Ogk} = \mathcal{L}^W_{OgIk} \cup \mathcal{L}^W_{OgOk}$, and $\mathcal{L}^U_{rgk}$ for the nonworking residence locations. Let i_ℓ denote the residence location in pair ℓ and, for workers, j_ℓ its workplace location. The residence location determines the area, with $i_\ell \in \mathcal{I}_I = \mathcal{J}$ for CZ residents and $i_\ell \in \mathcal{I}_O = \{o\}$ for outside-area residents. Nonworking choices retain sector g, but their systematic utility and conditional residence probabilities are identical across sectors.

Working and nonworking utility. Using the housing share α_ℓ defined in Appendix B.1, systematic utility is

$$\Phi_\ell = \begin{cases} \dfrac{\mathcal{B}_\ell y_\ell}{(q^R_{i_\ell})^{\alpha_\ell}(P^\rho_{i_\ell} P^{1-\rho}_{j_\ell})^{(1-\alpha_\ell)s_N}}, & \text{if working,} \\ \dfrac{\mathcal{B}_\ell y_\ell}{(q^R_{i_\ell})^{\alpha_\ell} P^{(1-\alpha_\ell)s_N}_{i_\ell}}, & \text{if nonemployed.} \end{cases} \tag{B.7}$$

Disposable income y_ℓ is given by (B.37). Outside income and prices are fixed except that an outside resident working inside the CZ receives the endogenous wage at that workplace. The values use the expenditure-share normalization derived in Appendix B.3.

Working, nonworking, and sector indices. The index $\mathcal{W}_{Igk}$ summarizes CZ residents' working location pairs and $\mathcal{W}_{Ogsk}$ those of outside-area residents with job location s. The index $\widetilde{\mathcal{W}}_{Ogk}$ combines the two job locations, and $\mathcal{U}_{rgk}$ summarizes nonworking residence locations,

$$\begin{aligned} \mathcal{W}_{Igk} &= \sum_{\ell \in \mathcal{L}^W_{Igk}} \Phi^\chi_\ell, & \mathcal{W}_{Ogsk} &= \sum_{\ell \in \mathcal{L}^W_{Ogsk}} \Phi^\chi_\ell, \\ \widetilde{\mathcal{W}}_{Ogk} &= \sum_{s \in \{I,O\}} \overline{B}^O_{gsk} \mathcal{W}^{\psi/\chi}_{Ogsk}, & \mathcal{U}_{rgk} &= \sum_{\ell \in \mathcal{L}^U_{rgk}} \Phi^\chi_\ell, \end{aligned} \tag{B.8}$$

with Φ_ℓ from (B.7). Working and nonworking opportunities combine within sector, and sectors combine within each area,

$$\begin{aligned} \mathcal{S}_{Igk} &= \overline{B}^E_{Igk} \mathcal{W}^{\eta/\chi}_{Igk} + \overline{B}^U_{Igk} \mathcal{U}^{\eta/\chi}_{Igk}, \\ \mathcal{S}_{Ogk} &= \overline{B}^E_{Ogk} \widetilde{\mathcal{W}}^{\eta/\psi}_{Ogk} + \overline{B}^U_{Ogk} \mathcal{U}^{\eta/\chi}_{Ogk}, \\ \mathcal{Q}_{rk} &= \sum_{g \in \mathcal{G}} \overline{B}^S_{rgk} \mathcal{S}^{\nu/\eta}_{rgk}. \end{aligned} \tag{B.9}$$

Because CZ residents have no job-location choice, ψ does not enter their indices.

CZ resident notation. Dropping the area subscript for CZ residents gives the main-text notation,

$$\begin{aligned} \mathcal{W}_{gk} &= \mathcal{W}_{Igk}, & \mathcal{U}_{gk} &= \mathcal{U}_{Igk}, & \mathcal{S}_{gk} &= \mathcal{S}_{Igk}, \\ \overline{B}^{E}_{gk} &= \overline{B}^{E}_{Igk}, & \overline{B}^{U}_{gk} &= \overline{B}^{U}_{Igk}, & \overline{B}^{S}_{gk} &= \overline{B}^{S}_{Igk}, \\ N_k &= N_{Ik}, & N_{gk} &= N_{Igk}, & E_{gk} &= E_{Igk}, \end{aligned} \tag{B.10}$$

with choice sets $\mathcal{L}^{W}_{gk} = \mathcal{L}^{W}_{Igk}$ and $\mathcal{L}^{U}_{gk} = \mathcal{L}^{U}_{Igk}$. The first line of (B.9) is then the main-text sector index (7).

Regional migration and residence. This paragraph and the next describe the low-skill ordering, so $k = L$ and the migration parameter is μ_L. The fixed national contribution is $\overline{\mathcal{O}}_k = \overline{B}^{\text{nat}}_k (V^{\text{nat}}_k)^{\mu_L}$, and V^{reg}_k is given in (B.5). The regional share and the share living in each area are

$$\begin{aligned} b^{\text{reg}}_k &= \frac{\overline{B}^{\text{reg}}_k \sum_{r\in\{I,O\}} \overline{B}^{R}_{rk} \mathcal{Q}^{\mu_L/\nu}_{rk}}{\overline{\mathcal{O}}_k + \overline{B}^{\text{reg}}_k \sum_{r\in\{I,O\}} \overline{B}^{R}_{rk} \mathcal{Q}^{\mu_L/\nu}_{rk}}, \\ \pi(r \mid \text{reg}, k) &= \frac{\overline{B}^{R}_{rk} \mathcal{Q}^{\mu_L/\nu}_{rk}}{\sum_{r'\in\{I,O\}} \overline{B}^{R}_{r'k} \mathcal{Q}^{\mu_L/\nu}_{r'k}}. \end{aligned} \tag{B.11}$$

Multiplying these shares gives

$$\begin{aligned} N_{rk} &= \mathcal{N}_k b^{\text{reg}}_k \pi(r \mid \text{reg}, k) = \mathcal{N}_k \frac{\overline{B}^{\text{reg}}_k \overline{B}^{R}_{rk} \mathcal{Q}^{\mu_L/\nu}_{rk}}{\overline{\mathcal{O}}_k + \overline{B}^{\text{reg}}_k \sum_{r'\in\{I,O\}} \overline{B}^{R}_{r'k} \mathcal{Q}^{\mu_L/\nu}_{r'k}}, \\ N^{\text{reg}}_k &= \sum_r N_{rk} = \mathcal{N}_k b^{\text{reg}}_k, \qquad N^{\text{nat}}_k = \mathcal{N}_k (1 - b^{\text{reg}}_k). \end{aligned} \tag{B.12}$$

Both migration and the choice between the two areas use μ_L. For the main-text equation, define

$$\begin{aligned} \overline{B}^{R}_{L} &= \overline{B}^{\text{reg}}_{L} \overline{B}^{R}_{IL}, \\ \mathcal{O}_L &= \overline{\mathcal{O}}_L + \overline{B}^{\text{reg}}_{L} \overline{B}^{R}_{OL} \mathcal{Q}^{\mu_L/\nu}_{OL}. \end{aligned} \tag{B.13}$$

The term $\mathcal{O}_L$ includes both the fixed national contribution and living in the outside area. It is endogenous because outside-area residents can work at CZ workplaces. Setting $r = I$ in (B.12) and substituting $\mathcal{Q}_{IL} = \mathcal{Q}_L = \sum_g \overline{B}^{S}_{gL} \mathcal{S}^{\nu/\eta}_{gL}$ gives the CZ's low-skill population N_L.

Sector allocation. The share of area-r residents choosing sector g is $\overline{B}^{S}_{rgk} \mathcal{S}^{\nu/\eta}_{rgk} / \mathcal{Q}_{rk}$, so

$$N_{rgk} = N_{rk} \frac{\overline{B}^{S}_{rgk} \mathcal{S}^{\nu/\eta}_{rgk}}{\mathcal{Q}_{rk}}. \tag{B.14}$$

For $r = I$, this gives the low-skill case of (8). The allocation includes both employed and nonemployed individuals. Nonemployment income, prices, and amenities are identical across sectors at each residence location. The baseline allocation and its normalization are derived below.

High-skill ordering. High-skill individuals choose a sector first and then a region within it. Let $\overline{B}^S_{gH}$ be the preference parameter for sector g, common to both areas, $\overline{B}^{\text{nat}}_{gH}$ and $\overline{B}^{\text{reg}}_{gH}$ those for living elsewhere in the country or in the region within sector g, and $\overline{B}^R_{rgH}$ that for area r. The value of sector g combines the fixed national value of that sector elsewhere in the country, V^{nat}_{gH}, with the two areas,

$$\begin{aligned} \mathcal{V}_{gH} &= \overline{\mathcal{O}}_{gH} + \overline{B}^{\text{reg}}_{gH} \sum_{r\in\{I,O\}} \overline{B}^R_{rgH} \mathcal{S}^{\mu_H/\eta}_{rgH}, \qquad & \overline{\mathcal{O}}_{gH} &= \overline{B}^{\text{nat}}_{gH} (V^{\text{nat}}_{gH})^{\mu_H}, \\ (V^{\text{all}}_H)^{\nu} &= \sum_{g\in\mathcal{G}} \overline{B}^S_{gH} \mathcal{V}^{\nu/\mu_H}_{gH}, \end{aligned} \tag{B.15}$$

with $\mathcal{S}_{rgH}$ from (B.9). The sector share, the regional share within the sector, and the area share are

$$\begin{aligned} \pi(g \mid H) &= \frac{\overline{B}^S_{gH} \mathcal{V}^{\nu/\mu_H}_{gH}}{\sum_{g'\in\mathcal{G}} \overline{B}^S_{g'H} \mathcal{V}^{\nu/\mu_H}_{g'H}}, \\ b^{\text{reg}}_{gH} &= \frac{\overline{B}^{\text{reg}}_{gH} \sum_r \overline{B}^R_{rgH} \mathcal{S}^{\mu_H/\eta}_{rgH}}{\mathcal{V}_{gH}}, \qquad & \pi(r \mid \text{reg}, g, H) &= \frac{\overline{B}^R_{rgH} \mathcal{S}^{\mu_H/\eta}_{rgH}}{\sum_{r'} \overline{B}^R_{r'gH} \mathcal{S}^{\mu_H/\eta}_{r'gH}}, \end{aligned} \tag{B.16}$$

so the numbers by area and sector are

$$\begin{aligned} N_{rgH} &= \mathcal{N}_H \, \pi(g \mid H) \, b^{\text{reg}}_{gH} \, \pi(r \mid \text{reg}, g, H) = \mathcal{N}_H \, \pi(g \mid H) \, \frac{\overline{B}^{\text{reg}}_{gH} \overline{B}^R_{rgH} \mathcal{S}^{\mu_H/\eta}_{rgH}}{\mathcal{V}_{gH}}, \\ N_{rH} &= \sum_{g\in\mathcal{G}} N_{rgH}, \qquad N^{\text{nat}}_H = \mathcal{N}_H \sum_{g\in\mathcal{G}} \pi(g \mid H)\big(1 - b^{\text{reg}}_{gH}\big). \end{aligned} \tag{B.17}$$

For the main-text equation, define

$$\overline{B}^R_{gH} = \overline{B}^{\text{reg}}_{gH} \overline{B}^R_{IgH}, \qquad \mathcal{O}_{gH} = \overline{\mathcal{O}}_{gH} + \overline{B}^{\text{reg}}_{gH} \overline{B}^R_{OgH} \mathcal{S}^{\mu_H/\eta}_{OgH}, \tag{B.18}$$

so that $\mathcal{V}_{gH} = \mathcal{O}_{gH} + \overline{B}^R_{gH} \mathcal{S}^{\mu_H/\eta}_{gH}$. Setting $r = I$ in (B.17) gives the high-skill case of (8). As for low-skill individuals, $\mathcal{O}_{gH}$ is endogenous because outside-area residents can work at CZ workplaces. Each sector has its own national option, and within sector g the migration elasticity is $\mu_H(1 - b^{\text{reg}}_{gH})$.

Labor-force participation. The employed share of those choosing sector g is the working term of $\mathcal{S}_{rgk}$ relative to its total,

$$\frac{E_{Igk}}{N_{Igk}} = \frac{\overline{B}^{E}_{Igk}\mathcal{W}^{\eta/\chi}_{Igk}}{\mathcal{S}_{Igk}}, \qquad \frac{E_{Ogk}}{N_{Ogk}} = \frac{\overline{B}^{E}_{Ogk}\widetilde{\mathcal{W}}^{\eta/\psi}_{Ogk}}{\mathcal{S}_{Ogk}}. \tag{B.19}$$

The first gives (10). The nonemployed population in sector g is $N_{rgk} - E_{rgk}$.

Job location and location pairs. An outside-area resident who works in sector g chooses job location s and then a location pair $\ell \in \mathcal{L}^{W}_{Ogsk}$, with shares

$$\pi(s \mid O, g, W, k) = \frac{\overline{B}^{O}_{gsk}\mathcal{W}^{\psi/\chi}_{Ogsk}}{\widetilde{\mathcal{W}}_{Ogk}}, \qquad \pi_{\ell|O,g,W,s,k} = \frac{\Phi^{\chi}_{\ell}}{\mathcal{W}_{Ogsk}}. \tag{B.20}$$

A CZ resident chooses a location pair directly. The conditional location-pair probabilities of workers are therefore

$$\pi_{\ell|I,g,W,k} = \frac{\Phi^{\chi}_{\ell}}{\mathcal{W}_{Igk}}, \qquad \pi_{\ell|O,g,W,k} = \frac{\overline{B}^{O}_{gsk}\mathcal{W}^{\psi/\chi}_{Ogsk}}{\widetilde{\mathcal{W}}_{Ogk}}\frac{\Phi^{\chi}_{\ell}}{\mathcal{W}_{Ogsk}}, \tag{B.21}$$

and the main-text probability abbreviates $\pi_{\ell k} = \pi_{\ell|I,g,W,k}$. Multiplying the area–sector population N_{rgk}, from (B.14) for low-skill and (B.17) for high-skill individuals, by the employment and location shares gives the working population, for CZ residents and outside-area residents respectively,

$$\begin{aligned} F_{ijgk} &= N_{Igk}\underbrace{\frac{\overline{B}^{E}_{Igk}\mathcal{W}^{\eta/\chi}_{Igk}}{\mathcal{S}_{Igk}}}_{\text{work}}\underbrace{\frac{\Phi^{\chi}_{\ell}}{\mathcal{W}_{Igk}}}_{\text{location pair}}, \qquad i \in \mathcal{J}, \\ F_{ojgk} &= N_{Ogk}\underbrace{\frac{\overline{B}^{E}_{Ogk}\widetilde{\mathcal{W}}^{\eta/\psi}_{Ogk}}{\mathcal{S}_{Ogk}}}_{\text{work}} \\ &\quad \times \underbrace{\frac{\overline{B}^{O}_{gsk}\mathcal{W}^{\psi/\chi}_{Ogsk}}{\widetilde{\mathcal{W}}_{Ogk}}}_{\text{job location}}\underbrace{\frac{\Phi^{\chi}_{\ell}}{\mathcal{W}_{Ogsk}}}_{\text{location pair}}. \end{aligned} \tag{B.22}$$

The first line gives (11). Substituting the area–sector populations gives the unconditional number choosing each pair.

Nonemployment and population accounting. For a nonworking choice $\ell \in \mathcal{L}^U_{rgk}$ with residence location i and sector g, the nonemployed population is

$$U_{igk} = (N_{rgk} - E_{rgk})\frac{\Phi^\chi_\ell}{\mathcal{U}_{rgk}}, \qquad U_{ik} = \sum_{g\in\mathcal{G}} U_{igk}. \tag{B.23}$$

An unavailable nonworking choice has $U_{igk} = 0$. Thus $\mathcal{U}_{rgk}$ is a utility index, while U_{igk} and U_{ik} count individuals. The allocations satisfy

$$\begin{aligned} \sum_g N_{rgk} &= N_{rk}, & \sum_{\ell\in\mathcal{L}^W_{rgk}} F_{i_\ell j_\ell gk} &= E_{rgk}, \\ \sum_{i\in\mathcal{I}_r} U_{ik} &= \sum_g (N_{rgk} - E_{rgk}), & \sum_r N_{rk} + N^{\text{nat}}_k &= \mathcal{N}_k. \end{aligned} \tag{B.24}$$

Conditional shares apply to nests with positive value; an empty nest has zero population. Write $\pi^{\text{all}}_{\ell k}$ for the unconditional probability, distinguished from the conditional location-pair probability $\pi_{\ell k}$ in the main text. For a regional choice,

$$\begin{aligned} \pi^{\text{all}}_{\ell L} &= b^{\text{reg}}_L \pi(r \mid \text{reg}, L)\pi(g \mid r, L)\pi(e \mid r, g, L)\pi_{\ell|r,g,e,L}, \\ \pi^{\text{all}}_{\ell H} &= \pi(g \mid H)\, b^{\text{reg}}_{gH} \pi(r \mid \text{reg}, g, H)\pi(e \mid r, g, H)\pi_{\ell|r,g,e,H}. \end{aligned}$$

For workers the last factor is (B.21); for nonworkers it is the residence-location share in (B.23). The national choice has probability $1 - b^{\text{reg}}_L$ for low-skill and $\sum_g \pi(g \mid H)(1 - b^{\text{reg}}_{gH})$ for high-skill individuals. For the full feasible choice set $\mathcal{A}_k$,

$$n_{\ell k} = \mathcal{N}_k \pi^{\text{all}}_{\ell k}, \qquad \sum_{\ell\in\mathcal{A}_k} \pi^{\text{all}}_{\ell k} = 1. \tag{B.25}$$

These probabilities determine equilibrium populations. Individual transitions across equilibria are derived in Appendix B.12.

B.2.3 Derivation from the taste distribution

For any positive vector of choice values x, let $\mathcal{V}_k(x)$ denote the all-location value obtained from the indices above, with each location-pair value Φ_ℓ replaced by x_ℓ and the national value replaced by x_{nat}. Define $G_k(x) = \mathcal{V}_k(x)^{\theta^{\text{all}}_k}$, where the shape of the top nest is $\theta^{\text{all}}_L = \mu_L$ and $\theta^{\text{all}}_H = \nu$. The nested Fréchet joint distribution in (5) is therefore

$$\Pr\{z_\ell \le t_\ell \text{ for every } \ell\} = \exp\left\{-\left[\mathcal{V}_k\big((t_\ell^{-1})_\ell\big)\right]^{\theta^{\text{all}}_k}\right\}. \tag{B.26}$$

The ordering in (B.4) ensures that this is a valid nested extreme-value distribution. Since each value index is homogeneous of degree one, maximized utility has distribution

$$\Pr\{\max_{\ell} z_\ell \Phi_\ell \le u\} = \exp\left[-\left(\frac{V_k^{\text{all}}}{u}\right)^{\theta_k^{\text{all}}}\right]. \tag{B.27}$$

Let $\partial_\ell G_k$ denote the derivative of G_k with respect to x_ℓ. The probability that ℓ maximizes utility integrates the density that its utility equals u while every other choice gives at most u:

$$\begin{aligned}\pi_{\ell k}^{\text{all}} &= \int_0^\infty \Phi_\ell \, \partial_\ell G_k(\Phi) \, u^{-\theta_k^{\text{all}}-1} \exp[-G_k(\Phi) u^{-\theta_k^{\text{all}}}] \, du \\ &= \frac{\Phi_\ell \, \partial_\ell G_k(\Phi)}{\theta_k^{\text{all}} G_k(\Phi)} = \frac{\partial \log V_k^{\text{all}}}{\partial \log \Phi_\ell}.\end{aligned} \tag{B.28}$$

At any nest,

$$\frac{\partial \log V_{vk}}{\partial \log V_{ak}} = \frac{\overline{B}_{va,k} V_{ak}^{\theta_v}}{\sum_{u \in \mathcal{C}(v)} \overline{B}_{vu,k} V_{uk}^{\theta_v}} = \pi_{a|v,k}. \tag{B.29}$$

The chain rule therefore gives the product of conditional shares in (B.25) and the preceding probability expression. For two choices a and d in the same nest, $\pi_{a|v,k}/\pi_{d|v,k} = (\overline{B}_{va,k}/\overline{B}_{vd,k})(V_{ak}/V_{dk})^{\theta_v}$. Hence θ_v is the elasticity of conditional odds with respect to relative value. Holding other values fixed, the own-share elasticity is $\theta_v(1 - \pi_{a|v,k})$. In particular, for low-skill individuals,

$$\frac{\partial \log N_L^{\text{reg}}}{\partial \log V_L^{\text{reg}}} = \mu_L(1 - b_L^{\text{reg}}), \tag{B.30}$$

and for high-skill individuals the corresponding elasticity within sector g is $\mu_H(1 - b_{gH}^{\text{reg}})$.

B.2.4 Baseline normalization and sectors

Let $\pi_{a|v,k}^0$ denote the model's baseline conditional probability. At the location-pair margin, it is predicted by the calibrated amenities, commuting costs, and baseline economic conditions:

$$\pi_{\ell|\mathcal{D},k}^0 = \frac{(\Phi_\ell^0)^\chi}{\sum_{m \in \mathcal{D}^0} (\Phi_m^0)^\chi}, \qquad \ell \in \mathcal{D}^0, \tag{B.31}$$

where $\mathcal{D}^0$ contains the choices available at baseline. Internal commuting probabilities come from fitted gravity flows, so an observed zero on a feasible link need not imply a zero probability. Broader conditional probabilities reproduce the corresponding calibrated population, sector, and participation totals. They are distinct from amenities and from preference parameters in levels.

At a broader nest, the preference parameters satisfy

$$\overline{B}_{va,k} = \pi_{a|v,k}^0 \left(\frac{V_{vk}^0}{V_{ak}^0}\right)^{\theta_v}. \tag{B.32}$$

At the location-pair margin, $\overline{B}_{va,k} = 1$ and this identity follows from (B.31). Writing $\widehat{V}_{ak} = V_{ak}/V^0_{ak}$ yields, on the baseline choice set,

$$\widehat{V}_{vk} = \left[\sum_a \pi^0_{a|v,k} \widehat{V}^{\theta_v}_{ak}\right]^{1/\theta_v}, \qquad \pi_{a|v,k} = \frac{\pi^0_{a|v,k} \widehat{V}^{\theta_v}_{ak}}{\sum_u \pi^0_{u|v,k} \widehat{V}^{\theta_v}_{uk}}. \tag{B.33}$$

In particular, when the feasible bilateral choice set is unchanged,

$$\pi^1_{\ell|\mathcal{D},k} = \frac{\pi^0_{\ell|\mathcal{D},k}(\Phi^1_\ell/\Phi^0_\ell)^\chi}{\sum_{m\in\mathcal{D}^0} \pi^0_{m|\mathcal{D},k}(\Phi^1_m/\Phi^0_m)^\chi}. \tag{B.34}$$

The fixed amenities and commuting costs cancel from the utility ratios and remain embodied in the baseline predicted probabilities. No additional bilateral preference parameter enters this update. Baseline counts and amenity calibration are described in Appendix C.6.1.

For area r, let E^0_{rgk} be employment in sector g, $E^0_{rk} = \sum_g E^0_{rgk}$, and U^0_{rk} nonemployment. The sector allocation of nonworkers is

$$a^0_{rgk} = E^0_{rgk}/E^0_{rk}, \qquad U^0_{igk} = a^0_{rgk} U^0_{ik}, \qquad i \in r. \tag{B.35}$$

All sectors at a nonworking residence have the same income, prices, and amenities. Summing over sectors recovers U^0_{ik}, and the baseline work probability within every sector is $\pi^0_{rk} = E^0_{rk}/(E^0_{rk}+U^0_{rk})$. With hats denoting ratios to baseline values of the indices in (B.8) and (B.9),

$$\widehat{\mathcal{S}}_{rgk} = \pi^0_{rk}\widehat{\widetilde{\mathcal{W}}}^{\eta/\psi}_{rgk} + (1-\pi^0_{rk})\widehat{\mathcal{U}}^{\eta/\chi}_{rgk}, \qquad \widehat{\mathcal{Q}}_{rk} = \sum_g a^0_{rgk}\widehat{\mathcal{S}}^{\nu/\eta}_{rgk},$$

$$\pi(g \mid r,k) = \frac{a^0_{rgk}\widehat{\mathcal{S}}^{\nu/\eta}_{rgk}}{\widehat{\mathcal{Q}}_{rk}}, \qquad \pi(W \mid r,g,k) = \frac{\pi^0_{rk}\widehat{\widetilde{\mathcal{W}}}^{\eta/\psi}_{rgk}}{\widehat{\mathcal{S}}_{rgk}}. \tag{B.36}$$

The parameter ν therefore governs sector choice, including the associated value of nonemployment; η governs work versus nonwork within a sector. An unconditional employed-sector share also reflects participation and location responses. When $\eta = \nu$, these two value indices combine into a single employment and sector index after aggregating the identical nonemployment choices. For low-skill individuals, the national choice is not divided across sectors. For high-skill individuals, the sector shares update in the same way with $\widehat{\mathcal{V}}_{gH}$, and each sector keeps its own fixed national option.

B.3 Disposable income and consumption demand

Disposable income is the wage or nonemployment income plus the residential dividend,

$$y_\ell = \begin{cases} w^g_{jk}, & \text{if working,} \\ y^U_k, & \text{if nonemployed,} \end{cases} + \mathbf{1}\{i \in \mathcal{J}\}D_k + \mathbf{1}\{i = o\}\overline{D}_{ok}. \tag{B.37}$$

Nonemployment income is externally financed. For CZ residents, this reduces to (12). Outside dividends are fixed. An outside resident employed inside the CZ receives that workplace's endogenous wage. A feasible consumption choice requires $y_\ell > 0$.

B.3.1 Expenditure shares and home and work shopping

A worker's service composite is $n = n_i^\rho n_j^{1-\rho}$. For fixed locations, the consumption problem is

$$\begin{aligned} \max_{h,n_i,n_j,x>0} \quad & \alpha_\ell \log h + \gamma_\ell \rho \log n_i + \gamma_\ell (1-\rho) \log n_j + \beta_\ell \log x, \\ \text{subject to} \quad & q_i^R h + P_i n_i + P_j n_j + x \le y_\ell. \end{aligned} \tag{B.38}$$

The amenity and taste terms are constant within this problem. The first-order conditions equate each good's expenditure to its utility exponent divided by the marginal utility of income. Summing these conditions and using the budget constraint shows that the marginal utility of income is $1/y_\ell$. Consequently,

$$\begin{aligned} q_i^R h_\ell &= \alpha_\ell y_\ell, & P_i n_i &= \rho \gamma_\ell y_\ell, \\ P_j n_j &= (1-\rho)\gamma_\ell y_\ell, & x_\ell &= \beta_\ell y_\ell. \end{aligned} \tag{B.39}$$

Housing demand is

$$h_\ell = \frac{\alpha_\ell y_\ell}{q_i^R}, \tag{B.40}$$

and spending on housing, services, and the tradable good is $\alpha_\ell y_\ell$, $\gamma_\ell y_\ell$, and $\beta_\ell y_\ell$. For CZ residents, $\alpha_\ell = \alpha_H$ and $\gamma_\ell = \gamma = (1-\alpha_H)s_N$, giving (13). A nonworker buys services only from home, so $P_i n_i = \gamma_\ell y_\ell$, with the same housing and tradable demands. Thus a dollar of additional disposable income raises service expenditure by γ_ℓ, allocated between home and work according to employment status.

B.3.2 Indirect utility and the expenditure-share normalization

Define $R_W = \rho^\rho (1-\rho)^{1-\rho}$ and $R_U = 1$, with endpoint values interpreted by continuity. Substitution of the demands into (B.1) gives

$$\begin{aligned} U_\ell^*(\omega) &= z_\ell(\omega) K_\ell \frac{\mathcal{B}_\ell y_\ell}{(q_i^R)^{\alpha_\ell} \mathcal{P}_\ell^{\gamma_\ell}}, & K_\ell &= \alpha_\ell^{\alpha_\ell} \gamma_\ell^{\gamma_\ell} \beta_\ell^{\beta_\ell} R_e^{\gamma_\ell}, \\ \mathcal{P}_\ell &= \begin{cases} P_i^\rho P_j^{1-\rho}, & e = W, \\ P_i, & e = U. \end{cases} \end{aligned} \tag{B.41}$$

The housing share, nonhousing shares, and shopping allocation are fixed within each entry comparison. The constant K_ℓ depends only on area r and employment status e, so write it as K_{re}. It is common across location pairs within each such group, although it can differ between workers

and nonworkers and across areas. Homogeneity of the working and nonworking indices allows these constants to be absorbed into the participation shifters:

$$\overline{B}^{E}_{rgk} = \overline{B}^{E,\text{raw}}_{rgk} K^{\eta}_{rW}, \qquad \overline{B}^{U}_{rgk} = \overline{B}^{U,\text{raw}}_{rgk} K^{\eta}_{rU}. \tag{B.42}$$

Thus the normalized systematic value in (4) produces the same probabilities and inclusive values as the raw indirect utility when the participation shifters are expressed in the corresponding units. This fixed taste normalization preserves the original Cobb–Douglas preferences and expenditure shares without a bilateral preference parameter. Each shopping counterfactual exercise uses the same normalization in its entry and no-entry equilibria.

B.4 Population, employment, and labor supply

Multiplying the conditional shares yields a regional choice probability $\pi^{\text{all}}_{\ell k} = b^{\text{reg}}_{k} \pi(r \mid \text{reg}, k)\pi(g \mid r, k)\pi(e \mid r, g, k)\pi_{\ell|r,g,e,k}$. For outside-area workers, the last probability includes the job-location choice. The national share is $1 - b^{\text{reg}}_{k}$. Conditional shares sum to one at every margin, so $\sum_{\ell} \pi^{\text{all}}_{\ell k} = 1$ and $n_{\ell k} = \mathcal{N}_k \pi^{\text{all}}_{\ell k}$ exhausts the fixed national population.

Let F_{ijgk} denote working counts and U_{igk} nonworking counts by sector. The relevant employment and population totals are

$$\begin{aligned} U_{ik} &= \sum_{g} U_{igk}, & L^{g}_{jk} &= \sum_{i} F_{ijgk}, \\ N_k &= \sum_{i \in \mathcal{J}} \left(\sum_{j,g} F_{ijgk} + U_{ik} \right), & N &= \sum_{k} N_k. \end{aligned} \tag{B.43}$$

Workplace employment includes inbound commuters; resident employment includes outbound commuters. These sums distinguish changes in local jobs from changes in residents' employment. In particular, entry can raise workplace employment through commuting without changing the number of employed CZ residents.

B.5 Individual values and choice changes

Let $y_{\ell k}$ be disposable income in choice ℓ and $q_i = (1+t) r^{R}_{i}$ the residential rent for a local resident. Housing has the Cobb–Douglas expenditure share α_ℓ. For an employed individual living in i and working in j, the change in the systematic utility is

$$d_{\ell k} = \log \frac{y_{\ell k}}{y^{0}_{\ell k}} - \alpha_\ell \log \frac{q_i}{q^{0}_{i}} - (1 - \alpha_\ell) s_N \left[\rho \log \frac{P_i}{P^{0}_{i}} + (1-\rho) \log \frac{P_j}{P^{0}_{j}} \right]. \tag{B.44}$$

Income includes the current skill dividend alongside wages or nonemployment income. For a nonworker the shopping term is $\log(P_i / P^{0}_{i})$. Outside wages, nonemployment income, dividends, and prices are fixed. An outside resident employed inside the CZ receives the endogenous local wage.

Fixed amenities and commuting costs remain embodied in the baseline predicted probabilities.

For each bilateral choice, $d_{\ell k} = \log(\Phi_\ell / \Phi_\ell^0)$.

Set $v_{\ell k} = d_{\ell k}$. Each nest v has log inclusive-value change and conditional probabilities

$$v_{vk} = \frac{1}{\theta_v} \log \sum_{a \in \mathcal{C}(v)} \pi^0_{a|v,k} \exp(\theta_v v_{ak}), \tag{B.45}$$

$$\pi_{a|v,k} = \pi^0_{a|v,k} \exp\{\theta_v (v_{ak} - v_{vk})\}. \tag{B.46}$$

Multiplying the conditional probabilities along the nests containing a choice, and then by $\mathcal{N}_k$, gives its population. The baseline probabilities carry the calibrated amenities and commuting costs, so no additional amenity parameters enter. Unavailable choices have $v_{ak} = -\infty$. A nest with no feasible choices has zero value in levels, log value $-\infty$, and zero choice probability. A choice with nonpositive disposable income is unavailable. The baseline calibration requires positive income in every choice with positive baseline mass.

For two members u and z of the same nest v with positive shares in both equilibria,

$$\log \frac{\pi_{u|v,k}}{\pi_{z|v,k}} - \log \frac{\pi^0_{u|v,k}}{\pi^0_{z|v,k}} = \theta_v (v_{uk} - v_{zk}). \tag{B.47}$$

Thus a scale is a conditional log-odds response to the relative value. For a binary choice with share π, the elasticity of that share to its own value is $\theta_v(1-\pi)$. This distinction applies to participation and inward commuting as well as regional attachment. The two migration choices, between the region and the rest of the country and between residence inside and outside the CZ, share μ and therefore combine into a single nest.

B.6 Service demand and shopping price indices

The service bundle combines varieties from the two service sectors $g \in \mathcal{G}_N = \{N_H, N_L\}$. These sectors differ in skill intensity; each can employ both worker skills k. Appendix C.5.1 gives their sector classifications. Firms form a continuum, each supplying one differentiated variety under monopolistic competition with free entry. The mass m_{jg} is a continuous measure of the number of active firms in sector g and tract j, each charging p_{jg}. All varieties enter the same CES bundle with elasticity $\sigma_N > 1$, so the sector grouping introduces no additional consumption nest. Sector prices and establishment masses are determined separately by production costs and free entry.

Sector-specific shopping demand. Shopping and commuting share the per-minute travel-cost parameter κ. The shopping preference parameters satisfy $\zeta_{ijg} = \zeta_{ij}\xi_{jg}$. The bilateral shifter ζ_{ij} is common to both industries, while ξ_{jg} measures destination–industry appeal. Thus $T_{ijg} = \zeta_{ij}\xi_{jg} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}$.

Each delivered variety price is $p_{jg} \exp(\kappa t_{ij})$; summing the dual price terms over varieties gives $T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N}$, while fixed outside opportunities contribute T_i^O. CES expenditure minimization

at origin i therefore gives

$$P_i = \left[T_i^O + \sum_{j \in \mathcal{J}} \sum_{g \in \mathcal{G}_N} T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N} \right]^{1/(1-\sigma_N)}. \tag{B.48}$$

Shephard's lemma gives the expenditure share received by each industry and destination:

$$\frac{\partial \log P_i}{\partial \log p_{jg}} = \frac{T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N}}{P_i^{1-\sigma_N}} = \pi_{ijg}^I. \tag{B.49}$$

Individuals can also purchase services outside the CZ. The term T_i^O summarizes the contribution of outside varieties, prices, shopping preferences, and travel costs to the service price index at origin i. These outside opportunities are held fixed across equilibria. The outside shopping share and the expenditure adding-up condition are

$$\pi_i^O = \frac{T_i^O}{P_i^{1-\sigma_N}}, \qquad \sum_{j \in \mathcal{J}} \sum_{g \in \mathcal{G}_N} \pi_{ijg}^I + \pi_i^O = 1. \tag{B.50}$$

Although T_i^O is fixed, the outside expenditure share responds to local prices and entry through P_i. Lower local prices or additional local varieties reduce the share spent outside. Expenditure $\mathcal{E}_i \pi_i^O$ accrues to businesses outside the CZ and does not enter local nontradable revenue.

A worker living in i and employed at j allocates $\rho \gamma_\ell y_{ijgk}$ to purchases from home and $(1-\rho)\gamma_\ell y_{ijgk}$ to purchases from work. Summing the first budget over F_{ijgk} and the second over F_{jigk} gives workers' spending from origin i. Nonemployment income and expenditure shares are identical across sectors at a given residence and skill, so nonworkers contribute $\gamma U_{ik}(y_k^U + D_k)$ after summing their counts over sectors. These three contributions give (14). The sums include feasible residence and workplace locations inside and outside the CZ. Inbound commuters contribute spending from work and outbound commuters contribute spending from home. Spending originating in the outside location accrues outside the CZ. Revenue at local destinations equals the spending directed to them from CZ origins.

Aggregation by tract. The main-text shopping equations combine industries within each tract. For a tract with positive variety mass, define total varieties and the CES price index per variety by

$$\begin{aligned} m_j &= \sum_{g \in \mathcal{G}_N} m_{jg}, \\ p_j^N &= \left[\sum_{g \in \mathcal{G}_N} \xi_{jg} \frac{m_{jg}}{m_j} (p_{jg})^{1-\sigma_N} \right]^{1/(1-\sigma_N)}. \end{aligned} \tag{B.51}$$

Let $T_{ij} = \zeta_{ij} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}$. Since $T_{ijg} = T_{ij}\xi_{jg}$, these definitions imply

$$T_{ij} m_j (p_j^N)^{1-\sigma_N} = \sum_{g \in \mathcal{G}_N} T_{ijg} m_{jg} (p_{jg})^{1-\sigma_N}. \tag{B.52}$$

Substituting into (B.48) gives (15). A tract with no varieties contributes zero to the price-index sum. Expenditure shares and revenue aggregate as

$$\pi_{ij}^I = \sum_{g \in \mathcal{G}_N} \pi_{ijg}^I, \qquad R_j^N = \sum_{g \in \mathcal{G}_N} R_{jg}. \tag{B.53}$$

Summing expenditure shares gives (16). Total nontradable revenue in tract j is

$$R_j^N = \sum_{i \in \mathcal{J}} \mathcal{E}_i \pi_{ij}^I. \tag{B.54}$$

All service revenue comes from the spending of individuals in the model. The tract price index reflects both industry prices and the composition of available varieties; the industry prices and entry responses remain distinct.

Changes in shopping shares and revenue. Define destination attraction and the origin denominator in changes by

$$\Omega_{jg} = \widehat{m}_{jg} (\widehat{p}_{jg})^{1-\sigma_N}, \qquad D_i = \sum_j \sum_{g \in \mathcal{G}_N} \pi_{ijg}^{I,0} \Omega_{jg} + \pi_i^{O,0}, \tag{B.55}$$

$$\pi_{ijg}^I = \frac{\pi_{ijg}^{I,0} \Omega_{jg}}{D_i}, \qquad \pi_i^O = \frac{\pi_i^{O,0}}{D_i}, \qquad \widehat{P}_i = D_i^{1/(1-\sigma_N)}. \tag{B.56}$$

Individual NT budgets follow the Cobb–Douglas demands and are assigned to home and work origins using ρ. Nonworkers shop from home. Revenue in each industry and tract is

$$R_{jg} = \sum_{i \in \mathcal{J}} \mathcal{E}_i \pi_{ijg}^I, \tag{B.57}$$

where spending $\mathcal{E}_i$ includes each individual's income and dividends.

B.7 Nontradable production, costs, and entry

The model contains two service sectors, $g \in \mathcal{G}_N = \{N_H, N_L\}$. They differ in skill intensity, but each employs both worker skills k. Productivity, skill shares, and fixed input requirements may differ across tracts and sectors. Wages, prices, and entry are determined separately for each sector. The parameters ϵ_m and σ_N are common to both, and the quantification gives the two sectors the same cost shares.

B.7.1 Production and unit cost

A firm in sector $g \in \mathcal{G}_N$ and tract j produces output Y from labor $\mathcal{L}$, commercial space H, and intermediate inputs X, omitting firm subscripts,

$$Y = A_{jg}\, \mathcal{L}^{\alpha_L^g} H^{\alpha_C^g} X^{\alpha_X^g}, \qquad \alpha_L^g + \alpha_C^g + \alpha_X^g = 1. \tag{B.58}$$

Here A_{jg} is productivity and α_L^g, α_C^g, and α_X^g are the expenditure shares on labor, commercial space, and intermediates. If L_k is the firm's employment of skill k, the two skills form a Cobb–Douglas labor bundle with fixed expenditure shares λ_{jgk}. Minimizing the wage bill required to produce one unit of this bundle gives its unit cost:

$$\mathcal{L} = \prod_k L_k^{\lambda_{jgk}}, \qquad c_{jg}^L = \prod_k \left(\frac{w_{jk}^g}{\lambda_{jgk}} \right)^{\lambda_{jgk}}, \qquad \sum_k \lambda_{jgk} = 1. \tag{B.59}$$

A worker of skill k costs the firm the wage w_{jk}^g. Minimizing the cost of output gives

$$c_{jg} = \frac{(c_{jg}^L)^{\alpha_L^g} (q_j^C)^{\alpha_C^g} (p^X)^{\alpha_X^g}}{A_{jg} (\alpha_L^g)^{\alpha_L^g} (\alpha_C^g)^{\alpha_C^g} (\alpha_X^g)^{\alpha_X^g}}, \qquad p_{jg} = \frac{\sigma_N}{\sigma_N - 1}\, c_{jg}. \tag{B.60}$$

Here $q_j^C = (1+t) r_j^C$ and $p^X = 1$. The constant in the denominator follows from the Cobb–Douglas production normalization. Monopolistic competition under CES demand implies the constant markup shown above. Differentiating unit cost with respect to input prices shows that labor, commercial space, and intermediates receive the constant shares α_L^g, α_C^g, and α_X^g of total cost, whatever the factor prices. Variable production and fixed inputs use the same technology, so these shares apply to their combined expenditure C_{jg}.

B.7.2 Factor demand and free entry

Cost minimization within the Cobb–Douglas labor bundle gives $w_{jk}^g L_k = \lambda_{jgk} c_{jg}^L \mathcal{L}$. Shephard's lemma applied to c_{jg} gives the same constant shares. Aggregating variable and fixed costs across firms within sector g gives the input demands

$$L_{jk}^g = \frac{\alpha_L^g \lambda_{jgk} C_{jg}}{w_{jk}^g}, \qquad H_{jg}^C = \frac{\alpha_C^g C_{jg}}{q_j^C}, \qquad X_{jg} = \frac{\alpha_X^g C_{jg}}{p^X}. \tag{B.61}$$

At the CES markup, variable cost is $(\sigma_N - 1) R_{jg}/\sigma_N$, leaving revenue less variable costs of R_{jg}/σ_N. Each firm also incurs an annual fixed cost f_{jg}. With m_{jg} active firms, free entry drives profits to zero:

$$f_{jg} = \varphi_{jg} c_{jg} m_{jg}^{1/\epsilon_m - 1}, \qquad \frac{R_{jg}}{\sigma_N} = m_{jg} f_{jg}. \tag{B.62}$$

Fixed inputs use the same technology as output, so their quantity per firm is $f_{jg}/c_{jg} = \varphi_{jg} m_{jg}^{1/\epsilon_m - 1}$. The parameter φ_{jg} scales this requirement and is held fixed across equilibria. These inputs are fixed

with respect to an individual firm's output; their cost depends on factor prices and the number of firms, which each firm takes as given. The elasticity of fixed inputs per firm with respect to the number of firms is $1/\epsilon_m - 1$. Thus, when $\epsilon_m < 1$, additional entry raises each firm's fixed input requirement; when $\epsilon_m = 1$, that requirement is constant.

The zero-profit condition implies that revenue covers variable and fixed costs: $C_{jg} = R_{jg}$. Substitution gives firm entry and the employment response within each industry:

$$m_{jg} = \left(\frac{R_{jg}}{\sigma_N \varphi_{jg} c_{jg}} \right)^{\epsilon_m}, \qquad d \log L^g_{jk} = d \log R_{jg} - d \log w^g_{jk}. \tag{B.63}$$

Holding φ_{jg} fixed, ϵ_m is the elasticity of the number of firms with respect to revenue relative to unit cost. At given revenue and unit cost, the elasticity with respect to φ_{jg} is $-\epsilon_m$. Higher fixed input requirements therefore support fewer firms. Skill and cost shares are fixed, so the employment response combines additional revenue with the change in the wage. Payroll in sector g is $\sum_k w^g_{jk} L^g_{jk} = \alpha^g_L R_{jg}$. The calibration of cost shares and fixed input requirements is in Appendix C.6.3.

B.7.3 Aggregation across service sectors

The production and entry equations apply separately to each service sector. Total nontradable employment by skill and commercial-space demand in tract j are

$$L^N_{jk} = \sum_{g \in \mathcal{G}_N} L^g_{jk}, \qquad H^{C,N}_j = \sum_{g \in \mathcal{G}_N} H^C_{jg}. \tag{B.64}$$

These totals enter property-market clearing and the employment measures in Section 4.6. Total varieties are $m_j = \sum_{g \in \mathcal{G}_N} m_{jg}$ and total revenue is $R^N_j = \sum_{g \in \mathcal{G}_N} R_{jg}$. The shopping price p^N_j is the CES price index per variety in (B.51), whereas each p_{jg} is the price charged by firms in sector g. Without a sector index, nontradable employment, commercial-space demand, varieties, and revenue denote the corresponding tract totals. These sums preserve the separate costs and entry responses of the two service sectors.

B.8 Tradable production and the entrant

For a tradable sector $g \in \mathcal{G}_T$ with output Y_{jg}, minimizing cost with the Cobb–Douglas technology gives

$$c_{jg} = \frac{(q^C_j)^{\alpha^g_C} (p^X)^{\alpha^g_X}}{A_{jg}} \prod_k \left(w^g_{jk} \right)^{\alpha^g_L \lambda_{jgk}}. \tag{B.65}$$

Fixed expenditure-share constants are absorbed into A_{jg}, and the intermediate input price is normalized to $p^X = 1$. Differentiating total cost $c_{jg} Y_{jg}$ with respect to each input price gives the labor

demand in (24),

$$L_{jk}^{g} = \frac{\alpha_{L}^{g} \lambda_{jgk} c_{jg} Y_{jg}}{w_{jk}^{g}}, \tag{B.66}$$

and the remaining input demands,

$$H_{jg}^{C} = \frac{\alpha_{C}^{g} c_{jg} Y_{jg}}{q_{j}^{C}}, \qquad X_{jg} = \frac{\alpha_{X}^{g} c_{jg} Y_{jg}}{p^{X}}. \tag{B.67}$$

The skill shares satisfy $\sum_{k} \lambda_{jgk} = 1$. Because output sells at the national price, the zero-profit condition (23) ties the skill-weighted wage a producer can pay to its productivity and commercial rent,

$$\alpha_{L}^{g} \sum_{k} \lambda_{jgk} \, d \log w_{jk}^{g} = d \log A_{jg} - \alpha_{C}^{g} \, d \log q_{j}^{C}. \tag{B.68}$$

Output Y_{jg} then adjusts until employment equals the labor supply at these wages, so a producer shrinks when its workers are offered higher wages elsewhere. Profits are zero in every location and sector.

Entry raises the productivity $A_{j_E g_E}$ of the entrant's location-sector by the factor that raises its employment by J^{E} in the entry equilibrium. The entrant is not distinguished from incumbent producers in its location-sector. It pays the sector–skill wages in its tract, and its demands for labor, commercial space, and intermediate inputs follow (B.66) and (B.67), so it competes with other producers for local commercial space.

B.9 Property supply, residential dividends, and property taxes

Residential and commercial supply follow (26), independently at each tract. Residential and commercial rents paid by users include the property tax: $q_{i}^{R} = (1+t) r_{i}^{R}$ and $q_{j}^{C} = (1+t) r_{j}^{C}$. Housing demand sums the individual demand $h_{\ell} = \alpha_{\ell} y_{\ell} / q_{i}^{R}$. Commercial demand sums the tradable and nontradable demands above. Equating each demand to supply gives (27).

Residential rental income $\mathcal{R} = \sum_{i} r_{i}^{R} H_{i}^{R}$ is distributed using fixed skill ownership shares. Hence

$$D_{k} = \frac{\omega_{k}^{0} \mathcal{R}}{N_{k}}, \qquad \sum_{k} N_{k} D_{k} = \mathcal{R}, \qquad d \log D_{k} = d \log \mathcal{R} - d \log N_{k}. \tag{B.69}$$

The differential applies to positive ownership shares. Dividends accrue to current residents, including renters and new arrivals; commercial rents accrue to absentee landlords.

The tax applies to pretax rent on all residential and commercial floor space, including the space used by the entrant's location-sector. With commercial rental income $\mathcal{R}^{C}$,

$$\mathcal{R}^{C} = \sum_{j \in \mathcal{J}} r_{j}^{C} H_{j}^{C}, \qquad \mathcal{T} = t \left(\mathcal{R} + \mathcal{R}^{C} \right). \tag{B.70}$$

Tax revenue leaves the modeled economy. It is not rebated to residents or spent locally. Nonemployment income comes from outside the CZ. The award is a payment to the entrant that does not change its hiring, and the model does not solve how it is financed.

B.10 Equilibrium accounting and feasible choices

For each entrant specification, equilibrium determines

$$\begin{aligned}
&\{w^g_{jk}, r^R_i, r^C_j, p_{jg}, m_{jg}\}_{i,j,k,g},\\
&\{F_{ijgk}, U_{ik}, n_{\ell k}, N^{\text{reg}}_k, N^{\text{nat}}_k, N_k, N\},\\
&\{L^g_{jk}, H^C_{jg}, R_{jg}\},\\
&\{h_\ell, P_i, \pi^I_{ijg}, \pi^O_i, \mathcal{E}_i, D_k\}.
\end{aligned}$$

These quantities satisfy individual choice and consumption, firms' input and entry conditions, labor and property-market clearing, the revenue account, finite-population accounting, and the residential portfolio described in Section 4.4.

In particular, the sector-by-sector labor-market conditions underlying (32) are

$$\sum_i F_{ijgk} = L^g_{jk}, \qquad g \in \mathcal{G},\ j \in \mathcal{J},\ k \in \{H, L\}. \tag{B.71}$$

The tradable conditions include the entrant's location-sector, whose labor demand reflects its higher productivity. The service conditions hold separately for each $g \in \mathcal{G}_N$ and each skill k. Summing commercial-space demands over all four sectors gives (27).

Modeled service destinations are tracts with positive pre-entry nontradable employment. Workplace–sector choices use the observed baseline employment support. The entrant's location-sector has positive baseline employment in every event, so entry opens no new choices. Primitives are calibrated jointly to pre-entry prices, employment, costs, spending, and budgets. Each skill has a positive current resident population, feasible consumption choices, and a housing expenditure share strictly between zero and one. Individual disposable income is positive for every available choice.

B.11 Place-based welfare and employment accounting

For a place a, either a county or the CZ, let V^s_{ak} be the nested inclusive value over choices with residence in a. Amenities, tastes, and preference parameters are common to the entry and no-entry calculations. Let Λ^0_ℓ collect the amenity, commuting, and price factors multiplying y_ℓ in (4), evaluated at no-entry prices. Equivalent variation EV_{ak} solves

$$\begin{aligned}
&V^0_{ak}(\text{EV}_{ak}) = V^1_{ak},\\
&\quad \Phi^0_\ell(x) = \mathbf{1}\{\ell \in \mathcal{A}^0_k\}\Lambda^0_\ell \max\{y^0_\ell + x, 0\}, \qquad i \in a.
\end{aligned} \tag{B.72}$$

Thus EV_{ak} is the uniform transfer, evaluated at no-entry prices and opportunities, that reproduces the entry inclusive value. Here $\mathcal{A}_k^0$ is the structurally available no-entry choice set. A transfer can restore an income-infeasible choice within this set. Place-level equivalent variation weights these skill-specific transfers by no-entry resident populations: $\mathrm{EV}_a = \sum_k N_{ak}^0 \mathrm{EV}_{ak}$ in total dollars and EV_a/N_a^0 per individual. The tables report these aggregates and both skill-specific transfers. This measure values residential opportunities in a place; individual welfare incidence can also be studied using matched transitions.

Workplace employment changes include the entrant's jobs, the adjustment at other tradable producers, and nontradable adjustment. The annual gross fiscal price for place a is $S^E/\Delta L_a$ when $\Delta L_a > 0$. The annual net fiscal measure uses $(\frac{1}{2}S^E - \Delta\mathcal{T})/\Delta L_a$, where the county pays half of the annual award S^E. The full commitment per net job is $\overline{S}^E/\Delta L_a$, where $\overline{S}^E$ is the full award commitment. The government account is CZ-wide, including when the employment denominator is a county.

B.12 Individual transitions and welfare

Equilibrium counts follow the nested shares in (B.25). Individual transitions use common taste draws across the no-entry ($s = 0$) and entry ($s = 1$) equilibria. Let $\mathcal{A}_k^s$ contain the choices with feasible consumption under specification s. Chosen locations and employment, and maximized indirect utility, are

$$\ell_k^s(\omega) = \arg\max_{a \in \mathcal{A}_k^s} z_a(\omega)\Phi_a^s, \qquad v_k^s(\omega) = \max_{a \in \mathcal{A}_k^s} z_a(\omega)\Phi_a^s. \tag{B.73}$$

The taste distribution and potential location–industry choices are fixed across counterfactuals. Feasibility responds to equilibrium prices and income.

The skill-k individuals moving from choice a to choice b number

$$T_{ab,k} = \mathcal{N}_k \Pr\{\ell_k^0 = a,\ \ell_k^1 = b\}, \qquad \sum_b T_{ab,k} = n_{ak}^0, \qquad \sum_a T_{ab,k} = n_{bk}^1. \tag{B.74}$$

The joint probabilities use the same draws in both equilibria. Summing them by sector, workplace, residence, or nonemployment gives movements between these categories. Employment changes are net stock changes; tradable worker transitions are measured at the workplace level.

Welfare comparisons use $v_k^0(\omega)$ and $v_k^1(\omega)$ for the same individuals. Each utility includes the dividend associated with the chosen current residence. Welfare averages among current residents also reflect the changes in composition generated by migration and sorting.

C Quantification and Equilibrium Computation

All monetary flows are annual and expressed in constant 2023 dollars. Superscript 0 denotes the pre-entry equilibrium. A subscript c identifies CZs in the calibration; it is omitted from the single-

CZ model in Section 4 and Appendix B.

This appendix supports Section 5 in the order of its subsections. Appendix C.1 collects the parameters, Appendices C.2–C.4 support the individual choices and budgets in Section 5.1, Appendix C.5 supports Section 5.2, and Appendix C.6 supports the construction of the pre-entry economy in Section 5.3. Appendix C.7 describes the equilibrium computation, and Appendix C.8 the exposure used in the validation of Section 5.4.

C.1 Common and CZ-specific parameters

Table 4 reports the parameters. The tax, travel-cost, technology, establishment-entry, and commercial-supply parameters are common across CZs. Residential supply elasticities vary by tract. Housing expenditure shares, shopping elasticities, ownership shares, and location-specific shifters are calibrated using the corresponding CZ data. The model uses migration scales $\mu_H = 2.32$ and $\mu_L = 1.51$, sector choice $\nu = 1.64$, and participation $\eta = 3.32$. The boundary scale varies across event baselines according to $\psi_e = q_{\text{size}(e)}\chi_{c(e)}$, where size$(e)$ is the event's baseline CZ-size group and $\chi_{c(e)}$ is the fixed residence–workplace scale of the event's CZ. The ratios for small, medium, and large CZs are $(0.67, 0.81, 1.00)$. Both worker skills face the same boundary scale within an event. Baseline wages, nonemployment rates, and preference parameters may differ by skill.

The nontradable basket comprises private retail, education, health, accommodation and food, and other services (NAICS 44–45, 61, 62, 72, and 81). Appendix C.5.2 derives the production cost shares of the two tradable groups and the nontradable composite, and Appendix C.6.3 describes nontradable fixed input requirements.

C.2 Sector choice, migration, and recruitment

This subsection supports the sector-choice, migration, and participation parameters in Section 5.1.

C.2.1 A public benchmark for the sector-choice elasticity

I discipline the common sector-choice elasticity, ν, using public Quarterly Workforce Indicators (QWI) employment and earnings data for 2012–2019. The preferred pooled employment-stock response is 1.64. The estimation compares tradable and nontradable employment, pooling the model's high- and low-skill industry categories within each broad sector. It provides an external benchmark for the common sector scale; its structural interpretation requires the restrictions discussed below.

Data and specification. Employment is full-quarter jobs (`EmpS`) and earnings are average monthly earnings of full-quarter employees (`EarnS`), for private-sector workers aged 25 and older. Nontradables comprise NAICS 44–45, 61, 62, 72, and 81; tradables comprise the remaining covered private industries. I use published broad-industry totals and sum employment and earnings

bills across counties and industries within each CZ, education group, and sector before calculating average earnings.[36]

For employment L_{cks} and average annualized earnings w_{cks} in CZ c, education group k, and sector $s \in \{T, N\}$, I estimate

$$\Delta \log L_{cks} = \beta^{\text{QWI}} \Delta \log w_{cks} + \alpha_{ck} + \delta_{d(c)s} + \gamma_{ks} + \varepsilon_{cks}, \tag{C.1}$$

where Δ denotes the 2012–2019 change and $d(c)$ is the Census division containing CZ c. The fixed effects absorb CZ-by-education, division-by-sector, and education-by-sector differences in growth. Education distinguishes workers with and without a bachelor's degree; the pooled specification imposes a common coefficient. Each sector receives half its CZ–education group's fixed 2012 benchmark employment weight, held constant across specifications.[37] With equal sector weights within each CZ–education group, the coefficient also equals the IV estimate from the relative-outcome regression

$$\Delta \log\left(\frac{L_{ckN}}{L_{ckT}}\right) = \beta^{\text{QWI}} \Delta \log\left(\frac{w_{ckN}}{w_{ckT}}\right) + \widetilde{\delta}_{d(c)} + \widetilde{\gamma}_k + \varepsilon_{ckN} - \varepsilon_{ckT}, \tag{C.2}$$

where $\widetilde{\delta}_d = \delta_{dN} - \delta_{dT}$ and $\widetilde{\gamma}_k = \gamma_{kN} - \gamma_{kT}$.

Instrument and identification. Because earnings and employment are jointly determined, I estimate equation (C.1) by weighted two-stage least squares. The instrument combines pre-period industry employment shares with subsequent employment growth outside the focal CZ:

$$a_{cksi} = \frac{\overline{L}_{cki}}{\sum_{j \in \mathcal{I}_s} \overline{L}_{ckj}}, \qquad Z_{cks} = \sum_{i \in \mathcal{I}_s} a_{cksi} G_{i,-c}, \tag{C.3}$$

$$G_{i,-c} = \log\left(\frac{L_{i,-c,2019}}{L_{i,-c,2012}}\right), \tag{C.4}$$

where $\overline{L}_{cki}$ is mean industry employment over 2007–2011, $\mathcal{I}_s$ contains the broad industries in sector s, and $L_{i,-c,t}$ pools employment across education groups outside the focal CZ. External growth uses the fixed source sample of 2,365 counties in 30 states, yielding 19 broad-industry shocks. Both stages use the same fixed effects and weights; the instrument for equation (C.2) is $Z_{ckN} - Z_{ckT}$.

Identification requires instrument relevance and orthogonality between residualized exposure and unobserved changes in relative labor supply, conditional on the fixed effects. For a labor-supply interpretation, external growth must affect relative employment through earnings rather than through correlated changes in sector amenities, nonwage compensation, or worker composition, and average earnings must track the opportunities of comparable workers. Predetermined shares and the leave-out construction do not establish these restrictions by themselves (Goldsmith-

[36] A county–industry–education–year observation is retained when at least three quarters have positive, finite employment and earnings with accepted QWI release codes (1, 6, or 10). Annual employment averages those quarters, and the annualized earnings bill averages $12 \times$ `EmpS` $\times$ `EarnS`. Suppressed observations are not imputed.

[37] The benchmark weight is the 2012 employment total in NAICS 31–33, 44–45, 51–55, 61, 62, 72, and 81. It is not recalculated when the covered tradable industries are expanded.

Pinkham et al., 2020; Borusyak et al., 2022, 2025). Standard errors are clustered by CZ. Anderson–Rubin confidence intervals address weak instruments under this covariance assumption; they do not account for dependence across CZs arising from shared industry exposures (Adão et al., 2019).

Estimates. The preferred pooled estimate is $\widehat{\beta}^{\mathrm{QWI}} = 1.64$, with a CZ-clustered standard error of 0.51, first-stage $F = 15.03$, and an Anderson–Rubin 95 percent confidence interval of $[0.79, 3.24]$. Table C.1 compares specifications using identical observations, instruments, and weights. Removing regional sector controls yields 0.80 (SE 0.37; $F = 36.68$). Additionally replacing CZ-by-education effects with plain CZ effects yields 1.16 (SE 0.32; $F = 37.05$) and allows differences in growth between education groups within a CZ to contribute to identification. The estimates therefore remain sensitive to the treatment of regional and education-specific growth.

Table C.1: Employment-stock responses across tradable and nontradable industries

Specification	Estimate	SE	First-stage F	AR 95% interval
Preferred controls	1.64^{***}	0.51	15.03	$[0.79, 3.24]$
No regional sector effects	0.80^{**}	0.37	36.68	$[0.05, 1.60]$
CZ and education-by-sector effects	1.16^{***}	0.32	37.05	$[0.57, 1.90]$

Notes: All rows pool education groups and use 2012–2019 changes, 2007–2011 exposure shares, and identical observations, instruments, and weights. Preferred controls are CZ-by-education, division-by-sector, and education-by-sector fixed effects. The second row removes division-by-sector effects; the third also replaces CZ-by-education effects with CZ effects. Standard errors are clustered by CZ. The first-stage statistic is the squared cluster-robust t statistic on the excluded instrument. AR intervals invert the Anderson–Rubin test using the same clustering and an $F(1, C-1)$ reference distribution, where C is the number of CZs. Stars indicate $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$ for two-sided CZ-clustered Wald tests of a zero coefficient, using t_{C-1} with $C = 378$; they do not use the AR tests.

Interpretation and the four-sector model. I use the pooled estimate as an external benchmark for the common sector scale $\nu = 1.64$ in Appendix C.2.2. The regression measures an employment-stock response that can include migration, commuting, participation, and composition. In the four-sector model, ν governs choice among sector opportunity values. Mapping the estimated coefficient into ν therefore requires restrictions connecting those values to observed earnings and accounting for the other adjustment margins. A structural calibration could instead target the model's analogue of equation (C.1), computed with the same aggregation, weights, and horizon.

The four-sector extension yields 0.80 (SE 0.36; first-stage $F = 18.50$; AR 95 percent interval $[0.13, 1.77]$). It uses more detailed industry data and a slightly smaller complete-choice sample, so the comparison also changes data coverage. Pairwise diagnostics vary substantially: the N_H–N_L estimate is -1.46 (SE 0.40; $F = 9.88$), with an AR interval of $[-3.02, -0.80]$. These findings qualify the interpretation of a common elasticity across all four choices. Dynamic worker-transition estimates, such as Artuç et al. (2010) and Caliendo et al. (2019), concern sector transitions governed by moving costs and continuation values. They provide context for sector mobility, but their parameters are not directly comparable to this seven-year employment-stock response.

C.2.2 Recruitment calibration

The recruitment calibration sets only the participation elasticity and the size-group border ratios reported in Table 4, holding sector choice and migration at their estimates. Section 5.1 describes the targets, the twelve calibration events, and the fit.

C.2.3 Migration elasticity

This subsection derives the shift-share regression behind the migration elasticities in Section 5.1. For high-skill individuals, who choose a sector before a region, it treats the CZ's skill-level population as one regional choice, which approximates the within-sector margin when sector shares change little.

From the model to the estimating equation. Equation (B.11) gives the regional share of skill-k individuals. Its log odds are

$$\log \frac{b_k^{\text{reg}}}{1-b_k^{\text{reg}}} = \log \frac{\overline{B}_k^{\text{reg}}}{\overline{B}_k^{\text{nat}}} + \mu_k \left(\log V_k^{\text{reg}} - \log V_k^{\text{nat}}\right). \tag{C.5}$$

Treating each commuting zone c as a region and the rest of the country as the national choice, the odds are $P_{ck}/(P_k^{US} - P_{ck})$. Differencing between 2010 and 2018 removes the preference parameters, and the change in the national value, which is common to all CZs within a skill, is absorbed by the division-by-skill effects:

$$\Delta \log \frac{P_{ck}}{P_k^{US} - P_{ck}} = \delta_{d(c),k} + \mu_k \, \Delta \log V_{ck}^{\text{reg}} + u_{ck}, \tag{C.6}$$

where u_{ck} collects changes in preference parameters and amenities relative to the division. Because the outcome is in log odds, its coefficient is μ_k itself; the population elasticity in equation (B.30) multiplies it by $1-b_k^{\text{reg}}$, which is close to one for a single CZ. The horizon is one eight-year difference, and the coefficient is not annualized.

By equation (B.3), the change in an inclusive value is, to first order, the share-weighted average of the changes in the values below it, so $\Delta \log V_{ck}^{\text{reg}}$ averages $\Delta \log \Phi_\ell$ over the CZ's choices. From equation (4), holding amenities and nonwage income fixed, the value of a working choice changes by

$$\Delta \log \Phi_\ell = \frac{w_{jk}^g}{y_\ell} \Delta \log w_{jk}^g - \alpha_{H,c(i)} \Delta \log q_i^R - (1 - \alpha_{H,c(i)}) s_N \, \Delta \log \mathcal{P}_\ell. \tag{C.7}$$

The estimating equation is

$$\Delta \log \frac{P_{ck}}{P_k^{US} - P_{ck}} = \delta_{d(c),k} + \mu_k \Delta \mathcal{V}_{ck} + u_{ck}, \qquad \mathcal{V}_{ck} = \log(e_{ck} w_{ck}) - 0.25 \log r_c. \tag{C.8}$$

Its value proxy replaces the average of these changes with the change in expected earnings, $e_{ckt} w_{ckt}$, less 0.25 times the change in rent. The coefficient on $\Delta \mathcal{V}_{ckt}$ therefore recovers μ_k when the proxy's

weights match the model's: a unit weight on earnings against w^g_{jk}/y_ℓ, and 0.25 on rent against $\alpha_{H,c(i)}$. Changes in service prices, dividends and nonemployment income enter the error. The sample also covers individuals aged 25–64 rather than everyone aged 18 or older.

Instrument and inference. Local earnings respond to local population through u_{ck}, so I instrument $\Delta\mathcal{V}_{ckt}$ with $Z_{ck} = \sum_n \omega_{ckn} g_{n,-c}$, where ω_{ckn} is the 2010 share of skill-k employment in commuting zone c in industry n and $g_{n,-c}$ is national CBP employment growth in industry n outside the CZ, with industries harmonized across classification vintages. The identifying assumption is $E[Z_{ck}u_{ck} \mid d(c)] = 0$: national industry growth outside the CZ affects local population only through the value of living there, conditional on division effects. Population growth can also reflect aging, mortality, and education transitions, which would violate this assumption if they were correlated with the industry mix. I estimate equation (C.8) by two-stage least squares with equal weights across commuting zones, using the 2010 and 2018 ACS five-year files and QWI earnings by skill. CZs with similar industry mixes are exposed to the same shocks, so Table C.2 reports standard errors clustered by three-digit industry shock as well as by commuting zone (Adão et al., 2019; Borusyak et al., 2022). The low-skill first stage is below ten, and both confidence intervals are wide. A matched 2002–2010 estimate is not available in comparable data.

Table C.2: Migration elasticity estimates

	High skill	Low skill
Estimate, μ_k	2.32^{***}	1.51^{***}
Standard error, clustered by CZ	0.75	0.51
Standard error, clustered by three-digit industry shock	1.13	1.24
First-stage Wald statistic	10.71	8.41
95% confidence interval	[1.35, 5.60]	[0.90, 4.40]
Commuting zones	558	559

Notes: Stars use the first reported standard errors, clustered by CZ, and two-sided t_{C-1} tests of a zero coefficient, where C is the skill-specific CZ count. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$. The industry-shock clustered standard errors and weak-instrument diagnostics are reported separately.

C.3 Commuting and shopping

This subsection supports the commuting and shopping parameters in Section 5.1.

C.3.1 Commuting

The commuting scale χ and the cost per minute κ are common to both skills. Within a lowest nest $\mathcal{B}$, equation (B.6) gives the share of the residence–workplace pair $\ell = (i, j)$ as Φ^χ_ℓ relative to the nest total, with no pair-specific preference parameter, and the conditional shares in equation (C.29) reproduce the baseline allocation. The baseline flows of skill k are the commuting gravity weights

balanced to the observed residence and workplace margins (Appendix C.6.1),

$$F^0_{ijk} = a_{ik}\, b_{jk} \exp(-\chi\kappa\, t_{ij}), \tag{C.9}$$

where a_{ik} and b_{jk} are the balancing factors that match the margins. Cobb–Douglas pair values factor into residence and workplace components as shown in Appendix C.6.2. These components determine the economic interpretation of the balancing factors. Changes in income and prices update the fitted shares through $\hat{\Phi}^{\chi}_{\ell}$ (Appendix B.5). Equation (C.9) is a gravity equation, and the corresponding estimating equation is

$$E[F_{ij}] = \exp\left(\alpha^C_i + \delta^C_j + \beta^C_c t_{ij}\right), \qquad \beta^C_c = -\chi\kappa, \tag{C.10}$$

where the residence and workplace effects α^C_i and δ^C_j take the place of the balancing factors.

I estimate equation (C.10) by PPML separately for each commuting zone, which accommodates the many pairs with no commuters (Silva and Tenreyro, 2006; Dingel and Tintelnot, 2026), using 2019 LODES residence–workplace flows of all workers, since the gravity weights are common to both skills, and 2020 driving times on routes of up to 180 minutes. The coefficient is identified from variation in travel time within residences and workplaces, and $\chi_c = -\beta^C_c/0.01$ under the normalization $\kappa = 0.01$ per minute. The scale χ governs where workers live and work within the CZ and is distinct from ψ_e, which governs whether residents of adjacent CZs work inside it.

C.3.2 Shopping

Using the industry aggregation in (B.52), spending from origin i at destination j in (16) is

$$\mathcal{E}_i \pi^I_{ij} = \underbrace{\mathcal{E}_i P_i^{\sigma_N - 1}}_{\text{origin}} \underbrace{\sum_{g \in \mathcal{G}_N} \xi_{jg} m_{jg} \big(p_{jg}\big)^{1-\sigma_N}}_{\text{destination}} \zeta_{ij} \exp\{-(\sigma_N - 1)\kappa t_{ij}\}. \tag{C.11}$$

Visits V_{ijc} measure trips rather than dollars. If spending per visit varies only by origin or by destination, the fixed effects absorb it. Writing the taste weight as $\zeta_{ij} = \exp\{\gamma_c \mathbf{1}[i = j] + \xi_c \mathbf{1}[\text{same county}]\}$ gives the estimating equation

$$E[V_{ijc} \mid \text{controls}] = \exp\{\alpha_{ic} + \delta_{jc} + \beta^S_c t_{ij} + \gamma_c \mathbf{1}[i = j] + \xi_c \mathbf{1}[\text{same county}]\}, \tag{C.12}$$

with $\beta^S_c = -(\sigma_{N,c} - 1)\kappa$.

I estimate equation (C.12) by PPML separately for each commuting zone, using 2019 visits and 2020 routes. The same-county term is included where supported, standard errors are clustered by origin and destination, and the routed support includes positive travel times up to 180 minutes, with five minutes assigned to trips within a tract. Visits are pooled over NAICS 44, 45, 61 excluding 6113, 722, and 81; education visits proxy for the health-care network and are counted once. The standard error of $\sigma_{N,c}$ is the coefficient's standard error divided by 0.01.

Origin and destination effects absorb how attractive each location is, but bilateral tastes and trip chaining can still covary with travel time. Within-CZ decay also does not identify how much spending leaks across commuting-zone boundaries, and baseline destination shares reflect the revenue balancing of Section 5.3 as well as the estimated decay.

From the coefficient to the elasticity. A pooled coefficient for each CZ's nontradable basket determines

$$\kappa = 0.01 \text{ per minute}, \qquad \sigma_{N,c} = 1 - \frac{\beta_c^S}{0.01}. \tag{C.13}$$

The coefficient κ is common to shopping and commuting and measures the exponential travel-cost increment per minute. Events in a CZ share $\sigma_{N,c}$, which governs substitution, markups, and free entry. The shifters T_{ij} and T_i^O incorporate shopping decay, and the destination productivities recovered in Appendix C.6.4 align spending flows with baseline revenue. Visit data identify destinations; expenditure accounts supply spending totals, with retail expenditure measured as the merchant margin. Shopping weights are fixed across counterfactuals.

C.4 Individual budgets

This subsection supports the individual budgets in Section 5.1.

C.4.1 Income and the housing share

Workers receive wage income w_{jk}^g. Nonemployment income y_k^U is measured using public assistance, Supplemental Security Income, and Social Security payments among nonworkers. It is paid during nonemployment. Housing expenditure measures rental-equivalent services for renters and owners. Income and housing measures are per individual and use survey weights.

The population comprises individuals aged 18 or older. Skill-specific nonworker-to-worker ratios from pre-entry ACS microdata scale the observed resident worker counts. The accessible outside area pools neighboring CZs. Its population ratios for individuals aged 25–64 are adjusted to cover everyone aged 18 or older using available neighboring-state microdata; available-state age adjustments also bridge the national population stocks. These geographic and demographic approximations preserve a common finite-population decomposition.

Outside housing dividends divide observed rental-equivalent residential rental income, net of the property tax, according to outside ownership value shares and populations. Outside property tax revenue is not rebated. These dividends, outside residential rents, and the outside housing expenditure share are fixed in the counterfactuals. The local portfolio account reconciles individual income and rental income.

For each CZ, let $\widehat{E}_{H,c}$ be the weighted total pre-entry rental-equivalent housing-service expenditure and $\widehat{Y}_c^{\text{inc}}$ the total after-tax worker earnings and nonemployment income in the same sample. The residential portfolio distributes $\widehat{R}_c = \widehat{E}_{H,c}/(1+t_c)$ to the sample's adults. Sum income and the

measured dividends within each observed household to obtain Y_{mc}. If E_m is its whole-household service bill and w_{mc} its survey weight times the PUMA-to-CZ fraction, the constant share and the implied service share are

$$\alpha_{H,c} = \text{wmed}_{m \in c}\left(\frac{E_m}{Y_{mc}}; w_{mc}\right), \qquad \gamma_c = s_N(1 - \alpha_{H,c}). \tag{C.14}$$

Each observed household receives its survey weight once when calculating the median, and the model applies the resulting share to each individual's budget. The median targets a typical household's expenditure burden, and aggregate expenditure remains a separate diagnostic (Appendix A.7). Events sharing a CZ use the same earliest-pre-entry sample and parameter. The outside expenditure share uses the corresponding household median in its existing neighboring-CZ sample and fixed outside resource account. In the model's baseline, with total local income $Y_c^{\text{inc},0}$, producer housing rental income satisfies

$$R_c^0 = \frac{\alpha_{H,c} Y_c^{\text{inc},0}}{1 + t_c - \alpha_{H,c}}. \tag{C.15}$$

The measured portfolio defines the empirical median's resource denominator; equation (C.15) determines rental income in the model's baseline. The portfolio divides this income according to ownership shares and current skill populations. Demanded housing at the observed residential rents anchors each tract's residential supply schedule. Constant expenditure shares approximate the housing Engel curve at the median household share, and the implied marginal service share is $s_N(1 - \alpha_{H,c})$. Appendix C.4.2 estimates that Engel curve.

C.4.2 The housing Engel curve

Housing expenditure determines how individual resources translate into demand for services and how residential rents enter location choices. The model uses a Cobb–Douglas housing share $\alpha_{H,c}$ for each CZ. The empirical calibration matches the household-weighted median housing-service expenditure share in the pre-entry sample. An auxiliary Engel-curve exercise describes the income gradient in housing shares and clarifies the distinction between a household budget share and a marginal expenditure parameter.

Estimating the Engel curve. The auxiliary exercise uses pre-entry ACS person and housing microdata for 15 CZs, selecting the earliest available pre-entry year within each CZ. Households are assigned fractionally from PUMAs to CZs; the crosswalk uses tract-population weights for 2010 PUMAs and area weights for 2000 PUMAs. All amounts are annual and expressed in 2023 dollars. Let n_m denote the number of adults in household m, ω_{mc} its household survey weight times the geographic allocation fraction, and $a_{mc} = n_m \omega_{mc}$ its adult weight. Per-adult resources in this auxiliary specification are reported household income less five percent of household wage income, divided by n_m. Positive-resource households below the within-CZ adult-weighted first percentile are trimmed. The fit uses owners and renters across the remaining income distribution, comprising

117,140 distinct households.

For renters, expenditure is annual contract rent. For owners, this particular auxiliary specification uses reported selected monthly owner costs when positive, with a rental-imputation fallback. Reported costs cover 99.97 percent of the included owner adult weight. The owner measure includes financing payments, including mortgage principal, and therefore differs from rental-equivalent service consumption. A survey-weighted renter regression of log rent on log rooms, bedrooms, structure type, and PUMA–year effects supplies the fallback and a common reference-dwelling price index q_m. Predicted rents are retransformed using a weighted smearing factor. Reference-dwelling rents are normalized to have an adult-weighted geometric mean of one.

Writing per-adult expenditure as e_m and resources as y_m, the auxiliary weighted least-squares regression is

$$\frac{e_m}{y_m} = \alpha^{SG} + \delta_c \frac{q_m}{y_m} + u_{mc}, \qquad \delta_c = (1-\alpha^{SG})\bar{h}_c. \tag{C.16}$$

The intercept is common across CZs and the price-to-income slopes are CZ-specific. The estimate is $\widehat{\alpha}^{SG} = 0.07$, with a conditional bootstrap standard error of 0.33 percentage points. The bootstrap resamples PUMA clusters; 195 of 199 draws yield usable estimates, holding the price and housing-imputation first stages fixed. This uncertainty measure omits first-stage and geographic-assignment uncertainty. The fitted intercept is the expenditure-share limit as q_m/y_m approaches zero, and hence an extrapolated limit rather than the observed average housing share. The corresponding expenditure relationship is

$$e_m = \alpha^{SG} y_m + (1-\alpha^{SG}) q_m \bar{h}_c, \tag{C.17}$$

$$\left.\frac{\partial e_m}{\partial y_m}\right|_{q_m} = \alpha^{SG}, \qquad \frac{e_m}{y_m} = \alpha^{SG} + (1-\alpha^{SG})\frac{q_m \bar{h}_c}{y_m}. \tag{C.18}$$

The derivative is structural only under the maintained demand and resource assumptions. Cross-sectional income variation can also reflect saving, measurement error, housing quality, financing arrangements, and household heterogeneity. Mean housing shares of approximately 24–31 percent in this auxiliary sample can therefore coexist with an estimated intercept near 6.70 percent.

Comparison with Tsivanidis. Tsivanidis (2026), Section 5.1, sets the long-run housing share $1-\beta = 0.24$ to match Bogotá's high-income expenditure pattern. His Appendix D.7 calibrates wages and the skill-productivity parameter to the wage premium, and then matches the GEIH average housing expenditure share of 0.31 using the housing minimum, jointly with the car-expenditure calibration. The reported post-period minimum is 1.21 in his model's units. My estimate of 0.07 is obtained from a pooled ACS regression with CZ-specific slopes. The estimators, housing measures, resource definitions, and populations differ; the estimates are not a controlled comparison of preferences between the United States and Bogotá.

Measurement and specification sensitivity are quantitatively material. A separate ACS exercise

measures owner housing with rental equivalents, reconciles the residential portfolio with alternative replacements of combined investment income, and reserves households for validation. Its pooled full-sample intercepts range from 1.56 to 2.39 percent across the resource conventions, while held-out renter comparisons can favor the fixed 0.07 value. Owner service expenditures are imputed, so the combined fit does not independently validate owner consumption. These results identify sensitivity to measurement and accounting; they do not isolate a single explanation for the gap with Tsivanidis. The auxiliary Engel curve is evidence on housing demand, with its assumptions and limits reported alongside the estimate.

C.4.3 Housing-value shares by skill

Let V_a be the reported housing value of owner-occupied household a, v_a its survey weight, and f_{ak} the fraction attributed to skill k. Value is split equally between the householder and a spouse or partner when both are present; a householder without one receives the full attribution. The pre-entry ownership share is

$$\omega_{ck}^0 = \frac{\sum_{a \in c,\, \text{owner}} v_a V_a f_{ak}}{\sum_{a \in c,\, \text{owner}} v_a V_a}, \qquad \sum_k \omega_{ck}^0 = 1. \tag{C.19}$$

Pre-entry ACS microdata discipline these shares, with PUMA-to-CZ allocation providing the geographic mapping. Value shares incorporate group sizes, homeownership rates, and property values. The weights remain fixed across equilibria and allocate each skill's share of current residential rental income equally among its current residents.

C.5 Firms and space

This subsection supports Section 5.2.

C.5.1 Industry groups and baseline allocation

Tradable industry type is defined by the 2000 Census bachelor's-degree employment share at four-digit NAICS. A tradable industry is high skill when its share exceeds the unweighted median across industries, 0.21. High-skill nontradables comprise private education and health (NAICS 61 and 62); low-skill nontradables comprise private retail, accommodation and food, and other services (NAICS 44–45, 72, and 81). This industry label is distinct from individual education. The productive sectors are T_H, T_L, N_H, N_L, and both worker skills are available in each. The tradable basket comprises NAICS 31–33, 51, 52, 54, and 55; private nontradables comprise 44–45, 61, 62, 72, and 81.

LODES supplies tract employment totals over event years $-4, -3, -2$. Within each broad tradable industry, I pool Infogroup establishment employment over these years and apply its high- and low-skill industry shares to the corresponding LODES mean. All classified industries in NAICS 51, 52, 54, and 55 belong to T_H; manufacturing employment is split using its local industry mix. Missing or ambiguous manufacturing allocations use the classified manufacturing mix in the same

tract, then the county, then the CZ. This rule applies to 2.89% of tradable employment across event baselines, and each allocation retains its source flag. Nontradable totals follow the fixed NAICS groups: private NAICS 61 and 62 enter N_H, and private NAICS 44–45, 72, and 81 enter N_L.

QWI private-sector employment and earnings provide the association between worker education and industry type. High-skill workers have a bachelor's degree or more; the remaining education categories form the low-skill group. QWI education categories describe workers aged 25 and above, so their industry association is used as a prior for the model's individual employment allocation. I retain industry-years with released information for both worker skills. Employment composition uses beginning-of-quarter employment; earnings use full-quarter employment to weight annualized monthly mean earnings, expressed in 2023 dollars. The wage and employment samples retain their separate disclosure flags. County estimates pool the three pre-entry years. Where a county lacks usable information in any year for either worker skill, the corresponding CZ and event window supply the prior. This geographic fallback accounts for 0.08% of productive employment across event baselines. Coverage files retain suppressed and unclassified industries and compare released detailed employment with broad-industry controls.

Within a tract and broad sector $b \in \{T, N\}$, let L^b_{jH} and L^b_{jL} denote the two industry totals, so that broad sector b and industry type $f \in \{H, L\}$ together identify sector $g = b_f$, and let L^b_{jk} denote the existing totals by worker skill. Iterative proportional fitting allocates the industry totals across workers while preserving both margins. If p^b_{jf} is the QWI high-skill worker share for industry type f, the allocation satisfies

$$\sum_k L^b_{jfk} = L^b_{jf}, \qquad \sum_f L^b_{jfk} = L^b_{jk}, \qquad \frac{L^b_{jHH} L^b_{jLL}}{L^b_{jHL} L^b_{jLH}} = \frac{p^b_{jH}/(1-p^b_{jH})}{p^b_{jL}/(1-p^b_{jL})}. \tag{C.20}$$

The odds-ratio condition applies to positive flows; zero employment margins remain zero. Each fitted home–work flow is divided using its workplace's industry shares conditional on worker skill. The accessible outside location uses Census-2000 national industry shares conditional on education.

Let $\widetilde{w}^b_{jfk}$ be the county or CZ QWI earnings prior, and $\overline{w}^b_{jk}$ the existing broad-sector wage. Initial wages retain the QWI ratio between industry types and match the broad-sector payroll:

$$w^{b,0}_{jfk} = a^b_{jk} \widetilde{w}^b_{jfk}, \qquad a^b_{jk} = \frac{\overline{w}^b_{jk} L^b_{jk}}{\sum_{f \in \{H,L\}} L^b_{jfk} \widetilde{w}^b_{jfk}}. \tag{C.21}$$

This normalization applies to positive broad-sector employment. It preserves workplace payroll and aggregate earnings at each residence. Wages subsequently clear distinct workplace–industry–worker-skill labor markets. Skill cost shares are calibrated from each industry's payroll. Initial nontradable revenue follows its payroll and the common labor cost share. Establishment masses and baseline destination expenditure shares are divided according to these revenue shares. The two industry masses sum to the tract's initial establishment mass, and both industries enter one CES shopping basket. Each sector uses the common sector scale ν; its work and nonemployment choices use the separate common participation scale η.

C.5.2 Production cost shares

Data and industry aggregation. I use the 2012 detailed supply–use accounts of the U.S. Bureau of Economic Analysis (BEA). The source is the 2012 worksheet in `Use_SUT_Detail.xlsx`, together with the industry's classification in the workbook's `NAICS Codes` concordance.[38] All source amounts are in millions of current 2012 dollars, and industry gross output is valued at basic prices.

Let $g \in \{T_H, T_L, N\}$ index the two tradable sectors and the nontradable composite, which pools N_H and N_L. The high-skill-intensive tradable group, T_H, contains NAICS 51, 52, 54, and 55. The low-skill-intensive group, T_L, contains manufacturing, NAICS 31–33. These labels describe broad production composites: both firm types employ both worker skills, and the grouping does not separately identify high-technology manufacturing. The nontradable composite contains retail trade, education, health care and social assistance, accommodation and food services, and other services (NAICS 44–45, 61, 62, 72, and 81). Real estate is excluded as an output sector but remains an input to the modeled industries. Retail output is measured by retail margins rather than the full value of merchandise sold.

Purchased intermediate inputs. For each industry i, let Y_i denote gross output, X_i total purchased intermediate inputs, and R_i purchases of real-estate services. These correspond to BEA rows `T018`, `T005`, and `531ORE`, respectively. For the set of industries $\mathcal{I}_g$ assigned to group g, define

$$Y_g = \sum_{i \in \mathcal{I}_g} Y_i, \qquad X_g = \sum_{i \in \mathcal{I}_g} X_i, \qquad R_g = \sum_{i \in \mathcal{I}_g} R_i. \tag{C.22}$$

The intermediate-input share is

$$\alpha_X^g = \frac{X_g - R_g}{Y_g}. \tag{C.23}$$

I sum monetary amounts before taking ratios, so the resulting share is weighted by industry gross output. Subtracting R_g keeps purchased real-estate services out of the outside-input composite and avoids assigning the same expenditure to both intermediate inputs and commercial space. The aggregation uses the BEA concordance, including the nonnumeric industry code `4B0000` for other retail activities.

Allocation between labor and commercial space. I allocate the remaining production-cost share according to

$$\alpha_L^g = 0.8(1 - \alpha_X^g), \qquad \alpha_C^g = 0.2(1 - \alpha_X^g). \tag{C.24}$$

Thus, $\alpha_L^g + \alpha_C^g + \alpha_X^g = 1$ and $\alpha_C^g/(\alpha_L^g + \alpha_C^g) = 0.2$ for each group. The common 80/20 split draws on the labor–floor-space calibration in Ahlfeldt et al. (2015), Hoelzlein (2023), and Miyauchi et al. (2025). Here, I apply that benchmark to the share remaining after purchased non-real-

[38] The workbook is distributed in BEA's supply–use data package: https://apps.bea.gov/industry/release/zip/SUPPLY-USE.zip. The version retrieved on October 28, 2026, contains historical 2012 estimates organized using the workbook's 2017-NAICS concordance. The archived source and replication files preserve the data vintage.

estate intermediates have been deducted. This retains sectoral differences in intermediate-input use without introducing additional group-specific labor–space parameters.

Purchased real-estate services alone do not measure the full economic cost of commercial premises, particularly when businesses own rather than rent their space. Conversely, gross operating surplus is not itself a purchase of intermediate inputs. These considerations motivate retaining the measured intermediate-input share while using a separate benchmark for the labor–space allocation.

The allocation is nevertheless a maintained approximation. The residual $Y_g - X_g + R_g$ contains employee compensation, purchased real-estate services, gross operating surplus, and net production taxes, subject to published rounding. Operating surplus also includes depreciation and returns to capital other than commercial premises. Allocating this entire residual between labor and space therefore does not recover their separately observed national-account payments or imply that all operating surplus and taxes accrue to local workers and property owners. It maps the accounting residual into the model's restricted set of production factors.

Resulting shares. Table C.3 reports the source totals and implied allocations. I retain the unrounded ratios in the calibration.

Table C.3: 2012 BEA inputs and calibrated production shares

	T_H	T_L	N
Number of detailed industries	38	232	41
Gross output, Y_g	5,730,400	5,746,010	4,831,575
Purchased intermediates, X_g	2,376,329	3,855,036	1,943,393
Real-estate purchases, R_g	163,670	24,482	305,066
Non-real-estate inputs, $X_g - R_g$	2,212,659	3,830,554	1,638,327
Labor share, α_L^g	0.4911	0.2667	0.5287
Commercial-space share, α_C^g	0.1228	0.0667	0.1322
Intermediate-input share, α_X^g	0.3861	0.6666	0.3391
Space share within labor–space bundle	0.2000	0.2000	0.2000

Notes: Monetary amounts are millions of current 2012 dollars. T_H comprises NAICS 51, 52, 54, and 55; T_L comprises NAICS 31–33; and N comprises NAICS 44–45, 61, 62, 72, and 81. Intermediate-input shares are calculated from BEA accounts. The 80/20 allocation of the remaining share is imposed.

Interpretation in the model. The model treats the non-real-estate intermediate composite as supplied from outside the CZ at a fixed price. This is a sourcing assumption, not a geographic fact identified by the national BEA tables: observed purchases can include inputs supplied within the same CZ. Consequently, α_X^g measures purchased-input intensity in the data and expenditure leakage only under the model's sourcing closure.

For the tradable Cobb–Douglas technologies, the calibrated shares are production exponents. For the nontradable labor–space CES nest, the 80/20 allocation is a baseline cost-share target

used to recover the distribution parameter at observed factor prices; expenditure shares can change when relative prices change. The allocation between the two worker skills within labor is calibrated separately.

With incumbent tradable output, commercial rents, and outside-input prices fixed, a proportional increase in both skill wages implies

$$\frac{\partial \log L_g}{\partial \log w} = -(1 - \alpha_L^g). \tag{C.25}$$

The corresponding elasticity magnitudes are approximately 0.509 for T_H and 0.733 for T_L. These are implications of the calibrated technology and fixed-output closure, rather than elasticities estimated from the BEA accounts.

C.5.3 Establishment entry

I set $\epsilon_m = 0.40$, in line with the OLS coefficient of 0.39 from the county regression below. The regression is descriptive rather than causal, for the reasons given at the end of this subsection, but it shows that the maintained value matches how establishment counts have moved with service receipts across counties.

The free-entry condition in (20) gives the number of varieties in tract j, suppressing the sector index g,

$$m_j = \left(\frac{R_j^N}{\sigma_N \varphi_j c_j^N} \right)^{\epsilon_m}. \tag{C.26}$$

Aggregating to counties and taking logs and five-year differences,

$$\Delta \log m_{jt} = \epsilon_m \left(\Delta \log R_{jt}^N - \Delta \log c_{jt}^N \right) - \epsilon_m \Delta \log \varphi_{jt}. \tag{C.27}$$

I measure varieties by establishments, revenue by receipts, and unit-cost growth by wage growth, and I absorb common movements in input prices and entry costs with end-year effects δ_t:

$$\Delta \log m_{jt} = \epsilon_m \left(\Delta \log R_{jt}^N - \Delta \log w_{jt} \right) + \delta_t + e_{jt}. \tag{C.28}$$

The regression uses 9,798 five-year county–window observations, after aggregating the five service sectors and excluding NAICS 53, and gives a historical descriptive coefficient of 0.39 with a conventional OLS standard error of 0.01. The error contains $-\epsilon_m \Delta \log \varphi_{jt}$ and the gap between unit-cost and wage growth. Because entry also raises revenue by attracting spending, and receipts differ from the model's merchant-margin output, I read the estimate as descriptive rather than causal.

C.5.4 Accounting units and the property tax

Amounts are expressed in constant 2023 dollars. Individual income, rents, production, and tax revenue are measured over the same one-year accounting period. The baseline property tax rate is

$t_a^0 = 0.28$. After entry, county rates adjust to finance service needs. Tax revenue purchases goods outside the modeled economy rather than providing cash rebates to residents. The event-specific local-benefit exercise is described in Appendix F.1.

I measure a sample-specific tax-to-housing-cost benchmark by adapting Couture et al. (2024, Section 4.1.3 and online Appendix C). They divide a population-weighted median of local median property-tax bills by a population-weighted median home value converted to an annual user cost. I apply this construction at the county level to the 44 distinct host counties of the 55 model events, counting each county once. The 2014 ACS five-year release covers 2010–2014; its end year equals the median event year, so this is a common-period benchmark rather than a uniformly pre-entry measure. County population, median annual real-estate taxes, and median owner-occupied home value come from ACS tables B01003, B25103, and B25077, respectively. All sample counties have valid estimates. Following the authors' conversion for 2014, I annualize home values at 4.6%:

$$\widehat{t}^{\,\mathrm{ACS}} = \frac{\mathrm{wmed}_{c,\,\mathrm{pop}_c}(\text{median annual tax}_c)}{0.046\ \mathrm{wmed}_{c,\,\mathrm{pop}_c}(\text{median home value}_c)} = \frac{\$2{,}432}{\$7{,}700.40} = 31.6\%.$$

The underlying population-weighted median county home value is \$167,400. The numerator and denominator are aggregated separately; this is not a median of household tax-to-cost ratios. The common dollar deflator cancels from the ratio. Using aggregate taxes and aggregate home values instead (ACS B25090 and B25082), with the same annualization, gives 26.1%. For comparison, taxes are 17.4% of observed annual owner costs (twelve times ACS B25089); those costs include mortgage payments, taxes, utilities, insurance, and relevant fees, so this is a different denominator. The ACS measures owner-occupied housing and taxes paid to all taxing jurisdictions, not county-government receipts alone or commercial tax rates. These measurements are empirical benchmarks; the equilibrium calculations use $t_a^0 = 0.28$ on pretax rent as the baseline rate, with counterfactual rates determined by the county budget.

C.6 Building the pre-entry economy

This subsection supports Section 5.3.

C.6.1 Baseline allocations and preference parameters

The baseline allocation covers individuals in private tradable and service employment and in nonemployment. The observed residence and workplace margins are balanced using the commuting gravity weights. Inbound and outbound flows connect the CZ to the neighboring region. LODES does not report commuting flows by education, so the boundary flows by skill use LODES earnings bands as a proxy: the top band, with monthly earnings above \$3,333, stands in for high-skill workers and the two lower bands for low-skill workers. For each tract and skill, the earnings-band flows set the share of jobs held by residents of the CZ rather than inbound commuters and the share of residents who work outside the CZ. These shares are applied to the QWI workplace totals by skill, which are measured by education. The additional outside-residence/outside-workplace worker stock

is neighboring-CZ employment net of existing inbound workers. Nonworker counts and national population stocks complete the finite-population account.

Let $n^0_{\ell k}$ be the baseline mass of skill k in choice ℓ, including the national choice. For a nest v whose members $\mathcal{C}(v)$ are location pairs or broader choices, write M^0_{vk} for the baseline mass of all choices in the nest. Let $\mathcal{I}(\ell)$ collect the conditional choices specifying ℓ. The nest's baseline conditional probabilities are

$$\pi^0_{u|v,k} = \frac{M^0_{uk}}{M^0_{vk}}, \qquad M^0_{vk} = \sum_{u \in \mathcal{C}(v)} M^0_{uk}, \qquad n^0_{\ell k} = \mathcal{N}_k \prod_{(u,v)\in\mathcal{I}(\ell)} \pi^0_{u|v,k}. \tag{C.29}$$

On choices with positive baseline mass, these conditional probabilities reproduce participation, sector, area, and commuting allocations. Location-pair commuting probabilities within the CZ are predicted by the fitted gravity system; they do not reproduce every realized bilateral count. The amenities recover these fitted probabilities in levels. At broader nests, (B.32) recovers the preference parameters given the baseline inclusive values. The shape parameters govern responses to relative utility. The scale order and the placement of the inbound commuting nest are in Appendix B.2.

Location amenities. Let a^0_{ik} and b^0_{jk} be the fitted residence and workplace factors for skill k, and ϑ^0_{jgk} the share of workplace employment in sector g. Within the CZ, fitted commuting flows are allocated across industries using these workplace shares:

$$F^0_{ijgk} = a^0_{ik} b^0_{jk} \vartheta^0_{jgk} \exp(-\chi\kappa t_{ij}), \qquad i, j \in \mathcal{J}. \tag{C.30}$$

The workplace shares sum to one across employment sectors. Because industry allocation depends on workplace and skill, the residential factor remains common across sectors. Let $\gamma = (1-\alpha_H)s_N$ and let $y^0_{jgk} = w^{g,0}_{jk} + D^0_k$ be disposable income for a CZ resident employed at j in sector g. One normalization of the location amenities is

$$\begin{aligned} B^R_{ik} &= (a^0_{ik})^{1/\chi} (q^{R,0}_i)^{\alpha_H} (P^0_i)^{\rho\gamma}, \\ B^W_{jgk} &= (b^0_{jk}\vartheta^0_{jgk})^{1/\chi} \frac{(P^0_j)^{(1-\rho)\gamma}}{y^0_{jgk}}. \end{aligned} \tag{C.31}$$

Substitution gives $(\Phi^0_\ell)^\chi = F^0_{ijgk}$ on internal working choices. Hence normalized utility powers reproduce the fitted location-pair probabilities without a further bilateral preference parameter. The broader preference parameters are expressed in the corresponding value units using (B.32).

Boundary flows retain their separately measured totals. For an existing working choice involving the outside location, define its fixed amenity composite by

$$\mathcal{B}_\ell = (F^0_{ijgk})^{1/\chi} \frac{(q^{R,0}_i)^{\alpha_\ell} (\mathcal{P}^0_\ell)^{\gamma_\ell}}{y^0_\ell}, \qquad i = o \text{ or } j = o. \tag{C.32}$$

These are the outbound, inbound, and outside–outside composites in (B.1). They index only the

local origin or local destination, or the single outside worker stock. They introduce no unrestricted residual for an internal residence–workplace pair. For nonemployment, choose B^U_{ik} so that $(\Phi^0_\ell)^\chi = U^0_{ik}$ for each nonworking choice at i; conditional residential probabilities then equal $U^0_{ik}/\sum_{a\in\mathcal{I}_r} U^0_{ak}$ in every sector. Broader nest probabilities allocate the nonworkers across sectors as in (B.35).

These inversions apply to choices with positive baseline mass.

C.6.2 Commuting gravity

Consider internal commuters in one skill–sector group and suppress its indices. The residence and workplace amenities are the fitted factors recovered in (C.31); the workplace amenity includes the sector dimension suppressed here. Let $T = D_{ck}$ be the common local dividend and set $\gamma_c = (1-\alpha_{H,c})s_N$. Cobb–Douglas maximization gives

$$\Phi_{ij} = \underbrace{\frac{B^R_i}{(q^R_i)^{\alpha_{H,c}} P_i^{\rho\gamma_c}}}_{\mathcal{R}_i} \underbrace{\frac{B^W_j (w_j + T)}{P_j^{(1-\rho)\gamma_c}}}_{\mathcal{W}_j} e^{-\kappa t_{ij}}. \tag{C.33}$$

The normalized-power probabilities within the bilateral choice group are proportional to Φ^χ_{ij}. Thus its internal flows have the gravity representation

$$F_{ij} = a_i b_j e^{-\chi\kappa t_{ij}}, \qquad \log F_{ij} = \delta_i + \zeta_j - \chi\kappa t_{ij}. \tag{C.34}$$

Writing M for the group's commuter mass and $Z = \sum_{a,b}(\mathcal{R}_a\mathcal{W}_b)^\chi \exp(-\chi\kappa t_{ab})$, the factors can be chosen as $a_i = M\mathcal{R}^\chi_i/Z$ and $b_j = \mathcal{W}^\chi_j$. At baseline, positive fitted factors solve the margin equations

$$O^0_i = a^0_i \sum_j b^0_j e^{-\chi\kappa t_{ij}}, \qquad D^0_j = b^0_j \sum_i a^0_i e^{-\chi\kappa t_{ij}}, \tag{C.35}$$

where O^0_i and D^0_j are the calibrated resident-worker and workplace margins for this group. One normalization fixes their otherwise arbitrary relative scale. Given baseline budgets and prices, the fitted factors identify the corresponding residence and workplace amenity composites. The fitted gravity system reproduces these tract margins and predicts the bilateral counts. An observed zero on a feasible link is compatible with a positive fitted expected flow.

To see how the model solves a counterfactual, let π^0_{ij} be the normalized baseline fitted flows. Within this group,

$$\pi^1_{ij} = \frac{\pi^0_{ij}\exp(\chi\Delta v_{ij})}{\sum_{a,b}\pi^0_{ab}\exp(\chi\Delta v_{ab})}, \tag{C.36}$$

$$\Delta v_{ij} = \log\frac{w^1_j + T^1}{w^0_j + T^0} - \alpha_{H,c}\log\frac{q^{R,1}_i}{q^{R,0}_i} - \gamma_c\left[\rho\log\frac{P^1_i}{P^0_i} + (1-\rho)\log\frac{P^1_j}{P^0_j}\right]. \tag{C.37}$$

The common commute cost and fixed amenities cancel in the value change; they remain encoded in π_{ij}^0. Market clearing determines wages, rents, shopping prices, and dividends, while the higher nests determine the group's counterfactual mass. This update uses fitted probabilities, so a zero realized survey or administrative count need not prevent a positive model flow after entry.

Residence and workplace factors incorporate income, prices, amenities, and group normalization. Higher choice nests determine the group's mass; boundary-flow calibration connects it to the outside economy. Baseline internal commuting probabilities are fitted to travel times and tract margins. Bilateral counts are predictions, with positive expected flows on feasible links between positive margins. This covariate-based approach follows the calibration principle examined by Dingel and Tintelnot (2026), who distinguish estimated probabilities from noisy realized counts in granular commuting data. Exact-hat updates can use these fitted probabilities.

The functional-form restriction can be seen by substituting the auxiliary Stone–Geary demand into the same joint-choice structure. Resources in (C.33) would become $w_j + T - q_i^R \bar{h}_c$, and log flows would contain the additional term $\chi \log(w_j + T - q_i^R \bar{h}_c)$. This origin–destination interaction generally cannot be absorbed by separate fixed effects. It arises from purchasing power: the same minimum housing bill absorbs a larger fraction of resources at a lower-paying workplace. The Cobb–Douglas specification makes that purchasing-power relationship multiplicatively separable while retaining individual and ownership budget consistency.

C.6.3 Production costs and payroll

Appendix B.7 derives unit cost and factor demands. The following calibration applies to each service sector $g \in \mathcal{G}_N$.

The baseline shares $(\alpha_L^N, \alpha_C^N, \alpha_X^N) = (0.53, 0.13, 0.34)$ are the nontradable-composite shares in Table C.3, common to N_H and N_L. They are the shares α_L^g, α_C^g, and α_X^g of both service sectors $g \in \mathcal{G}_N$ in (17). Because production is Cobb–Douglas, these shares hold in every tract and at any factor prices, so no tract-level inversion is needed. The outside category is purchased intermediate inputs other than real-estate services. Property taxes enter through the tax-inclusive commercial rent.

The model's labor bill equals payroll:

$$W_{jg} = \sum_k w_{jk}^g L_{jk}^g = \alpha_L^g C_{jg} = \alpha_L^g R_{jg}. \tag{C.38}$$

The same wage determines firms' labor costs and individuals' employment income. The labor share above is applied directly to payroll.

Observed revenue, unit cost, and establishment mass determine the baseline fixed-input shifter through the zero-profit condition:

$$\varphi_{jg} = \frac{R_{jg}^0}{\sigma_N c_{jg}^0 (m_{jg}^0)^{1/\epsilon_m}}. \tag{C.39}$$

The inversion uses the specified shopping elasticity, production technology, and property tax. Reported entrant wages provide a validation comparison for the common workplace–skill tradable wages used in (24).

C.6.4 Service revenue and productivity inversion

The inversion recovers the service productivity and operating-cost shifters that make household spending, the shopping network, and service revenue consistent in the baseline equilibrium. It adapts the approach of Oh and Seo (2023) and proceeds in four steps. Revenue targets are inferred from payroll. The service expenditure share is set so that household spending covers these targets in aggregate. Destination productivities allocate that spending across tracts. Operating-cost shifters then reconcile revenue per establishment with free entry. The inversion is performed separately for each event's commuting zone, and I suppress the CZ index below. Superscript 0 denotes the baseline, i indexes shopping origins, j indexes destinations, and $g \in \mathcal{G}_N = \{N_H, N_L\}$ indexes the two service sectors.

Revenue targets. Service sales are not observed by tract, so I infer them from payroll. Let W_{jg}^0 denote annual service payroll and m_{jg}^0 the number of establishments in tract j and sector g. Variable production and establishment operation use the same technology, so under free entry labor payments equal the labor share α_L^N of revenue (equation (C.38)). The revenue targets are therefore

$$R_{jg}^0 = \frac{W_{jg}^0}{\alpha_L^N}, \qquad R_j^0 = \sum_{g \in \mathcal{G}_N} R_{jg}^0, \qquad R^0 = \sum_j R_j^0. \tag{C.40}$$

These targets follow from payroll and the production specification; they are not independently observed sales.

Aggregate expenditure. Let Q_i^0 denote the income remaining after housing, including dividends, that is assigned to shopping origin i, before the service expenditure share is applied. An employed individual assigns the fraction ρ of this income to her residence and $1-\rho$ to her workplace, and a nonworker assigns all of it to her residence. Inward commuters therefore contribute to the budgets of their workplace origins, while income assigned to origins outside the CZ is excluded from Q_i^0. Service expenditure originating at i is $\mathcal{E}_i^0 = s_N Q_i^0$.

Let $\bar{\pi}^O$ be the measured share of these budgets spent outside the CZ, weighted by spending. The service expenditure share at which household spending covers aggregate revenue is

$$s_N^{\text{req}} = \frac{R^0}{(1-\bar{\pi}^O)\sum_i Q_i^0}. \tag{C.41}$$

Because individuals cannot spend more than their income after housing on services, I cap the share

at one and define the pooled external budget $\mathcal{E}^{\text{ext}}$ as the remaining shortfall,

$$s_N = \min\{s_N^{\text{req}}, 1\}, \qquad \mathcal{E}^{\text{ext}} = R^0 - (1 - \bar{\pi}^O) s_N \sum_i Q_i^0. \tag{C.42}$$

Thus $\mathcal{E}^{\text{ext}} = 0$ whenever household spending can cover the revenue targets. Otherwise, $\mathcal{E}^{\text{ext}}$ is the smallest additional budget that reconciles the aggregate accounts. It represents purchases financed outside the modeled household budgets.

Spatial allocation and productivity. Given total spending, the inversion recovers where it goes. Let Γ_{ij} and Γ_i^O denote the initial shopping weights for destination j and for the outside alternative. These weights carry the shopping network's travel-cost structure and its baseline normalization of destinations. For a vector of destination multipliers $\delta = (\delta_j)_j$, the shares of origin i's spending that go to j and outside the CZ are

$$\pi_{ij}(\delta) = \frac{\Gamma_{ij}\delta_j}{\Gamma_i^O + \sum_{j'} \Gamma_{ij'}\delta_{j'}}, \qquad \pi_i^O(\delta) = \frac{\Gamma_i^O}{\Gamma_i^O + \sum_{j'} \Gamma_{ij'}\delta_{j'}}. \tag{C.43}$$

The outside weight is fixed, which pins down the level of the destination multipliers.

The external budget follows the distribution of household purchases, so households finance the fraction $1 - \mathcal{E}^{\text{ext}}/R^0$ of each tract's revenue. I recover δ by solving

$$\sum_i \mathcal{E}_i^0 \pi_{ij}(\delta) = \left(1 - \frac{\mathcal{E}^{\text{ext}}}{R^0}\right) R_j^0 \qquad \text{for each active destination } j. \tag{C.44}$$

Summing these equations across destinations shows that

$$\frac{\sum_i \mathcal{E}_i^0 \pi_i^O(\delta)}{\sum_i \mathcal{E}_i^0} = \bar{\pi}^O, \tag{C.45}$$

so the inversion preserves aggregate shopping leakage while allowing each origin's outside share and its bilateral destination shares to change.

I solve the system by iterative proportional fitting. Each iteration scales every destination multiplier by the ratio of its target to its predicted household revenue, with a common normalization that keeps the outside weight fixed. The calculation keeps the shopping network's existing support, so every destination with a positive revenue target must be reachable from origins with positive budgets.

The multipliers map into productivity through CES demand. With elasticity $\sigma_N > 1$, a destination's attractiveness is proportional to its price raised to the power $1 - \sigma_N$, and at fixed input prices and establishment counts, higher productivity lowers price proportionally. Relative to the

initial normalization, the recovered multiplier therefore corresponds to

$$\frac{A^0_{jg}}{A^{\text{ref}}_{jg}} = \delta_j^{1/(\sigma_N-1)}, \qquad \frac{p^0_{jg}}{p^{\text{ref}}_{jg}} = \delta_j^{-1/(\sigma_N-1)}. \tag{C.46}$$

A destination that receives more revenue than the shopping network alone would send it is thus inferred to have more productive service firms, whose lower prices draw more spending. The productivity adjustment is common to the two service industries within a tract, and their baseline shares of the tract's spending are set separately as

$$\vartheta^0_{jg} = \frac{R^0_{jg}}{R^0_j}. \tag{C.47}$$

Together, the tract multipliers and industry spending weights reproduce each tract–industry revenue target. The recovered productivities are conditional on the maintained shopping preferences and cost specification, and they do not separate physical productivity from unobserved attractiveness of a destination.

Operating costs and baseline consistency. With CES markup pricing, variable profits equal revenue divided by σ_N, so free entry determines baseline operating expenditure per establishment,

$$f^0_{jg} = \frac{R^0_{jg}}{\sigma_N m^0_{jg}}. \tag{C.48}$$

I recover the operating-cost shifter that delivers this expenditure at baseline input prices and productivity (equation (C.39)), which preserves the observed establishment counts. Baseline amenity normalizations absorb the resulting change in service price levels, so the inversion also preserves baseline worker allocations. The baseline equilibrium therefore reproduces the employment, payroll, revenue, and establishment-count targets.

Counterfactual expenditure and revenue. The recovered productivity and operating-cost shifters, the service expenditure share, and the total external budget are held fixed in counterfactuals, while establishment counts and prices are endogenous. Let hats denote ratios to baseline values, and let π^0_{ij} and $\pi^{O,0}_i$ be the fitted baseline shopping shares. The counterfactual share of origin i's spending received by tract j and sector g is

$$\pi'_{ijg} = \frac{\pi^0_{ij}\vartheta^0_{jg}\widehat{m}_{jg}\widehat{p}^{1-\sigma_N}_{jg}}{\pi^{O,0}_i + \sum_{j',g'}\pi^0_{ij'}\vartheta^0_{j'g'}\widehat{m}_{j'g'}\widehat{p}^{1-\sigma_N}_{j'g'}}. \tag{C.49}$$

Outside opportunities stay fixed, but the share spent outside the CZ changes as inside prices and varieties change.

Given counterfactual origin budgets $\mathcal{E}'_i$, household purchases and total revenue are

$$R_{jg}^{HH\prime} = \sum_i \mathcal{E}'_i \pi'_{ijg},$$
$$R'_{jg} = R_{jg}^{HH\prime} + \mathcal{E}^{\text{ext}} \frac{R_{jg}^{HH\prime}}{\sum_{j',g'} R_{j'g'}^{HH\prime}}. \tag{C.50}$$

The external budget is fixed in nominal terms, but its allocation follows current household purchases. The revenue targets are imposed only in the baseline. In counterfactuals, revenues are determined by equilibrium spending.

C.7 Equilibrium computation and verification

The current solver uses sector–region ordering for high-skill workers and region–sector ordering for low-skill workers, followed by the participation and workplace nests, with tract-specific residential supply.

Individual choices are evaluated using fixed baseline predicted probabilities and changes in utility, as in (B.33). Fixed amenities and commuting costs cancel from utility ratios and remain encoded in those probabilities. Equation (B.46) normalizes the available choices in each equilibrium.

Each equilibrium jointly solves for wages by sector and skill, residential and commercial rents, service establishment mass, and portfolio dividends, all represented in logs. A matrix-free Newton method uses GMRES steps and a line search. Continuation in the size of the entry is available when a direct full-entry solve fails. Reported equilibria satisfy the full counterfactual specification.

Saved reference states are re-verified before reuse, and newly solved states satisfy equilibrium and accounting residual tolerances of 10^{-8}. Each saved state allocates positive disposable income to every available individual choice.

Dollar equivalent variation is evaluated after the equilibrium solve. A bracketed scalar root-finder finds the transfer at reference prices that matches the entry opportunity index; this calculation does not solve the equilibrium again. Safeguarded interpolation and bisection give the same equivalent-variation solution. Initializing nearby counterfactual exercises from verified states reduces the equilibrium iterations without relaxing the residual checks.

No recorded cash award enters the current equilibria. The new-property exemption and endogenous county property-tax rates do enter the equilibrium. Shopping rules and the combined home-shopping/inbound counterfactual use their own no-entry equilibria.

Of 226 public events, 216 have certified observed-entry results. Nine remain blocked by source data or metadata and one lacks a certified baseline. The reported observed-entry sample therefore spans 95 CZs.

Numerical convergence establishes that the reported states satisfy the specified equations; it does not establish uniqueness or empirical validity.

C.8 Exposure for the validation scorecard

Updated results pending. A current-model scorecard matching the empirical worker-source and exposure definitions has not yet been produced. The available baseline-residence, neighboring-tract proxy is a preliminary diagnostic and is not substituted for that validation.

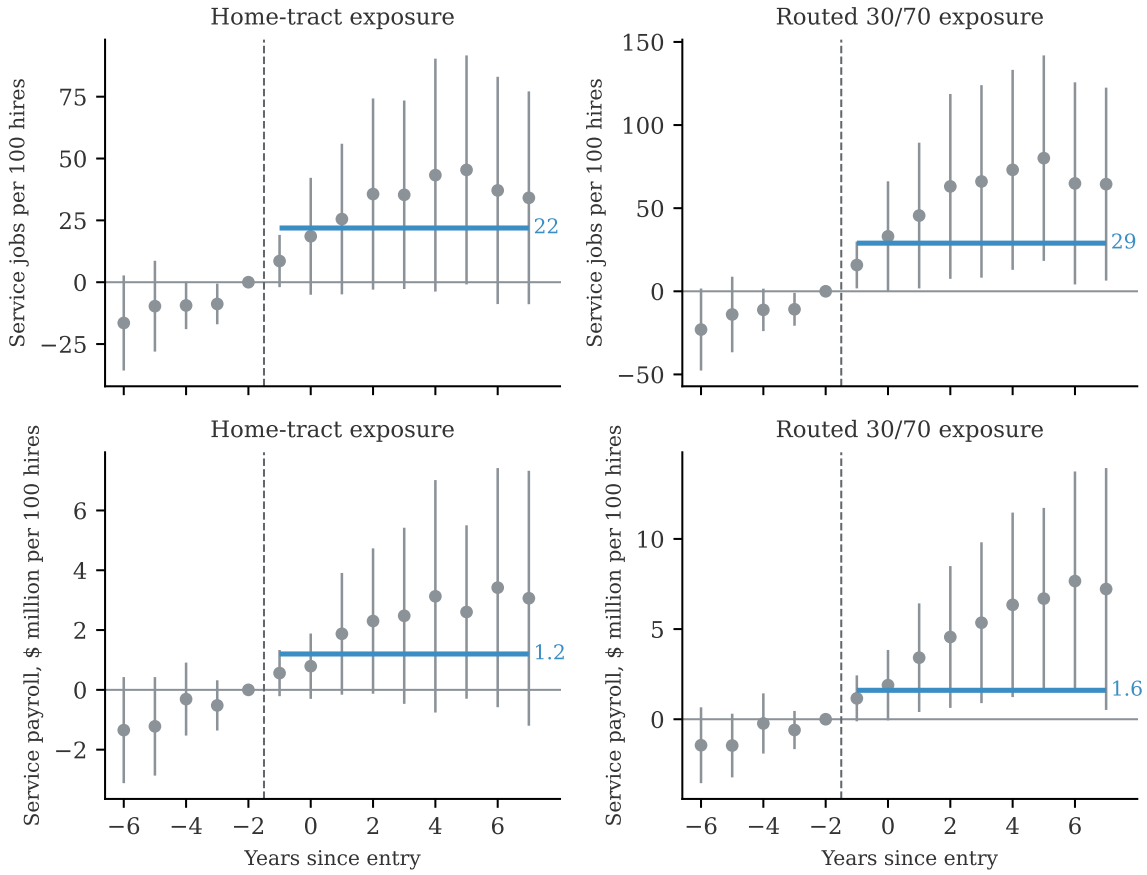
D Validation: Supporting Comparisons

This appendix reports the two comparisons that Section 5.4 cites but does not show, the Fact 3 event study for all nontradables and the stacked contrast between the top and bottom quartiles of exposed tracts.

All nontradables. Figure D.1 repeats Figure 7 with the broader service definition of the release, which adds education and health (NAICS 61 and 62) to retail, food and accommodation, and other services. The model's response is 22 jobs per 100 exposed hires under the home-tract exposure against a years-three-to-seven mean of 39 in the data, and 29 against 70 under the hyperlocal exposure; payroll is \$1.2 million against \$2.9 million and \$1.6 million against \$6.7 million. The ratios are 0.56 and 0.42 for jobs and 0.41 and 0.24 for payroll, against 0.91 and 0.68 for core jobs and 0.51 and 0.39 for core payroll on the same years-three-to-seven means in the main text. The model lies inside the annual 95 percent interval at every horizon under the home-tract exposure and at every horizon but year five of the payroll design under the hyperlocal exposure. The gap that opens when education and health are added is the one the main text attributes to how those sectors are funded and sited; the model treats them as demand-driven services and has no separate margin for them.

The stacked quartile contrast. The release also reports a stacked design that compares the change in service employment between the top and bottom quartiles of exposed tracts within each event. Table D.1 places the model's analog beside it. The model's contrast is a tenth of the data's under either exposure. The ratio decomposes exactly into the ratio of exposure gaps, how much more exposed the top quartile is than the bottom in the model relative to the data, and the ratio of responses per unit of gap. For the hyperlocal exposure at year seven the model-to-data ratio of 0.11 is 0.38 times 0.30; for the home-tract exposure at year three it is 0.10, equal to 0.29 times 0.34. Both factors fall short. The exposure gap is smaller in the model because the data concentrate hires' residences in a few tracts near each site while the model spreads them more evenly across the commuting zone, and the response per unit of gap is smaller for the same reason the continuous payroll response is, the model's service jobs per dollar of local spending. The stacked design therefore tests the concentration of residences as much as the demand mechanism, and the continuous design in the main text is the one that isolates the latter.

Figure D.1: All nontradables where the entrant's workers live: empirical event study and the model



Notes: As Figure 7, for all nontradables (NAICS 44 to 45, 61, 62, 72, 81). Circles are the released continuous event-study coefficients per 100 exposed hires with 95 percent confidence intervals; year −2 is the reference and hiring begins in year −1 (dashed line). The blue line is the model's static response from the same design on all 216 events.

Table D.1: Stacked top-minus-bottom quartile contrast: released estimates and the model

Exposure	Outcome	Data, mean of years 3–7	Model	Ratio
Home tract	Service jobs per 100 hires	39.2	3.6	0.09
Home tract	Service payroll ($ million)	3.39	0.20	0.06
Hyperlocal 30/70	Service jobs per 100 hires	30.5	4.0	0.13
Hyperlocal 30/70	Service payroll ($ million)	3.19	0.22	0.07

Notes: Released stacked estimates for all nontradables from files 1A and 1C (30 and 40 commuting zones), averaged over years three to seven; the release gives no cross-year covariance, so no interval is reported for the mean. Model values are equal-event averages of the top-minus-bottom quartile contrast on the 53 single-event commuting zones, for the model's full service sector, so they are broad-outcome analogs rather than core-sector tests. The decomposition in the text uses the released and model exposure gaps at the stated horizon.

E Employment, Welfare, and Incidence: Supporting Results

This appendix supports Section 6 in the order of its subsections. The observed sample contains 216 events in 95 CZs. Observed-event summaries weight events equally. All money is annual 2023 dollars; "resident" refers to the model's adult population, including nonworkers aged 25–64; other-age nonworkers are excluded. Appendix F reports the award-accounting and location comparisons for Section 7.

E.1 Employment and the payroll accounting chain

This subsection supports the employment and payroll results in Section 6. Let J_e be the promised job target and D_e the realized employment increase after the exemption in the entrant's tract and industry, $\Delta T_e^{\text{other}}$ the change in other tradable jobs, and ΔN_e the change in service jobs. The observed full model satisfies

$$\Delta L_e^{\text{net}} = D_e + \Delta T_e^{\text{other}} + \Delta N_e, \qquad M_e^N = 100\frac{\Delta N_e}{J_e}, \qquad M_e^{\text{net}} = 100\frac{\Delta L_e^{\text{net}}}{J_e}. \tag{E.1}$$

The service multiplier per net tradable job is $100\Delta N_e/(D_e + \Delta T_e^{\text{other}})$ when its denominator is positive. The job target is calibrated without the exemption; D_e can differ slightly from J_e. The reported median is taken over the event-level ratio, rather than a ratio of separately reported medians.

Let ΔP_e be the additional payroll at the entrant's site. Each ledger entry is $100\Delta X_e/\Delta P_e$. Resident earnings equal workplace payroll less inbound workers' earnings plus residents' earnings outside the CZ. Disposable income adds nonemployment payments and private rental dividends. Resident service budgets are $s_{N,c}(1-\alpha_i)$ times income at each residence. Home- and workplace-origin budgets can differ from resident budgets because workers cross the CZ border. Allocating these origin budgets over shopping destinations gives local spending and spending outside modeled destinations. Identities are checked in levels within events; medians across events are not an additive accounting chain.

Table E.1 reports the full chain per \$100 of additional payroll in the shocked tract and industry. Residential rental dividends add about \$9, and small changes in nonwork payments bring the increase in resident disposable income to about \$50. Housing expenditure rises by about \$10, and spending on tradable goods absorbs part of the remainder, leaving residents' service budget about \$28 higher. Net imports of workplace-origin shopping budgets add about \$6, while additional purchases from service firms outside the CZ absorb about \$10. Spending at service firms inside the CZ therefore rises by about \$25. These figures are medians of event-level ratios, so the reported amounts need not add together; the accounting identities hold within each event.

Table E.1: Payroll, resident income, and service spending per $100 of additional payroll at the entrant's site

Account	Median	P10	P90
CZ workplace payroll	78.07	68.22	89.12
Resident labor earnings	39.64	28.88	51.98
Nonemployment payments	0.53	0.13	0.99
Rental dividends	8.93	6.23	12.52
Resident disposable income	49.64	35.93	65.49
Housing expenditure	10.16	6.89	14.16
Resident service budget	28.38	19.52	42.43
Net workplace budget imports	6.16	3.81	8.91
Local-origin service budget	34.98	26.39	48.82
Spending outside	10.02	6.77	15.02
Spending at local service firms	25.11	15.93	35.62

E.2 Welfare and its components

This subsection defines the welfare measure used throughout Section 6. For skill k and place of residence a, a county or the CZ, let V_{ak}^t be the model's inclusive value over feasible residential and employment opportunities. The cash equivalent EV_{ak} solves

$$V_{ak}^0(y^0 + \mathrm{EV}_{ak}, p^0) = V_{ak}^1(y^1, p^1), \qquad \mathrm{EV}_a = \sum_k N_{ak}^0 \mathrm{EV}_{ak}, \qquad \overline{\mathrm{EV}}_a = \frac{\mathrm{EV}_a}{\sum_k N_{ak}^0}. \tag{E.2}$$

The transfer is uniform across the baseline alternatives of a given skill. Prices and primitives on the left are held at their baseline values. Reported welfare per entrant job is EV_c/J_e; its wage equivalent divides that ratio by the employment-weighted post-entry wage in the entrant's tract and industry. Both ratios are formed within events before reporting their medians. The measure values a place's opportunities and does not track the welfare of a fixed cohort. Regional indices are nonlinear, so host-county and rest-of-CZ welfare are evaluated directly, not obtained by subtracting or adding tract welfare totals.

Updated results pending. The current-model welfare decomposition has not yet been exported. It must include the endogenous property-tax rate alongside wages, dividends, housing prices, and service prices; the earlier four-block decomposition is not reported as a current result.

E.3 Commuting and shopping restrictions

This subsection supports the comparisons in Figure 8. Write Y_e^{ab} for the entry-minus-no-entry response, where $a = 1$ imposes home-only shopping and $b = 1$ holds inbound headcounts fixed by skill. Every specification receives the same calibrated full-model productivity shock. The original promised-job denominator is held fixed, and actual employment at the entrant's site can differ. The joint interaction is

$$I_e = Y_e^{11} - Y_e^{10} - Y_e^{01} + Y_e^{00}. \tag{E.3}$$

All contrasts are computed within the same event before taking medians or percentiles. Fixed inbound headcounts do not freeze workplace allocations. Home-only shopping has its own solved no-entry comparator; the joint specification uses that comparator's inbound totals. Restricted welfare uses the opportunity values of the corresponding specification.

Table E.2: Paired mechanism effects relative to the full model

Paired effect	Median	P10	P90
Home: Service jobs	-3.61	-5.46	-1.39
Home: Net jobs	-1.50	-2.93	0.02
Home: Welfare	-0.08	-0.52	-0.01
Inbound: Service jobs	-5.78	-8.95	-3.65
Inbound: Net jobs	-21.73	-34.81	-12.59
Inbound: Welfare	0.41	0.16	1.65
Both: Service jobs	-7.87	-11.16	-4.73
Both: Net jobs	-21.89	-34.31	-13.10
Both: Welfare	0.24	0.01	1.06
Interaction: Service jobs	1.59	0.84	2.49
Interaction: Net jobs	1.30	0.43	2.27
Interaction: Welfare	-0.03	-0.26	0.00

Notes: Employment is per 100 original promised jobs; welfare is annual dollars per baseline resident. Restriction rows subtract the full-model response. Interaction rows use equation (E.3). Percentiles describe the 216 paired event effects.

E.4 Additional supply and shopping restrictions

Tables E.3 and E.4 include every restriction with a certified result in the final November 3–4 export. They retain that batch's participation elasticity $\eta = 3.02$ and inbound-commuting coefficients; they do not combine these counterfactuals with the later recalibrated full-model outcomes. Each paired effect uses exactly the events completed for that restriction. The original productivity shock and added commercial capacity are held fixed, and all job changes retain the original promised-job denominator.

Fixed space shuts down the ordinary supply response, while preserving the entrant's explicit capacity addition. Fixed migration holds the inside-CZ and adjacent-region populations constant by skill; residential choices within each region remain free. Fixed participation and sector choice hold their respective conditional shares at the no-entry values. These labor restrictions are quantity constraints implemented with nonpecuniary access wedges, rather than newly estimated elasticities. The reported opportunity welfare includes the restricted choice system.

Home-origin shopping retains the destination network. Strict home-tract shopping removes other destinations and the outside option; it is reported only for the 59 certified events in this historical batch. There were 154 unsupported events and three failed solves for that restriction. This is distinct from the later home-or-nearest-service-tract exercise. No-outside shopping retains the 0.74/0.26 origin split and the within-CZ network but removes the outside destination for inside origins. Shopping restrictions use their own solved no-entry comparator, with no reinversion of

productivity or preference shifters.

Table E.3: Completed mechanism restrictions: median equilibrium responses

Specification	Events	Service jobs	Net jobs	EV/resident
Full model	216	22.46	74.32	10.32
Home-origin shopping	216	18.75	73.06	10.26
Fixed inbound commuters	216	16.18	50.05	10.98
Both restrictions	216	14.37	49.78	10.82
Fixed residential space	216	19.84	69.30	8.00
Fixed commercial space	216	22.49	75.24	10.62
Both space supplies fixed	216	19.76	70.86	8.32
Fixed regional populations	216	12.49	56.29	-0.01
Fixed participation shares	216	22.71	72.96	10.55
Fixed sector shares	215	22.56	74.02	10.38
Strict home-tract shopping	59	16.60	66.56	21.75
No outside shopping	215	29.42	76.96	10.06

Notes: Service and net jobs are per 100 original promised jobs. EV is annual dollars per baseline resident under the corresponding specification. Samples differ where cases are unsupported, failed, or unfinished; the strict home-tract row must not be compared directly with the full model's full-sample median. Table E.4 instead compares every restriction with the full model on its own identical event sample. These are the earlier-calibration November 3–4 results.

Table E.4: Completed mechanism restrictions: median paired changes from the full model

Restriction	Events	Δ service	Δ net	Δ EV/resident
Home-origin shopping	216	-3.61	-1.50	-0.08
Fixed inbound commuters	216	-5.78	-21.73	0.41
Both restrictions	216	-7.87	-21.89	0.24
Fixed residential space	216	-2.45	-4.66	-2.82
Fixed commercial space	216	0.08	0.75	0.04
Both space supplies fixed	216	-2.07	-3.34	-2.56
Fixed regional populations	216	-9.34	-17.48	-10.29
Fixed participation shares	216	0.04	-0.73	-0.02
Fixed sector shares	215	0.42	-0.02	0.00
Strict home-tract shopping	59	-2.79	-9.22	-0.40
No outside shopping	215	7.05	2.88	-0.11

Notes: Each restriction's entry-minus-no-entry response is compared with the same event's unrestricted response using the same parameter version. The table reports the median of these paired differences, not the difference of medians. Employment changes are per 100 original promised jobs; welfare changes are annual dollars per baseline resident. Restricted shopping uses its own no-entry equilibrium. No missing cases are assigned zero effects, and these separate restrictions are not additive components.

E.5 Spatial incidence, exposures, and regression checks

Table 6 uses baseline commuting and shopping probabilities to measure exposure to the expansion's payroll. For event e, let j_e and g_e denote the expanded workplace and industry, $\mathcal{I}_e$ the tracts in its CZ, and O the outside residence. Let L^0_{e,ij_eg_ek} be baseline workers of skill k who live in i and work in the target location-sector, and let w^0_{e,j_eg_ek} be their cash wage. The share of the target

location-sector's payroll received by residents of i is

$$p_{ei} = \frac{\sum_k w^0_{e,j_e g_e k} L^0_{e,ij_e g_e k}}{\sum_{r \in \mathcal{I}_e \cup \{O\}} \sum_k w^0_{e,j_e g_e k} L^0_{e,rj_e g_e k}}. \tag{E.4}$$

These are fitted baseline worker flows, not the expansion's realized hires. Including outside residents in the denominator preserves payroll leakage through inbound commuting.

Write $b_{ei} = s_{N,c(e)}(1 - \alpha^0_{H,ei})$ for the fraction of income allocated to services after housing at residence i, using its baseline housing expenditure share $\alpha^0_{H,ei}$. Let $S^0_{e,ij}$ be the baseline probability that shopping from origin i reaches tract j, after the service-productivity inversion. With home-origin share ρ, the two exposures are

$$H_{ej} = 100\rho \sum_{i \in \mathcal{I}_e} p_{ei} b_{ei} S^0_{e,ij}, \tag{E.5}$$

$$W_{ej} = 100(1-\rho) S^0_{e,j_e j} \sum_{i \in \mathcal{I}_e \cup \{O\}} p_{ei} b_{ei}. \tag{E.6}$$

Outside residents' home-origin purchases remain outside the CZ; their work-origin purchases enter through W_{ej}. Both measures are dollars reaching a destination per \$100 of gross baseline payroll in the target location-sector. They do not measure the actual increase in residents' income or service spending. The shopping probabilities reproduce baseline household-financed service revenue; any pooled external service demand is excluded when recovering that household revenue target.

The regressions use the same complete-case sample in every column, excluding the expansion tract and tracts without a positive measurable baseline service wage. Controls are log one plus service employment, log average service cash wages, and log residential rents. Average wages combine both service industries and both worker skills. Event-by-county fixed effects remove the county mean for each event; observations receive equal weight. Standard errors use CZ-clustered CR1 covariance, accounting for absorbed effects in the finite-sample correction, and two-sided t_{94} tests.

As a prediction check, I remove event-by-county means and estimate the slopes on all but one CZ, repeating for each CZ. Applying these slopes to the omitted CZ gives a pooled within-county R^2 of 77.18 percent for the full specification, compared with 77.57 percent in sample. This checks whether the relationship carries across simulated CZs; it does not constitute validation against independent observed outcomes.

Updated results pending. The illustrative spatial-incidence map and exact geographic service-job decomposition remain pending for the current model.

E.6 Standardized shocks and market characteristics

The 500-job and proportional-job designs each cover 216 events in 95 CZs and both industries, without a property exemption. The proportional target is 0.5 percent of baseline employed CZ residents. The payroll design retains the 162 events with completed admissible solves in both

industries; remaining cases are not imputed. Each outcome uses its own realized job increase in the target location-sector in the per-job denominator. These are static comparisons, not transition paths.

Table E.5: Standardized shocks by entrant industry

Design / industry	Events	Service jobs	Net jobs	EV/resident	Payroll/job
500 jobs TH	216	25.27	77.52	6.47	105,577
500 jobs TL	216	15.67	69.59	4.33	75,515
0.5 percent TH	216	28.02	79.91	37.83	116,185
0.5 percent TL	216	18.66	70.86	25.50	84,681
50 million dollars TH	162	30.97	90.53	9.51	99,846
50 million dollars TL	162	18.28	75.53	8.80	75,080

Notes: Event-weighted medians. Service and net jobs are per 100 jobs added in the target location-sector. Welfare is annual dollars per baseline resident; payroll per job is annual incremental payroll in the target location-sector divided by its job increase. TH and TL denote high- and low-skill-intensive tradables. Sample sizes are identical across industries within each design but differ across designs.

The cross-market regressions include industry effects and exclude the calibrated service expenditure share. Predictors are standardized after removing industry means. Shopping leakage is weighted by baseline shopping-origin budgets; residential supply elasticity is weighted by baseline residents. The inbound share is inbound workers as a fraction of jobs in the CZ, not the share of residents working outside it. Standard errors use CZ-clustered CR1 covariance with the absorbed effects in the finite-sample adjustment; significance uses two-sided t_{94} tests. These are descriptive projections, not uncertainty intervals for the structural parameters or the numerical solutions.

Table E.6: Service multipliers and market characteristics: 500-job shock

	(1)	(2)	(3)	(4)	(5)	(6)
Log baseline employed residents	-2.15***	-2.75***	-3.83***	-0.89	-2.79***	0.44
	(0.79)	(0.74)	(0.76)	(0.92)	(0.99)	(1.42)
Inbound employment share		-1.28**			-0.04	-0.12
		(0.62)			(0.78)	(0.78)
Shopping leakage			-2.63***		-2.37***	-1.31
			(0.63)		(0.79)	(0.84)
Residential supply elasticity				1.77**	1.26*	1.12*
				(0.74)	(0.72)	(0.67)
Shopping elasticity, $\sigma_{N,c}$						-2.64**
						(1.17)
Commuting elasticity, χ_{ce}						-0.60
						(0.77)
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes
Within-industry R^2 (%)	8.46	10.80	15.99	11.35	17.36	20.64
Observations	432	432	432	432	432	432
CZ clusters	95	95	95	95	95	95

Notes: Same definitions as Table 7. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

F Counterfactual Costs, Locations, and Entrant Types

Except for the fiscal comparisons in Section F.1, this appendix documents the November 3–4 batch used in Section 7. Observed-location outcomes cover 216 events. The model includes the 50-percent exemption on newly added property and finances county service needs through property tax rates, with no resident fiscal transfer. Dollar values are annual 2023 dollars. Welfare is residential-opportunity equivalent variation and is not additive across nested geographic areas. These experiments use the earlier participation elasticity $\eta = 3.02$ and the matching inbound-commuting coefficients, not the later 55-event fit. All observed-site comparators use that same earlier parameter version. The final export contains 7,742 certified runs; 60 failed, four unfinished, and 180 unsupported comparisons are not imputed. This is a completed-run report, not a claim that every planned experiment finished.

F.1 Fiscal returns

The fiscal exercise uses the November 5 labor calibration and 54 certified hypothetical entry scenarios, drawn from 70 records with verified local-benefit coverage. 15 records lack both a usable dollar amount and a documented percentage, and 1 lacks a supported baseline. None is assigned zero cost. The priced sample retains 3 duplicate references and treats retention-only, never-hired, and uncertain-site records as hypothetical expansions. It is a scenario summary, not an estimate for 70 independent realized awards. Existing baselines are reused; 33 priced records have revised hiring-year information and 1 has a revised site that is not installed in this exercise.

For 43 scenarios, let B_e be the annual local benefit in 2023 dollars. Sources include measured benefits, contractual amounts, and flagged monetary proxies. Source totals are divided by the documented term, with a ten-year default where the term is unavailable. Where annual observations are available, each is deflated using its own price year before averaging; missing years are not zeros. Two supplemental amounts use a property-level PILOT saving and a later-investment appropriation, respectively; neither is treated as verified event-specific realized savings. Remaining source and investment-vintage qualifications are retained in the underlying records.

Let K_e^{add} denote the new commercial capacity, measured using the baseline space requirement for the promised-job expansion. Let $j(e)$ and $a(e)$ denote its tract and county. At baseline prices, the equivalent exemption fraction is

$$\xi_e = \min\left\{1, \frac{B_e}{t^0_{a(e)} r^{C,0}_{j(e)} K_e^{\text{add}}}\right\}. \tag{F.1}$$

This caps 10 scenarios at a full exemption; the excess recorded amount is retained in the input audit but is not paid in the model. For 11 scenarios without dollar amounts, ξ_e is the documented percentage instead. These include conditional maxima, phased percentages averaged across their documented years, tax-linked grants, and personal-property exemptions applied as proxies to the model's commercial capacity. Their costs are model-implied, not observed tax savings.

Productivity is calibrated to the promised-job target without an exemption. That productivity and K_e^{add} remain fixed when the exemption is introduced; the labor parameters are not re-estimated. Commercial space has a common gross occupancy price, so the owner rent on exempted space is

$$r_e^{C,\text{new}} = \frac{1 + t_{a(e)}}{1 + (1 - \xi_e)t_{a(e)}} r_{j(e)}^{C}. \tag{F.2}$$

Writing $\mathcal{R}_{a(e)}^{C,\text{old}}$ for current rental income on nonexempt commercial space, the host county's budget becomes

$$t_{a(e)} \left[\mathcal{R}_{a(e)} + \mathcal{R}_{a(e)}^{C,\text{old}} + (1 - \xi_e) r_e^{C,\text{new}} K_e^{\text{add}} \right] = G_{a(e)}. \tag{F.3}$$

Service needs $G_{a(e)}$ follow equation (30). No government cash transfer is paid to residents. The annual tax expenditure is the revenue forgone on the exempt fraction of new capacity at equilibrium prices and the equilibrium county rate,

$$C_e = \xi_e t_{a(e)} r_e^{C,\text{new}} K_e^{\text{add}}. \tag{F.4}$$

Consequently C_e need not equal B_e, even when the baseline cap does not bind. This is a fixed exemption fraction, not a fixed-dollar payment.

For geography A, cost per net job is $C_e/\Delta L_{eA}$, where ΔL_{eA} includes the expansion, changes at other tradable producers, and service jobs relative to no entry. Welfare per public dollar is EV_{eA}/C_e, using annual residential-opportunity equivalent variation. These quantities use the same cost numerator for the CZ and host county. Welfare is evaluated separately for each geography and is not additive across them. Each scenario receives equal weight; quantiles of ratios are not ratios of the separately reported quantiles. All 54 priced solves passed the equilibrium and export checks.

Table F.1: Local-benefit scenarios by input basis

Input group	Scenarios	Annual cost per net job ($) CZ	County	Welfare per public dollar CZ	County
Monetary inputs, uncapped	33	1,652	1,215	8.24	1.94
Monetary inputs, capped	10	7,420	5,723	1.08	0.40
Documented percentages	11	5,310	5,235	1.45	0.74

Notes: Within-group medians; annual 2023 dollars. Monetary inputs include flagged proxies. Capped cases receive at most a full exemption on added capacity. Percentage-only cases use model-implied costs. The three groups partition the 54 priced scenarios.

On the same 43 monetary-input scenarios, dividing the full annual recorded amount B_e by net CZ jobs gives a median of $2,000, compared with $2,100 using C_e. The pooled modeled figure of $3,600 also includes the 11 percentage-only scenarios, so it is not a same-sample comparison with the recorded-dollar figure. No full state-and-local award or lifetime cost is inferred from the pooled annual result.

Additional property-tax receipts finance service needs and are not credited as unrestricted funds against C_e. The ratios condition on entry and therefore do not identify the subsidy-induced employ-

Table F.2: Fiscal scenarios on the same supported baselines

Measure	No exemption	50% exemption	Event-specific
Commuting zone			
Service jobs per 100 promised jobs	24.18	23.11	22.84
Net jobs per 100 promised jobs	77.11	74.01	74.46
Annual tax expenditure per net job ($)	0	3,267	3,563
Annual welfare per baseline resident ($)	20.89	18.13	16.92
Resident welfare per public dollar	–	2.99	2.82
Host county			
Service jobs per 100 promised jobs	16.86	15.36	14.92
Net jobs per 100 promised jobs	92.74	90.19	89.19
Annual tax expenditure per net job ($)	0	2,763	3,025
Annual welfare per baseline resident ($)	31.37	24.56	22.70
Resident welfare per public dollar	–	1.35	0.86

Notes: Equal-weight medians for the same 54 scenarios under no exemption, the previous common 50-percent exemption, and the event-specific exemption. Productivity shocks, capacity, baselines, and labor parameters are held fixed. All employment and welfare changes are relative to no entry. Welfare per public dollar is undefined when tax expenditure is zero. All dollars are annual 2023 dollars. Differences between column medians are not medians of within-scenario differences.

ment gain or a causal return to public spending. Additionality and fiscal break-even calculations under this rule remain pending.

F.2 Alternative sites and paired payroll comparisons

Each event has up to ten alternative tracts drawn in its host county, excluding its own observed tract. The seed is fixed at 20261028. Sites are drawn before checking labor support; unsupported sites remain unpriced and are not replaced. Events without complete supported pairs are excluded. Table 9 retains only the 213 events for which all supported alternatives have completed both the fixed-job and fixed-payroll solves. The sample contains 2,026 alternatives, plus each observed site once. These completed grids need not represent the full event sample.

For the job design, productivity is chosen to reach the promised-job target before the exemption, then held fixed when the exemption is applied. For the payroll design, additional payroll is matched to the tract–industry payroll at the observed location, with employment allowed to adjust. The industry, capacity rule, and exemption fraction are retained. The amount of added space and exempted tax can vary across locations; no fixed recorded cash award is added. Income and welfare statistics use annual flows.

Let Y_{es}^d be the outcome at site s for event e in design $d \in \{J, P\}$, including the observed site. The range statistic is

$$100 \frac{\max_s Y_{es}^P - \min_s Y_{es}^P}{\max_s Y_{es}^J - \min_s Y_{es}^J}.$$

The tables report its median across complete event grids. The reported range reduction is 100 minus this median; it may be negative. Best-site welfare gains compare the maximum evaluated CZ

welfare with observed-site welfare within the same event. Employment/welfare rank correlations use Spearman ranks within the fixed-job candidate set. These are descriptive comparisons of evaluated sites, not causal estimates or an exhaustive county optimum.

F.3 County service gains and baseline site characteristics

Table F.3 reports the completed site regressions on 2,239 sites in 213 event grids and 93 CZs. The dependent variable is the host county's change in service jobs per 100 original promised jobs. The original denominator is retained when payroll is fixed and realized employment changes. Each regression absorbs event effects and gives every event equal total weight.

Home- and work-route exposures measure baseline dollars reaching county service businesses per $100 of gross payroll in the target location-sector. They combine skill-specific wages, commuting, housing and service expenditure shares, the 0.74/0.26 home/work origin split, and the inverted shopping probabilities. Baseline site controls are the target location-sector's mean wage and employment, service employment in the tract, and the residential supply elasticity. The calibrated service expenditure share enters the exposure accounting but is not an additional regressor. Required payroll in column (3) is an equilibrium outcome, so this column is an accounting diagnostic rather than a causal estimate.

Table F.3: County service-job gains across alternative sites

	Fixed jobs (1)	Fixed jobs (2)	Fixed jobs (3)	Fixed payroll (4)	Fixed payroll (5)
Home-route exposure	0.71***	0.45***	0.56***	0.51***	0.40***
	(0.10)	(0.07)	(0.06)	(0.04)	(0.03)
Work-route exposure	0.25	0.87***	0.47***	0.35***	0.69***
	(0.19)	(0.14)	(0.14)	(0.08)	(0.06)
Log baseline target location-sector wage		20.59***			11.82***
		(2.47)			(0.66)
Log(1 + target location-sector jobs)		-1.06***			0.12
		(0.33)			(0.08)
Log(1 + site service jobs)		-1.59***			-1.23***
		(0.14)			(0.10)
Residential supply elasticity		0.04			0.47
		(1.39)			(0.57)
Required payroll (million dollars)			1.87***		
			(0.39)		
Within-event R^2 (percent)	13.18	56.93	62.58	28.78	70.31
Observations	2,239	2,239	2,239	2,239	2,239
Events	213	213	213	213	213
CZ clusters	93	93	93	93	93
Event effects	Yes	Yes	Yes	Yes	Yes

Notes: The same complete paired sites enter every column. Columns (1)–(3) hold promised jobs fixed; columns (4)–(5) hold observed-site incremental payroll fixed. Required payroll is millions of annual dollars per 100 original promised jobs. Event fixed effects and equal total event weights are used throughout. Parentheses contain CZ-clustered CR1 standard errors; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, from two-sided t_{92} tests. Reported R^2 is conditional on event effects. These are descriptive relationships among the earlier-calibration model outcomes.

F.4 Mechanism effects on site rankings

Updated results pending. The current batch applies mechanism restrictions at observed sites only. Alternative-site rankings under these restrictions and whole-CZ location searches are not available.

F.5 Industry swaps

Each of the 216 observed sites has completed expansions in both tradable industries, retaining its original promised-job target and exemption fraction. The original-industry result is the observed-location solve; the other is the industry-swap solve. Medians use identical paired event samples. The comparison reports economic outcomes; returns per public dollar remain pending because the tax expenditure can change with the industry and site.

F.6 The additionality calculation

Each of the 54 recorded-subsidy scenarios in Table 10 is solved three times: with the recorded exemption, without it, and without entry. Let W_{ex} and W_{no} be residents' annual welfare relative to no entry when the firm enters with and without the exemption. If the exemption causes a share s of the entries it subsidizes, offering it yields W_{ex} in every scenario, while withholding it yields W_{no} from the share $1 - s$ of firms that would enter anyway and nothing from the rest. The expected gain from offering it is $W_{\text{ex}} - (1 - s) \cdot W_{\text{no}}$, which Table 11 divides by the exemption's annual tax expenditure. The exemption pays off when s exceeds $1 - W_{\text{ex}}/W_{\text{no}}$. In the median scenario, this threshold is 12 percent for CZ residents and 15 percent for host-county residents. Expected net jobs are computed the same way from net workplace jobs.